

Compressor Rotor Joints and Interference Fit Study

Mechanical Engineering Senior Project

by

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Senior Project

Sponsor: *Solar Turbines*

California Polytechnic

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6/4/2026

Statement of Disclaimer

Since this project is a result of a class assignment, it has been graded and accepted as fulfillment of the course requirements. Acceptance does not imply technical accuracy or reliability. Any use of information in this report is done at the risk of the user. These risks may include catastrophic failure of the device or infringement of patent or copyright laws. California Polytechnic State University at San Luis Obispo and its staff cannot be held liable for any use or misuse of the project.

Contents of this Report

This report fulfills the Senior Project requirement for the Mechanical Engineering Department. This document contains:

1. Scope of Work (SOW)
2. Preliminary Design Review (PDR) Report
3. Critical Design Review (CDR) Report
4. Final Design Review (FDR) Report
5. Project Expo Poster

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Scope Of Work (SOW)

By:

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Senior Project

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November 7th, 2025

<u>Project Overview</u> Solar Turbines need a comprehensive study on the effects axial and radial inference fits have on turbine compressor rotor stiffness. The aft weldment of the compressor has press-fit dowel pins and pilot joints to connect one weldment section to another. These fits need to be assessed through a finite element modal analysis, as well as practically tested through modal impacts to investigate how the fits affect the stiffness of the turbine's connected weldments. The experimental data will be used to validate the finite element analysis (FEA) model.	<u>Key Specifications</u> The engineering specifications listed in the table below are made to define how we will achieve Solar Turbines' requests during this project. Each specification is made to be validated through either a specified target or completion. <table border="1" data-bbox="954 621 1425 1016"> <thead> <tr> <th>Specification</th><th>Target Value</th></tr> </thead> <tbody> <tr> <td>Frequency</td><td>> 186 Hz</td></tr> <tr> <td>Effective Stress</td><td>< 20 MPa</td></tr> <tr> <td>Heat</td><td>1400 °C</td></tr> <tr> <td>Modal Analysis</td><td>Completion</td></tr> <tr> <td>Iterations</td><td>3 Iterations</td></tr> <tr> <td>Physical Tolerance</td><td>>±0.0001"</td></tr> <tr> <td>Machining</td><td>Completion</td></tr> <tr> <td>Weight</td><td>50 lbs.</td></tr> </tbody> </table>	Specification	Target Value	Frequency	> 186 Hz	Effective Stress	< 20 MPa	Heat	1400 °C	Modal Analysis	Completion	Iterations	3 Iterations	Physical Tolerance	>±0.0001"	Machining	Completion	Weight	50 lbs.
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<u>Scope and Deliverables</u> • <i>CAD model of the rotor part we are working on</i> • <i>Researched material properties</i> • <i>CAD model of compressor assembly including multiple parts as needed</i> • <i>ANSYS model to assist in analyzing stiffness and natural frequency</i> • <i>Completed model test procedure on a physical part</i> • <i>Final results of natural frequency and stiffness of parts</i> • <i>A final report summarizing all findings</i>	<u>What's Next</u> The next step of our project is to provide a full preliminary design review (PDR) of our plan to conduct the press fit analysis study. Within the next four weeks, we will ideate action plans and use various decision matrices to finalize our testing procedure. From there, we will proceed with the testing plan and gather results to present in the critical design review (CDR) and final design review (FDR).																		

Overview

Solar Turbines is a company that specializes in creating turbine engines with the purpose of power generation for prolonged periods of time. The turbine engines are designed to run for ~30,000 hours non-stop before requiring a complete engine overhaul. Some phenomena affecting engine vibration that Solar Turbines is experiencing with their engines are shifting between the aft (AFT) weldment and forward (FWD) weldment, resulting in reduced stiffness of the compressor rotor. Due to these issues, Solar Turbines has asked us to investigate these issues and propose a potential solution. Our main objective is to analyze how different axial and radial interference fits affect the stiffness of the compressor. The goal is to reduce shifting between weldments. The stiffness in rotor system comes from two sources; the rotor geometry and the bearings. The rotor geometry provides stiffness but the joints at the curvics or the weldments introduce flexibility (hence reducing the stiffness). For us to determine if our calculations would increase rotor stiffness, we will require experimental data from Solar Turbines regarding their current FWD and AFT weldment and compare those values to our experimental data. Overall, the scope of our project is to increase rotor stiffness and reduce shifting between weldments. To ensure that we are well informed and produce a product that not only meets these requirements, but surpasses expectations, we researched topics regarding interference fits and rotor dynamic stiffness. Moreover, we meticulously analyzed our customer needs and engineering specifications to ensure our design is compatible with current production and meets requirements.

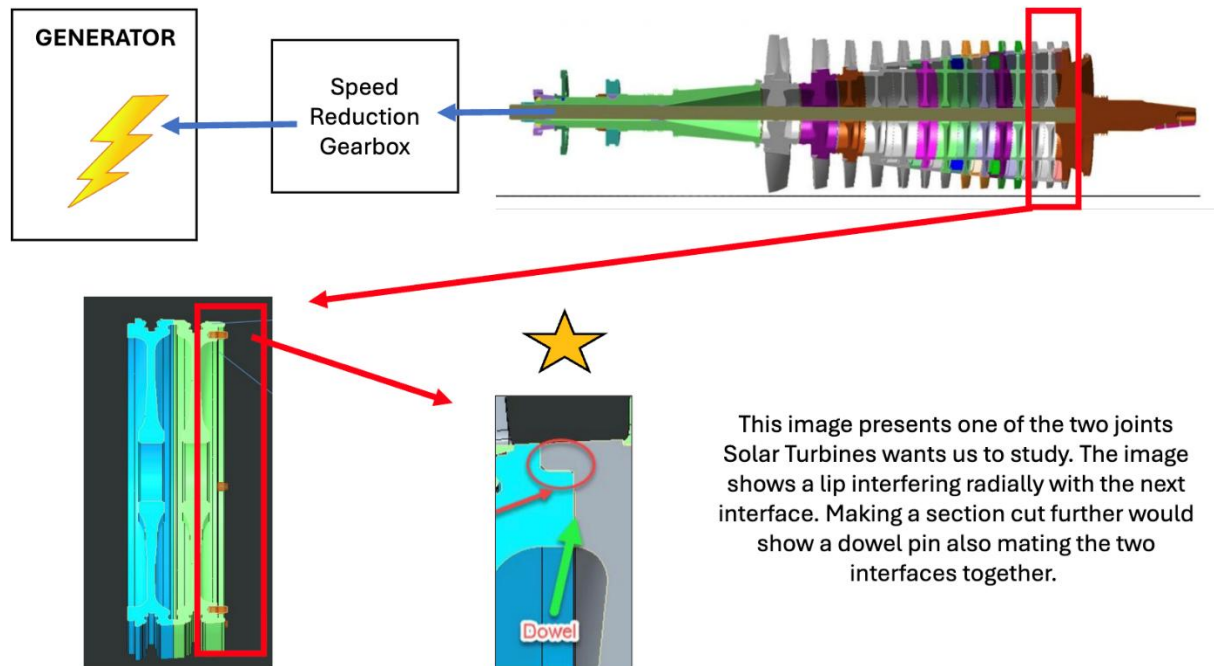


Figure 2. Boundary sketch illustrating project scope

Researched Topics and Existing Solutions

The main type of research we conducted for this project was technical research, as our project focuses on technical testing and analysis for Solar Turbines. Rather than having various other products to compare with, we researched the fit connections used by competing turbine companies. Our goal was to prepare ourselves for considering other mating connections for rotors and set up the testing plan for the prototype modal testing.

Table 1: Existing Solutions

This table details the different types of fits previously used to address issues most similar to that of Solar Turbines and provides examples of their usage.

Solution Type	Description	Examples
Turbine shroud with two axial pin mounting and cover plates	This solution consists of a turbine shroud assembly where ceramic shroud segments are attached to a metal carrier using axial pins and sealed cover plates to handle high temperatures and thermal expansion. The pins transfer loads and maintain alignment while allowing some movement for thermal expansion. The sealed cover plates close and seal the holes preventing the pins from ejecting and blocking gas leaks. Some improvements include reducing the number of parts and optimizing pin geometry.	[1]
Turbine shroud with spring-loaded axial pins	In this solution, the metal carrier and the ceramic blade track segment are connected by an axial pin and retainer plug. The retainer plug locks the pin in place. A “biasing” element or spring is placed between the head of the axial pin and carrier to keep the components pressed against each other. Many flanges are used to help with alignment and support for the blade track segment. Some drawbacks of this solution are that the tolerances for manufacturing are very high to ensure the correct preload of the springs, and the added complexity makes it difficult to assemble.	[2]
Finite Element Method – Based Interference Fit Design	This approach uses finite element analysis to model the three-dimensional stresses and deformations in interference-fitted mechanical assemblies. Unlike traditional methods based on Lamé’s equations, FEM accounts for non-linear contact behavior, friction, and complex geometries, resulting in more accurate predictions of stress, deformation, and safety factors. The study introduces two safety factors to guide design tolerance and improve reliability: one for structural strength and one for torque transmission to guide design tolerance and improve reliability.	[3], [4], [5]
Bolts and flanges	In this solution, the casing sections are joined with circumferential bolted flanges and horizontally split joints that keep the compressor and turbine aligned while allowing easy maintenance. These connections provide stiffness, sealing, and precise alignment between rotating and stationary components under high temperature and pressure. This relates to our project because, like interference and dowel-pin fits, these joints rely on controlled contact and preload to maintain structural	[6]

	integrity and stiffness under cyclic loading. This solution could be improved by strategically placing the bolts to make the turbine easier to assemble and disassemble.	
Tapered interference fit	This solution works by pressing two matching conical surfaces together to create frictional resistance through elastic deformation and contact pressure. They provide strong, reliable joints with high holding strength and resistance to loosening, especially when small taper angles and sufficient contact length are used. Their performance can be improved by optimizing taper geometry and surface friction to maximize efficiency while avoiding plastic deformation during assembly.	[7], [8], [9]
Curvic couplings	Curvic couplings work by using interlocking curved teeth on mating faces to precisely align and rigidly join rotating components, transmitting torque while maintaining accurate concentricity. They perform well in providing repeatable assembly, high load capacity, and excellent alignment in turbine and compressor rotors. They can be improved by refining tooth geometry to reduce stress concentrations, applying advanced coatings or materials for better wear resistance, and using modern manufacturing techniques to achieve tighter tolerances and smoother tooth surfaces.	[10], [11], [12]

Customer Needs

Our main customer, Solar Turbines, requested a stiffness analysis on their turbine rotor according to the press and pilot fits used to hold AFT and FWD weldment rotors together. Rather than having a list of vague needs/ wants, we are provided with a list of goals to reach over the course of the project. The customers' needs listed below are similar to the ones that can be seen in our QFD in Appendix A.

Table 2: Customer Needs

This table lists the identified customer requirements with justification and assigns a relevant, measurable engineering specification to be used to evaluate the adequacy of the prototype in addressing this need.

Customer Needs	Relevant Engineering Specifications (Spec #)	Justification
Prevent shifting	Spec 1: Frequency	We currently assume natural frequency will directly relate to the amount of shifting seen through the spring constant.
ANSYS model	Spec 2: Heat Consideration Spec 3: Stress Analysis Spec 4: Modal Analysis	The ANSYS model(s) developed must include considerations for heat transfer, stress, and natural frequencies as these are all important parameters in the system being studied.
Stiffness vs. Interferences	Spec 5: Multiple iterations	Multiple iterations will allow us to deliver different parts that the customer can choose from for their needs.
Machinable	Spec 6: Tolerances	The part we are studying calls for tight tolerances. Tight tolerances increase the difficulty in machining.
Able to Assemble	Spec 7: Assembly/ Disassembly Time	The part must not be overly complex to where technicians need intense training and time to complete task.
Maintain Production Cost	Spec 8: Machining	Machining will be the most expensive part of our project. Tight tolerances, long machining time, jig preparation and other factors must be heavily taken into consideration.
Physical Prototype Model	Spec 9: Weight	The part should not be overly heavy, but strong enough to hit stress specifications and factor of safeties.

Engineering Specifications

The engineering specifications table is a list of achievable goals meant to describe how we will achieve the end goal of providing Solar Turbines with a stiffness analysis on their AFT and FWD weldment rotors. These specifications are made to be specific thresholds on technical achievements while having a “compliance” column to explain how we plan to measure the achievements. A rough draft of these specifications can be seen in the QFD in Appendix A.

Table 4: Engineering Specifications

This table lists the identified engineering specifications and assigns a specific target value or ability to be used to evaluate the adequacy of the prototype in addressing each need.

Spec #	Spec	Specification Description	Target	Tolerance	Risk *	Compliance **
1	<i>Frequency</i>	Need to have a natural= frequency greater than 186hz	>186 hz	+inf, -0.0	H	A, T
2	<i>Heat</i>	New mating method must function at operating temperature	1400 C	±200	M	A
3	<i>Stress</i>	Lower stress caused by interference fits while maintaining sufficient mating strength	<20Mpa	±1 MPa	L	T
4	Modal	Create modal analysis model of the compressor assembly	complete	[-]	L	I
5	Iterations	Create a minimum of 3 different mating configurations	complete	[-]	L	A,T
6	Tolerance	Create tolerances that are machinable within reason	>0.0001”	±0.001	H	T
7	Assembly/ disassembly	Create mating configurations that can be assembled and disassembled with minimal tooling	complete	[-]	L	I
8	Machining	Create mating surfaces that are machinable without complex and unique tooling	complete	[-]	H	I

9	Weight	Prototype weldment should be a reasonable weight that could be easily transported by a person	50 lb.	± 5 lb.	L	A, I
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[5] S. P. Singh and R. K. Sharma, "Interference Fit Simulation on Pin Joint by Using ANSYS," *International Journal of Research in Engineering and Technology (IJRET)*, vol. 4, no. 5, May 2015.

(ijret.org) * Risk of meeting specification: (H) High, (M) Medium, (L) Low

** Compliance Methods: (A) Analysis, (I) Inspection, (S) Similar to Existing, (T) Test

Additional Background

In the turbine system, mass primarily comes from the disks, forward cone, aft hub, and shafts. The overall stiffness is determined by the rotor geometry, though joints such as weldments and curvic couplings introduce flexibility that reduces stiffness; bearings contribute additional stiffness to the system. Damping is mainly provided by an oil wedge in the bearings. At operating speeds up to 120,000 rpm (for the T130 turbine), the rotor exhibits several mode shapes, including banana and parabola modes, with small deflections of 1–3 thousandths of an inch. Most Solar Turbines machines use tilt-pad radial and thrust bearings, and rotor dynamic analyses model the system with simplified axi-symmetric and revolved components; blades are represented as point masses and bearings are modeled using hydrodynamic film theory. Critical modes for the T130 are around 2400 rpm (turbine overhung), 4500 rpm (compressor “banana” mode), and 9300 rpm (“spaghetti” mode near full speed). Depending on the rotor’s length-to-diameter ratio, either single-plane or two-plane balancing is used to improve dynamic imbalance and vibration. Sensors, such as proximity probes and accelerometers, are used to measure displacement, casing vibration, and overall dynamic behavior.

Annotated Bibliography

- [1] J. Freeman, A. D. Sippel, and D. J. Thomas, “Turbine Shroud Assembly With Sealed Pin Mounting Arrangement,” Apr. 30, 2025. Accessed: Oct. 23, 2025. [Online]. Available: <https://patentimages.storage.googleapis.com/30/92/a2/d49c5a0007e201/EP4095355B1.pdf>

This patent describes a turbine shroud assembly for a gas turbine engine, featuring a seal segment supported by a carrier and a mount system that uses pins and a cover plate to secure the seal while managing thermal-expansion stresses. It is relevant to our senior project because it offers a practical mechanical solution for high temperature sealing in turbine systems, which is directly applicable to the thermal management portion of our work.

- [2] T. J. Freeman, A. D. Sippel, and D. J. Thomas, “Turbine Shroud Assembly With Axially Biased Pin And Shroud Segment,” Apr. 30, 2025 Accessed: Oct. 23, 2025. [Online]. Available: <chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://patentimages.storage.googleapis.com/30/92/a2/d49c5a0007e201/EP4095355B1.pdf>

This patent describes a turbine shroud assembly designed for gas turbine engines, involving a carrier segment, shroud segment, and an axially biased pin/biasing member arrangement to maintain sealing engagement under thermal expansion. It is relevant to our project as it addresses maintaining tight seals and structural integrity under high temperature/thermal-cycling conditions, which is helpful for thermal conditions we are studying. From this document we learned how a biasing spring can apply preload to a shroud segment via a pin-mount arrangement, improving sealing at the gas path interface despite thermal expansion.

- [3] Zhang, Y., et al. “Design of Interference Fits via Finite Element Method.” *International Journal of Mechanical Sciences*, vol. 42, no. 9, 13 June 2000, pp. 1835–1850. *ScienceDirect*.

This article applies the finite element method to model interference fits and shows that traditional thick-wall cylinder theory can be inadequate. It is relevant because our project involves high-precision mechanical joints subject to thermal expansion and structural loads, and this work gives insight into how FEM improves fit-design accuracy. From the paper, we learned about the effects of three-dimensional stress distributions in interference fits and how design adjustments influence contact pressures and long-term reliability.

- [4] Karmankar, Rahul G. “Stress Analysis of Interference Fit by FEM.” *International Research Journal of Engineering and Technology*, vol. 02, no. 09, Dec. 2015, pp. 2641–2648.

This document presents a finite element analysis of interference fits using ANSYS to evaluate contact and Von Mises stresses for various allowances between a pin and hole. It is relevant to our turbine project because it demonstrates how FEA can be applied to study stress distribution and contact behavior in precision-fit mechanical assemblies. We learned how changes in interference allowance affect stress levels, which helps us better understand how to optimize fits and avoid material failure in turbine components.

- [5] Imran, Mohammed, et al. “Interference Fit Simulation on Pin Joint by Using Ansys.” *International Journal of Research in Engineering and Technology*, vol. 04, no. 05, May 2015, pp. 579–583.

This document presents an ANSYS-based finite element simulation of an interference fit in a pin and plate joint using steel and aluminum materials. It is relevant to our turbine project because it demonstrates how FEA can predict stress distribution, deformation, and contact behavior in precision mechanical joints. We learned how different materials respond to interference fits, helping us better understand how material choice and fit tolerances affect stress performance and joint reliability.

- [6] R. C. Stancliff, "The General Electric LM5000 Marine Gas Turbine," in *Proceedings of the ASME Turbo Expo*, Toronto, ON, Canada, Jun. 4–8, 1989, Paper No. 89-GT-13. [Online]. Available: <https://asmedigitalcollection.asme.org/GT/proceedings/GT1989/79146/V002T03A001/238533>.

This paper describes the General Electric LM5000 marine gas turbine, a high-power engine adapted from an aircraft design for marine applications. It is relevant to us because it demonstrates how precise casing joints and connections maintain stiffness, alignment, and sealing under high thermal and mechanical loads. We learned how bolted flanges are designed to manage stress and ensure reliability.

- [7] D. Bozkaya and S. Muftu, "Efficiency considerations for the purely tapered interference fit (TIF) abutments used in dental implants," *Journal of Biomechanical Engineering*, vol. 126, no. 4, pp. 393–401, Aug. 2004, doi: 10.1115/1.1784473.

This paper discussed how tapered interference fits create strong mechanical joints by using elastic deformation and frictional pressure between conical surfaces, offering high stability and resistance to loosening. This insight relates to your turbine project because similar principles can improve the reliability of tapered shaft or hub connections—optimizing taper angle and surface finish can enhance holding strength while preventing material yielding during assembly or operation.

- [8] D. Bozkaya and S. Muftu, "Mechanics of the tapered interference fit in dental implants," *Journal of Biomechanics*, vol. 36, no. 11, pp. 1649–1658, 2003, doi: 10.1016/S0021-9290(03)00177-5.

This paper discussed how tapered interference fits create strong, self-locking joints by generating frictional resistance through elastic or partially plastic deformation of the contacting conical surfaces. The study showed how taper angle, contact length, and interference depth control pull-out strength and stress distribution, with excessive interference leading to material yielding. This relates to our project because similar principles apply to press-fitted or tapered shaft–hub connections. Optimizing taper geometry and fit pressure can improve joint reliability while avoiding overstress or slip under dynamic loading.

- [9] N. Vishwakarma, A. Renuke, and V. M. Phalle, "Effect of tapered interference fit between impeller and shaft in turbo machines," *Journal of Mechanical Engineering – Strojnícky časopis*, vol. 68, no. 3, pp. 25–32, 2018, doi: 10.2478/scjme-2018-0024.

This journal article discusses how using a tapered interference fit between the impeller and shaft reduces stress concentrations and maintains strong contact pressure compared to a conventional cylindrical fit. This design improves torque transmission and allows for higher rotational speeds without the need for keys or splines. It relates to our project because similar principles can be applied to improve joint reliability, reduce fatigue at connection points, and ensure stable

operation under high-speed rotational loads.

- [10] “Back to Basics: Curvic Coupling Design,” *Gear Technology*, pp. 34–48, Nov.–Dec. 1986, Gleason Works, Rochester, NY.

This paper discussed how curvic couplings use precision-ground curved teeth to align and lock rotating components together, ensuring accurate centering and efficient torque transmission. They are highly effective in maintaining rigidity and repeatability in multi-disk assemblies like turbine rotors. This relates to our project because the same principles of precise alignment, torque transfer, and structural integrity are critical for connecting rotor sections and ensuring smooth, reliable turbine operation.

- [11] S. R. Pisani and J. J. Rencis, “Investigating CURVIC coupling behavior by utilizing two- and three-dimensional boundary and finite element methods,” *Engineering Analysis with Boundary Elements*, vol. 24, no. 4, pp. 271–275, 2000.

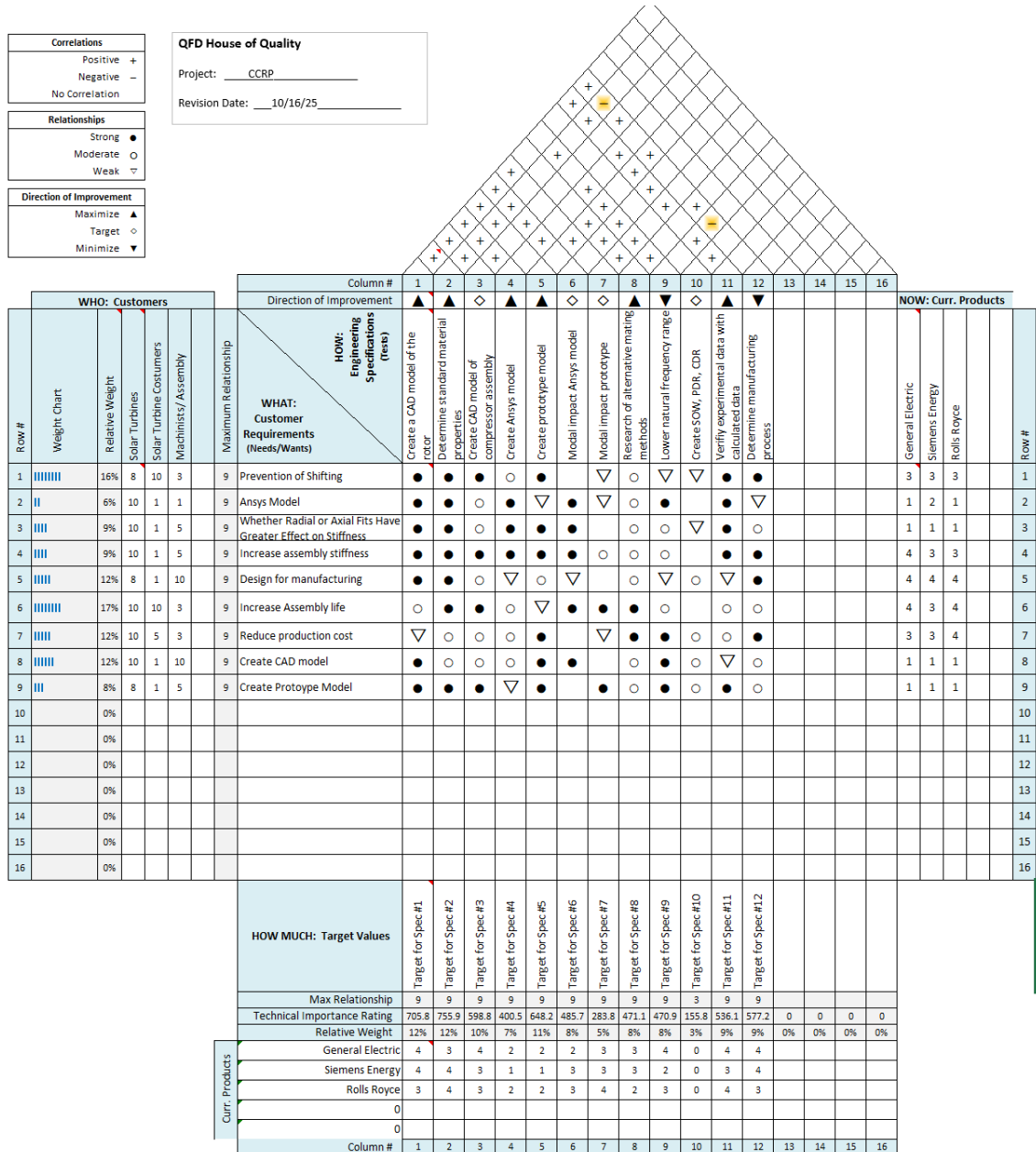
This article discusses how curvic couplings effectively transmit torque and maintain precise alignment between rotating components while highlighting the importance of analyzing stress concentrations at the tooth roots. The study showed that accurate 3D modeling is essential to predict these stresses and improve component life. It is relevant to our project because we will face similar challenges in ensuring reliable torque transfer, minimizing stress concentrations, and maintaining alignment in large rotating structures.

- [12] X. Jiang, Y. Zhu, J. Hong, and Y. Zhang, “Development and validation of analytical model for stiffness analysis of curvic coupling in tightening,” *Journal of Aerospace Engineering*, vol. 27, no. 4, p. 04014012, 2014, doi: 10.1061/(ASCE)AS.1943-5525.0000348.

This article discusses how the stiffness of a curvic coupling is strongly influenced by its geometry, friction characteristics, and the bolt tightening process, which together affect how torque and loads are transferred between rotating components. The study showed that optimizing parameters like tooth height and pressure angle can stabilize stiffness and reduce stress concentrations. This relates to our project because similar coupling behavior and load transfer mechanisms are critical for ensuring structural stability, precise alignment, and reliable torque transmission in large rotating assemblies.

Appendix A: QFD House of Quality

The QFD house of quality is an expansive diagram showcasing the design considerations and specifications we will try to achieve during this project. The diagram considers Solar Turbines as the main customer, and some of Solar Turbines' competitors as the "current products". As this diagram is often used for product creation, we used the "customer needs/wants" section as a list of tasks required of us by Solar Turbines. This diagram is what was used to make the engineering specifications seen in the Scope of Work.



Appendix B: Non-Cited Research

This appendix includes the citations for other research we completed but were not cited in this specific report. The other sources we found to help us with our general research are listed below.

- [13] Bonisoli, Elvio, et al. “Interference Fit Estimation through Stress-Stiffening Effect on Dynamics.” *Mechanical Systems and Signal Processing*, vol. 160, 16 Apr. 2021. *ScienceDirect*.
- [14] Sikanen, Eerik, et al. *Effect of Operational Temperature on Contact Dynamics of Shrink-Fitted Compressor Impeller Joint*. Cham Springer, 14 June 2019, pp. 3341–3351.
- [15] Erena, Diego, et al. “NUMERICAL STUDY on the INFLUENCE of ARTIFICIAL INTERNAL STRESS RELIEF GROOVE on FRETTING FATIGUE in a SHRINKFITTED ASSEMBLY.” *Depósito de Investigación de La Universidad de Sevilla* , vol. 151, Nov. 2020.
- [16] Leedham, L. H., and M. I. Mech.E. “Some Problems in the Manufacture of Experimental Gas Turbines.” *Proceedings of the Institution of Mechanical Engineers*, vol. 163, no. 1, Nov. 1950.

Compressor Rotor Joints and Fits Interference Study

Preliminary Design Review

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Four Panel Chart

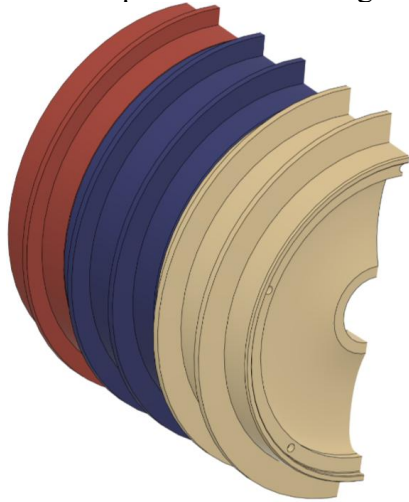
Project Overview Solar Turbines needs a comprehensive study tool on the effects axial and radial interference fits have on turbine compressor rotor stiffness. The weldments of the compressor have press-fit dowel pins and pilot joints to connect one weldment section to another. These fits need to be assessed through a finite element modal analysis as well as practically tested through modal impacts on manufactured prototypes to investigate how the fits affect the stiffness of the turbine's connected weldments. The experimental data will be used to validate the finite element analysis (FEA) model tool to eventually be presented to Solar Turbines.	Concept description Our Concept CAD is comprised of 3 individual components: aft hub, aft weldment, and forward weldment. These sections will be connected and analyzed on multiple different mating techniques. 
Concept justification The software we are using for our turbine analysis is Ansys. We are using Ansys because it is highly capable for finite element studies and is widely used in industry. In addition, our sponsor, Solar Turbines, emphasizes that the analysis to be with Ansys. Regarding our physical scaled down prototype, we are going to select pin and interference configurations that our finite element analysis determines to be most optimal for increasing assembly stiffness. The reason for this selection is to first validate that our finite element analysis can accurately predict physical characteristics like natural frequency and effective stiffness. We will then test the different configurations on the physical prototype to validate that the optimization technic utilized in our software matches real world results.	What's Next The next step for our project includes beginning development an ANSYS model that factors in thermal effects and contact stress to accurately solve for the natural frequencies of the design. The goal is to have a sufficient model by the end of winter break. We will also determine a schedule for when the prototype models should be accomplished. Other steps include repeated internal designs reviews, feasibility prototypes, and creating engineering drawings to prepare for manufacturing.

Figure 1. Four-Panel Chart for Preliminary Design Review

Overview

The system we will create to analyze the turbine compressor will consist of an accurate Ansys

simulation model used to determine the optimal press-fit types and configurations to best increase rotor stiffness. Our plan is to create an Ansys model that Solar Turbines can use to study these different configurations and press fit interferences to use in further design decisions. The general workflow of this Ansys model can be visualized in Figure 2z below. After studying the load paths and rotor dynamics of the Titan 130 gas turbine, we will study the methods of stress analysis with Ansys and create a physical model of the forward, aft, and aft hub weldments to find the best proposed mating technique. Our final product will be the complete Ansys model with a report on our findings and test data from interference fits, making an overall tool Solar Turbines can use in the future.

We plan to use the tool we make to find the best three mating configurations from Ansys to present to Solar Turbines. To validate our Ansys model, we plan to test the tool on a scaled-down physical model that will be machined out of steel for validation. So far, we have developed Pugh and decision matrices on different press fit configurations we will run in the Ansys simulation. From the Ansys simulation, we will find the optimal top three to test physically. We will then conceive a specific plan to test those three configurations on our physical model and refine any inaccuracies in the Ansys model. Validation of our ANSYS model will come through comparing the deflection and natural frequencies measured from the test set up, hand calculations, and having a percent error no greater than 10%.

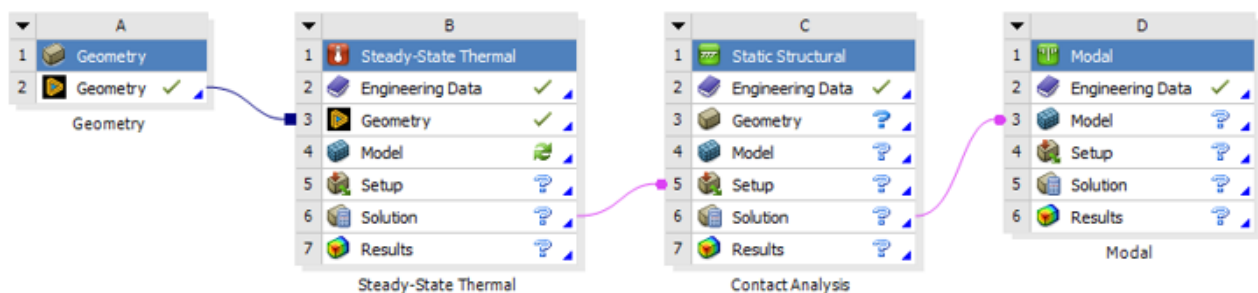


Figure 2. Ansys workflow from modeling physical geometry to finding results.

This report will cover our concept ideas for the Ansys simulation tool in greater detail, as well as the next steps in studying the interference fits and manufacturing the physical model. We will use this report to explain our intentions and ideas for this project, as well as gain feedback from our coaches and our sponsor about our research methods. This project overall is research focused and thus needs validation on whether our plans to create an Ansys study tool will be sufficient to meet Solar Turbine's needs. In the appendix of this report, we include various decision matrices, a preliminary analysis, and our project Gantt chart to provide resources on our overall design process so far. The decision matrices of appendices A and B were used to organize the many different mating methods that could be used for the rotors into reasonable options to be studied with the Ansys model. The preliminary analysis shown in appendix C was an initial mathematical analysis on the forces presented in the pins during the compressor's operation. The Gantt chart in appendix D shows our project timeline in the following weeks up to the CDR.

Concept Description and Justification

Considering the nature of our project, it is difficult to predict the exact interference fits and

configurations that our Ansys tool will consider the most optimal for increasing rotor stiffness. Thus, we will begin by analyzing the most common and widely used mating methods used in turbine engines. There is no exact certainty that our software will align with these current mating techniques being the most optimal, but to talk about different concepts for our project, we are forced to make certain assumptions. Those assumptions include that the most likely configurations that we are going to be testing and manufacturing are going to be a 5 and 7 dowel pin configuration with different pilot and dowel pin interference fits. We believe that these assumptions are relatively reasonable because they are tested configurations that are used in other turbines, giving these configurations merit and a decent probability that they will have similar performance characteristics to Solar Turbines' current configuration as shown in Figure 2. This helps us establish a safe and realistic starting point of what configurations work and which direction we can move our design to increase stiffness and performance characteristics.

In order to have a deeper understanding of the complexity and decision making regarding our project, we first must emphasize some driving engineering requirements that must be met by our design. First, Solar Turbines has a 30,000-hour operating time requirement for which their turbines must operate without any maintenance or inspection. To meet this requirement, it must be decomposed into several child requirements that must be met prior to make this requirement feasible. Some of those child requirements being that the mating method must not shear or slip during operation. In addition, the mating method must be sufficient at the operating temperature and pressure. Consequently, the mating surface must also not reach fatigue failure before the 30,000-hour requirement given by Solar Turbines. Another important requirement that must be met by our design is that it must be manufacturable and installable, preventing any highly advanced or extremely expensive manufacturing process that is not finically feasible for Solar Turbines. All of these requirements must be met while also increasing the joint stiffness, increasing the lifecycle and reliability of the turbine engine.

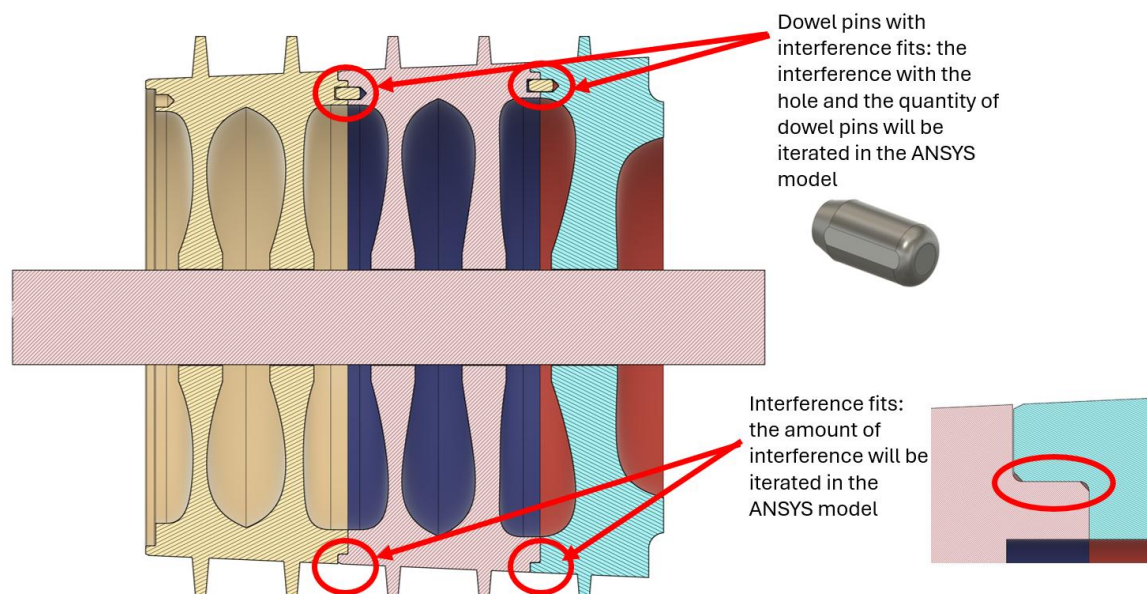


Figure 3. Base configuration mating method used by Solar Turbine's T130.

The first configuration that we will test will comprise of five solid steel dowel pins with an axial and radial interference fit that will be determined later by our Ansys model. This press fit configuration will be accompanied by a pilot fit that closely resembles Solar Turbines' current configuration with altered

interference values, which will also be determined later by our Ansys model. To guarantee that the shear and fatigue requirements associated with our mating joints are met, we will be using Solar Turbines' current dowel pin dimensions. We believe this configuration should be tested because it would provide us with similar data collected by Solar Turbines when they tested their current configuration. This is important because it will assist us by helping identify any issues that might arise with our Ansys software by pointing out any large discrepancies that should not exist. Another reason why this configuration should be tested is to acquire base line results, for which we will use to compare our other configurations. Lastly, this configuration acts as a safe first trial for us to test and ensure our manufacturing and testing processes are accurate and repeatable.

The second configuration that we will test will comprise of seven hardened steel dowel pins with an axial and radial interference fit that will be determined later by our Ansys model. This press fit configuration will also be accompanied by a pilot fit similar to the first configuration but with different interference values. The reason why we believe we should test this configuration is to study how having an increased number of pins and press fit stress along the weldments and hubs affect overall joint stiffness. This will also help us study how having increased dowel pins increased the complexity of manufacturability and assembly due to tolerance propagation. Having several high-tolerance holes becomes difficult to manufacture because slight errors that are within our tolerance callout may affect the precision of our part and affect the joint stiffness in ways we may have not considered. Important factors that need to be studied with this configuration are how having more dowel pins evenly spaced out throughout the pieces affects the stress distribution within the mating parts. This could have unforeseen implications by increasing the overall stress in the part and inducing premature stress cracking that is not experienced with the standard five pin configuration. Lastly, for this configuration we will also need to keep or exceed the same cross-sectional area that the original five dowel pins had but with seven dowel pins. This can be accomplished by downsizing the current dowel pins sizes to be comparable to a 7-pin configuration.

For our last configuration, we will use a fully welded assembly. This involves creating a model of the aft hub and aft and forward weldments that will be joined only by an outer weld where the pilot fit currently exists. The reason why we will be testing this configuration is to measure how a permanent bond between the two mating parts performs in comparison to our dowel pin and pilot joint configurations. The reason why we are using a welded joint configuration is because welded joints offer one of the highest joint stiffnesses possible for joining parts together. This will help us gauge how successful our pin configurations are at increasing joint stiffness by comparing it to the theoretical maximum joint stiffness. The maximum joint stiffness occurs when the two parts are fused together into one piece, so the stiffness is dependent on material and geometrical properties. The issue with making this configuration purely welded is that it is difficult and costly to fully weld the aft hub with the aft weldment, as both are steel. In addition, other issues associated with this process are potential warping due to heat concentrations which are difficult to correct. This is very detrimental to the assembly because it is rotating at high speed where any imbalance or warping in the part compounds exponentially, making the assembly unstable and prone to premature failure, which is unacceptable due to the turbine's long life requirement.

Alternative Solutions Table

If the chosen concept is not approved by the customer, there are several alternative options the team can propose. First, the number of dowel pins can be easily adjusted. Initially, there will be proposed configurations with five and seven dowel pins to hold the rotor plates together. If the number of dowel pins is too high, the CAD models can be modified to match the customer's recommendations and needs. Another concept that can be proposed is the use of different materials for the rotor plates. Currently, Solar Turbines uses a nickel alloy to manufacture their rotor plates. However, since the project aims to build a model and procedure that can accurately predict the stiffness of a turbine configuration, using less stiff materials could help produce larger deflections and more easily validate the data. If a material that is too stiff is chosen, the data may be difficult to read because the changes would be smaller. These different concepts are summarized below in Table 1.

The alternative solutions for our project will depend on information we will gain later during the project. For the dowel pins, there are more than just the 4 and 6 pin configurations that can be chosen. However, this design decision will be driven by the ANSYS model. Once the preloads and temperatures are taken into account for the natural frequency, then we will only decide what number of pin configurations we will want to try. For the material choices, we will first propose steel. Choosing steel means the model we build we resemble Solar Turbines' actual prototype more than aluminum or copper would. Steel would also be easier to weld for the last of our tested configurations.

Table 1: Alternative Solutions Considered

Solution Name	Description	Justification
4 dowel pins	4 dowel pins spaced equally about the center of the turbine shaft	Less dowel pins may be preferred to be explored to reduce manufacturing and assembly time
6 dowel pins	6 dowel pins spaced equally about the center of the turbine shaft	More pins can possibly help keep the system stiffer and result in less shifting
Material Choice	Materials like aluminum or copper are less stiff	Choosing a less stiff material can assist in noticing changes in deflection easier to validate the ANSYS model
Rotor Plate Testing	The testing will be completed using assuming there is no shaft that runs through the connected rotor plates	No shaft running through the rotor plates will be reduce the lumped mass component of the analysis and focus more on the rotor plates

Appendix A: Morphological Matrix

The morphological matrix was our second design matrix we used to narrow down ideas on which mating configurations we were going to analyze with the Ansys model. This matrix was used to not only display the various methods two weldments could be mated together, but also display which methods achieved which required design functions. The column on the far left contains the functions developed in our function decomposition, which explain the ways the project achieves its goals. The remaining columns contain the ideas thought of to complete each function.

Table 2: Morphological Matrix

Morphological Matrix					
Function	Option 1	Option 2	Option 3	Option 4	Option 5
Validate the ANSYS model	use another FEA software to perform the same analysis	hand calculations	physical model with different pin radial positions to perform modal impact testing	physical model with different interference fits to perform modal impact testing	physical model with different material variations
Improve axial Positioning	pilot fits	bolts	barbed dowel pins	welding	adhesives
Improve Rotational Positioning	bolts	Connex dowl pins	slotted dowel pins	hardened dowel pins	coiled spring pin
Improve configuration	2 pins	3 pins	4 pins	5 pins	6 pins
Improve installation	bolts	Flat vent dowel pins	adhesive	2 pins	Connex dowel pins
Ease of manufacturability	Hardened dowel pins	Welding	bolts	adhesives	pilot fit
Adjustable geometry	slotted dowel pins	Bolts	adhesive	welding	pilot fits
Joining force	Welding	solid barbed dowel pin	bolts	adhesives	Connex dowel pins
Adjustable interference fit	dowel pin diameters	Bolt diameters	Welding bead width	pilot diameter	adhesive thickness
Prolonged life cycle	heat treatment	Anodization	sacrificial pieces	lubrication	malleable
Inexpensive material selection	Aluminum	stainless steel	Inconel	Cres steel	Nickel Alloy
Affordable production methods	sheet metal	metal 3d printed	casted	machined	Water Jet

Appendix B: Weighted Decision Matrix

The weighted decision matrix was the last matrix used to finalize the configurations we planned to analyze and test. Included below is the weighted decision matrix along with our morphological matrix color coded to show the combinations of ideas we selected to consider for the Ansys modeling. Combined, the two matrices show the ideas we decided were the best to consider. The column on the far left of the weighted decision matrix shows the same functions shown in the morphological matrix, followed by a column of weights for each function, showcasing the importance of each function. From there, each string of color-coded ideas were assessed on how they handled each function. The bottom row contains the total score for each string of ideas, showing the best ideas to the worst.

Table 3: Weighted Decision Matrix

Weighted Decision Matrix					
Team: CCRP	11/9/2025	Idea 1	Idea 2	Idea 3	Idea 4
Specification	Weight	Red	Blue	Green	Orange
Validate the ANSYS model	0.15	5	4	4	2
Improve axial Positioning	0.1	4	3	2	1
Improve Rotational Positioning	0.05	3	3	2	2
Improve configuration	0.1	3	3	4	2
Improve installation	0.05	3	4	3	4
Ease of manufacturability	0.1	3	4	3	3
Adjustable geometry	0.05	5	4	3	3
Joining force	0.15	4	2	3	2
Adjustable interference fit	0.05	5	3	3	1
Prolonged life cycle	0.1	5	2	3	2
Inexpensive material selection	0.05	3	3	3	3
Affordable production methods	0.05	4	4	1	5
Total	1	4	3.15	3	2.3

Appendix C: Preliminary Analysis

Preliminary analysis on the pin and pilot joint loading was completed using hand calculations. This analysis was done to give an idea of the types of forces normally seen in a Solar Turbines' compressor and rotor. Using specific criteria from Solar Turbines, we conducted quick mathematical stress calculations.

We began our basic analysis by calculating the maximum torque and stress that can be translated through the aft hub, aft weldment, and forward weldment. By calculating the max torque, we needed to solve for the max shear force the current 5 dowel pin configuration can handle in addition to the torque that can be carried by the outer pilot fit. To calculate the torque that the outer pilot joint can carry we needed to solve the pressure at the interface between the aft weldment and the aft hub. This was accomplished by identifying the interference fit and material properties of the interfacing parts. Moreover, having calculated the max pressure and torque that can be transmitted through the assembly, we calculated the max stress that the assembly could experience. This helped us determine a more accurate interface configuration that will have to meet or exceed the torque and stress the current configuration can handle.

Preliminary analysis

1

TOPICS: / criteria

shear, press fit, radial stress, frequency;

done

done $\phi = 0.500 \pm 0.001$

length $L = 0.80 \pm 0.02$

done

incont X 75 D AMS 5049
hardness RC 32-39

Pin - aft weld - hub

5 pins

counter bore

hole

hole $\phi = 0.5180 \pm 0.0005$

length $L = 0.50 \pm 0.015$

} aft weld mit

• aft hub

CUP

• outer diam = 20.044 ± 0.001

$Z = 0.410 \pm 0.002$

done

hole $\phi = 0.4998 \pm 0.003$

length $L = 0.45 \pm 0.1$

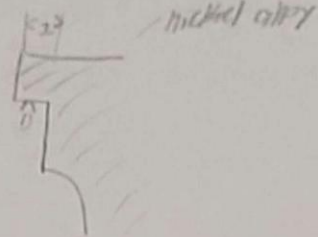
[outer dia = 19.6505 ± 0.0010]

Pin weld

[Diam = 19.684 ± 0.001]

• $F = \sqrt{\frac{h}{m}}$ Tbd

press fits



Doubt Pin Shear

5 pins

$$\text{net area} = 5 \cdot \frac{\pi D^2}{4} ; D_{\max} = 0.5003 \quad D_{\min} = 0.5001$$

$$\text{net area}_{\min} = 0.98214 \text{ in}^2$$

$$\eta = \frac{F}{A} ; \text{Assume Force Perpendicular/along moment}$$



$$\eta_{\max} = 75 \text{ ksi for Inconel}$$

$$F_{\max} = \eta \cdot A_{\min}$$

$$F_{\max} = 73660.5332 \text{ lbf}$$

@ 18.960" radius

$$\text{Torque} = F \cdot r$$

$$\text{Max torque} = 1396603.71 \text{ lbf-in}$$

through hub

$$\text{Pins alone} = 116383.642 \text{ lbf-ft}$$

in shear

$$\text{Press fit} \quad P = \frac{8}{d \left[\frac{1 - \gamma_s^2}{E_s} + \frac{1 - \gamma_h^2}{E_h} \cdot \frac{D_o^2 + d^2}{D_o^2 - d^2} \right]}$$

d = shaft diam

D_o = hub's outside diameter in in

γ = interference

Nickel 901 → γ = 0.32

$$E = 200-207 \text{ GPa} = 2.90 \text{ E7} - 3.002 \text{ E7 psi}$$

Same material

• hub to fit smallest 8

hub 20.044 ± 0.001

shaft 20.067 ± 0.001

$$20.045 - 20.066 = [-0.021"]$$

$$P_{\text{hub}} = \frac{0.021}{20.066} \left[\frac{1 - 0.32^2}{2.901E7} + \frac{1 - 0.32^2}{2.901E7} \cdot \left(\frac{0.021^2 + 20.066^2}{21 - 20.066^2} \right) \right]$$

$$P_{\text{hub}} = 1470.9 \text{ psi}$$

$$F_{\text{ric}} = N P \cdot A_{\text{cont}}$$

assume A_{cont} = middle ground between 20.045 & 20.066

$$A_{\text{cont}} = L \cdot \pi \cdot d \rightarrow L = 0.410 \pm 0.002 \text{ assume } 0.408$$

$$A = 0.408 \cdot \pi \cdot 20.0555$$

$$A_{\text{cont}} = 25.7065$$

assume $N_{\text{stat}} = 0.15$ dry steel

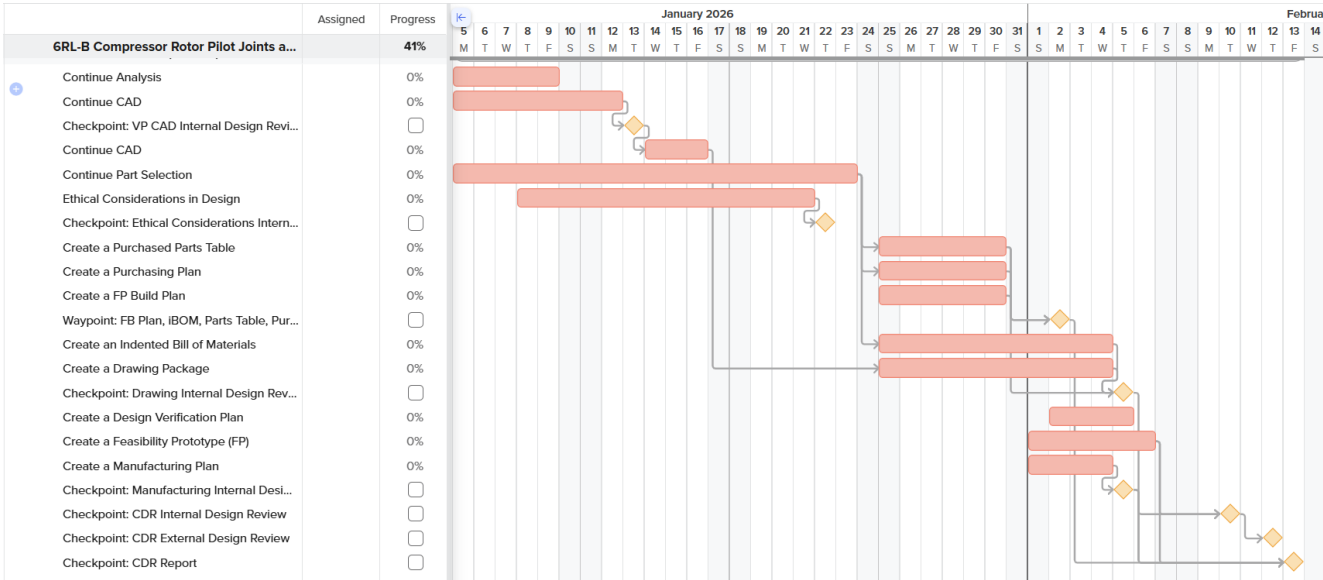
$$F_{\text{ric}} = 0.15 \cdot 1470.9 \cdot 25.7065 \text{ [psi} \cdot \text{in}^2]$$

$$F_{\text{ric}} = 5671.77 \text{ lbf}$$

$$\text{max torque} = F \cdot \frac{D_{\text{cont}}}{2}$$

$$\begin{aligned} T_{\text{max}} &= 56675.12 \text{ lbf-in} \\ &= 4739.594 \text{ lbf-ft} \end{aligned}$$

Our Gantt chart has been used from the beginning of this project to set out reasonable and actionable goals throughout the design and creation process. The next large checkpoint of this project is the Critical Design Review (CDR). The Gantt chart shown below displays the various tasks and waypoints we will complete from this PDR to the CDR.



Compressor Rotor Joints and Interference Fit Study

Critical Design Review

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2/13/2026

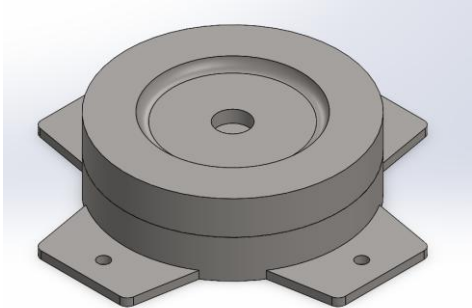
Project Overview Solar Turbines needs a comprehensive study tool on the effects radial interference fits have on turbine compressor rotor stiffness. The weldments of the compressor have press-fit dowel pins and pilot joints to connect one weldment section to another. These fits need to be assessed through a finite element modal analysis as well as practically tested through modal impacts on manufactured prototypes to investigate how the fits affect the stiffness of the turbine's connected weldments. The experimental data will be used to validate the finite element analysis (FEA) model tool to eventually be presented to Solar Turbines.	Concept Description Our project will include the creation of an Ansys model of Solar Turbine's rotor disks and three manufactured assemblies that will be tested to validate the Ansys model. These include two assemblies with dowel pin connections and one assembly welded together. Each will be mounted to a shake table and struck with an impact hammer to produce vibrations to be read with accelerometers. Each assembly is estimated to cost around \$120, with the majority of the funds being used on specialized tooling for the manufacturing process. 
Design Specifications The specifications for this design include the accurate modeling of radial interference effects on rotor stiffness from the pilot fit and dowel pins, along with qualitative verification from testing. The modeling and testing of only radial interferences was decided between Solar Turbines and the team due to complications that would arise with manufacturing axial interference in the physical testing model. Other simplifications made to the design for the models, such as dowel pin size and lack of contours, allow for feasible manufacturing, while also meeting the requirements on requested radial interference qualitative data.	What's Next The next steps include finalizing results from the Ansys model parameterization of multiple configurations, manufacturing and testing of the three physical verification configurations, and a final design review presentation to Solar Turbines to provide analysis results. The Ansys model results will be recorded by the end of the winter quarter. The first configuration of the verification testing will be manufactured also by the end of the winter quarter. During spring quarter, the remaining verification testing models will be created, and all three models will be tested in the vibrations laboratory. Finally, the final presentation of this project's results will be given to Solar Turbines in-person at San Diego.

Figure 1. Four-Panel Chart for Critical Design Review.

Overview

The goal of this project is to complete a comprehensive study on how the radial interference fit of the mating dowel pins and pilot joints affect the rotor stiffness on Solar Turbine's Titan 130 turbine compressor. An Ansys model has been developed, where dowel pin and pilot fit interference levels have been parameterized to find trends in the effected rotor stiffness via natural frequency. A study on pilot fit length has also started with the Ansys model, finding the stiffness of the assembly when the length of the pilot fit is increased. Three assemblies will be manufactured to validate the results of the Ansys model. Although the Ansys model will be used to find the best theorized configurations, two configurations that will be the easiest to manufacture will be chosen for two physical assemblies. In this way, the Ansys model can still be verified while keeping the manufacturing feasible given the resources at Cal Poly. The third physical assembly will be a rigid welded one to act as a datum for the qualitative data of the two other assemblies.

The goal of this report is to explain in detail the progress of the Compressor Rotor Joints and Fits Interference Study. This report will first contain our Concept Description, which will go into further detail on our Ansys model and respective manufactured verification prototypes. Next will be our Design Justification, which will explain the decisions made regarding the required design specifications. Then, the Manufacturing Plan section will describe the detailed procedure for manufacturing our verification prototypes. Next, the Design Verification section will talk about the testing plan for the assemblies. Finally, the Budget section will break down the cost analysis performed so far.

Concept Description

The objective of this project is to quantify how the stiffness of a rotor plate assembly changes as a function of interference fit parameters. Solar Turbines is interested in understanding how variations in dowel pin interference, pilot fit interference, and pilot contact length influence the global stiffness of a gas generator rotor plate interface. Assembly stiffness directly affects natural frequency, which is critical for vibration performance and avoidance of resonance in turbine systems. A clear understanding of this relationship allows for improved manufacturing and design decisions that enhance structural reliability and dynamic stability.

The theoretical foundation for this work comes from vibration analysis of a single degree of freedom system. The governing equation of motion shows that natural frequency is proportional to the square root of stiffness divided by mass. Rearranging this relationship shows that stiffness is equal to mass multiplied by the square of the natural frequency. This equation forms the basis of the methodology used in this study. By extracting natural frequencies from a preloaded finite element model and multiplying the squared frequency by the total mass of the assembly, the effective stiffness of the assembly can be determined. This approach directly links measurable vibratory behavior to structural stiffness generated by interference fits.

The final design consists of a scaled rotor plate assembly that represents the interface between turbine rotor components. The assembly includes two circular plates joined by a central pilot interface and constrained radially by dowel pins. Controlled interference fits are introduced at both the dowel pin interfaces and the pilot fit interface. When assembled, these interferences generate contact pressure between mating surfaces. The resulting compressive stresses increase resistance to relative motion, thereby increasing the effective stiffness of the structure. As stiffness increases, natural frequency correspondingly increases according to vibration theory. The design therefore enables controlled manipulation of preload to observe changes in structural stiffness.

Full-scale Solar Turbines rotor plates are approximately twenty inches in diameter and exceed the dimensional and weight limitations of the Cal Poly vibrations laboratory equipment. The laboratory shake table has size constraints and a maximum load capacity of one hundred pounds. For this reason, a geometrically scaled assembly with a diameter of seven inches and an approximate total weight of twenty pounds was developed. Although reduced in size, the scaled assembly preserves the essential contact mechanics and geometric relationships necessary to observe stiffness trends. The simplified design allows reliable laboratory testing while maintaining the physical mechanisms that govern stiffness generation in the full-scale assembly. Axial tolerances were removed from the study to isolate the effects of radial interference. By eliminating unnecessary variables, the experiment focuses strictly on how radial preload influences assembly stiffness.

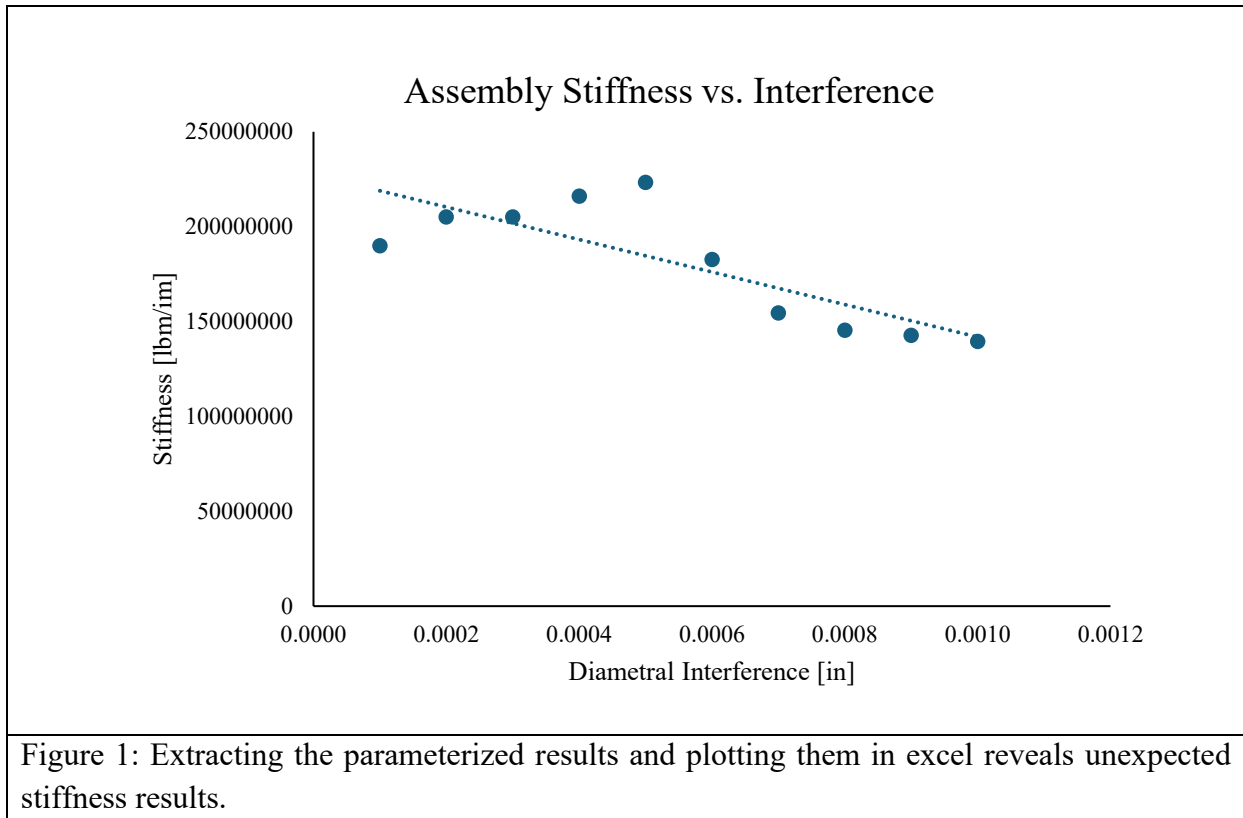
Finite element modeling was performed in Ansys Workbench to replicate the experimental configuration as accurately as possible. The geometry was constructed directly within Ansys DesignModeler to take advantage of full parameterization capabilities. Parameterization allows geometric dimensions such as dowel hole diameter, pilot bore diameter, interference magnitude, and pilot contact length to be defined as adjustable variables. These variables can then be systematically modified through a design point table. When a new design point is executed, Ansys automatically updates the geometry, resolves the static structural contact analysis, transfers the resulting stress state to a modal analysis, and outputs the updated natural frequencies. This automated workflow ensures consistency between iterations and significantly reduces manual modeling time.

The analysis sequence begins with a static structural contact simulation. Frictional contact conditions are applied at all interfaces where interference occurs. Large deformation effects are enabled to account for nonlinear contact behavior that arises from interference induced preload. A fixed boundary condition is applied to the bottom surface of one plate to replicate the experimental mounting condition used during testing. A refined tetrahedral mesh is generated with increased element density in contact regions to capture stress gradients accurately. Mesh refinement is continued until the maximum allowable node count for the student license is reached. The static solution captures the preload stresses generated by interference fits and establishes the initial stress state of the assembly.

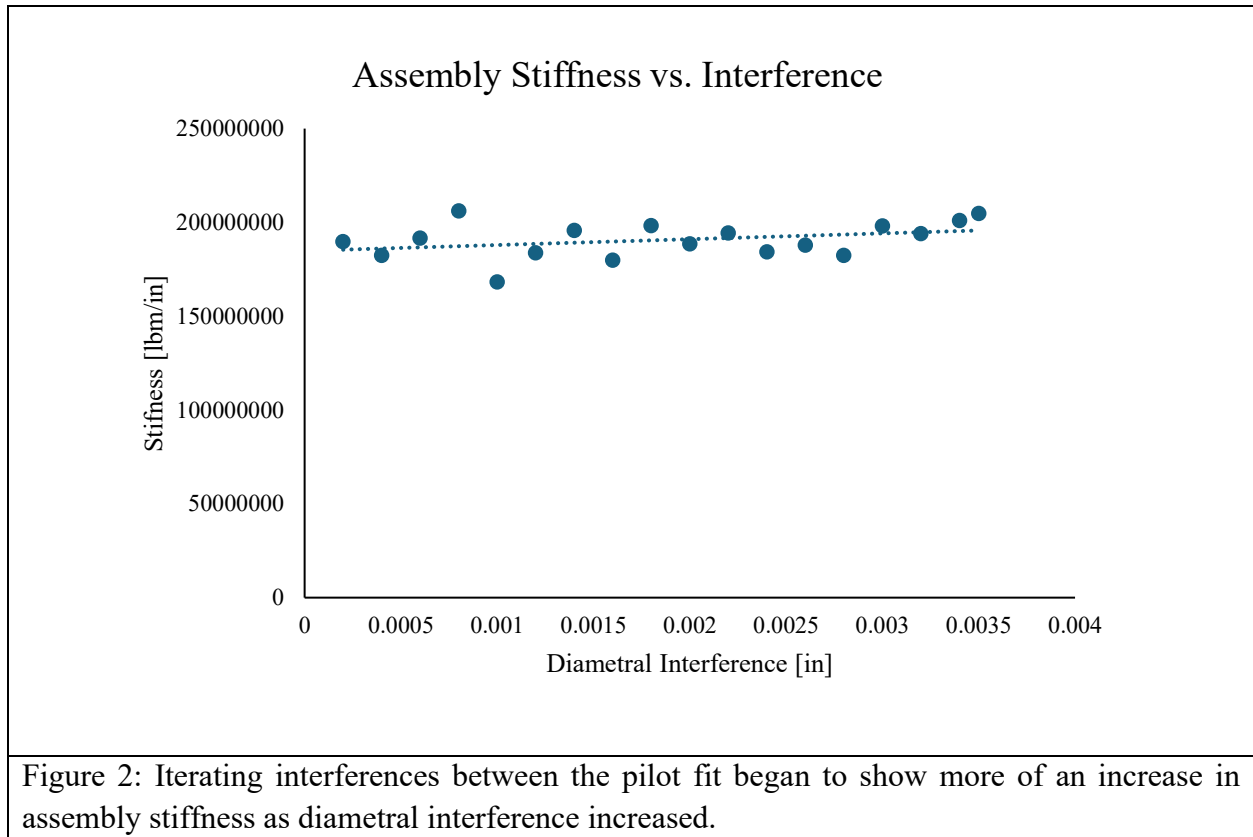
Following completion of the static analysis, a modal analysis is performed using the preloaded stress condition. This preloaded modal analysis is essential because it incorporates the stiffness contribution from contact pressure. Without preload, the modal solution would neglect the additional stiffness generated by interference fits. The first several natural frequencies are extracted, and the dominant bending mode is selected for stiffness calculation. The stiffness of each configuration is computed by multiplying the assembly mass by the square of the extracted natural frequency. Because mass remains constant between configurations, changes in natural frequency directly reflect changes in stiffness.

Interference values were selected based on ANSI B4.1 standards for locational interference fits. Locational fits are appropriate when positional accuracy between components is critical, which is the case for turbine rotor plates. The standard defines several classes of interference magnitude, including light, medium, and heavy drive fits. Solar Turbines currently operates within the lighter interference classes for dowel pins. Therefore, the parametric study was conducted across a range spanning from light to moderate interference values. Fits exceeding this range were not considered due to manufacturing difficulty and reduced practicality. A similar approach was applied to the pilot fit. The pilot contact length was treated as a separate variable with an upper limit defined as one half of the total plate thickness. This constraint ensured that the design remained realistic and manufacturable.

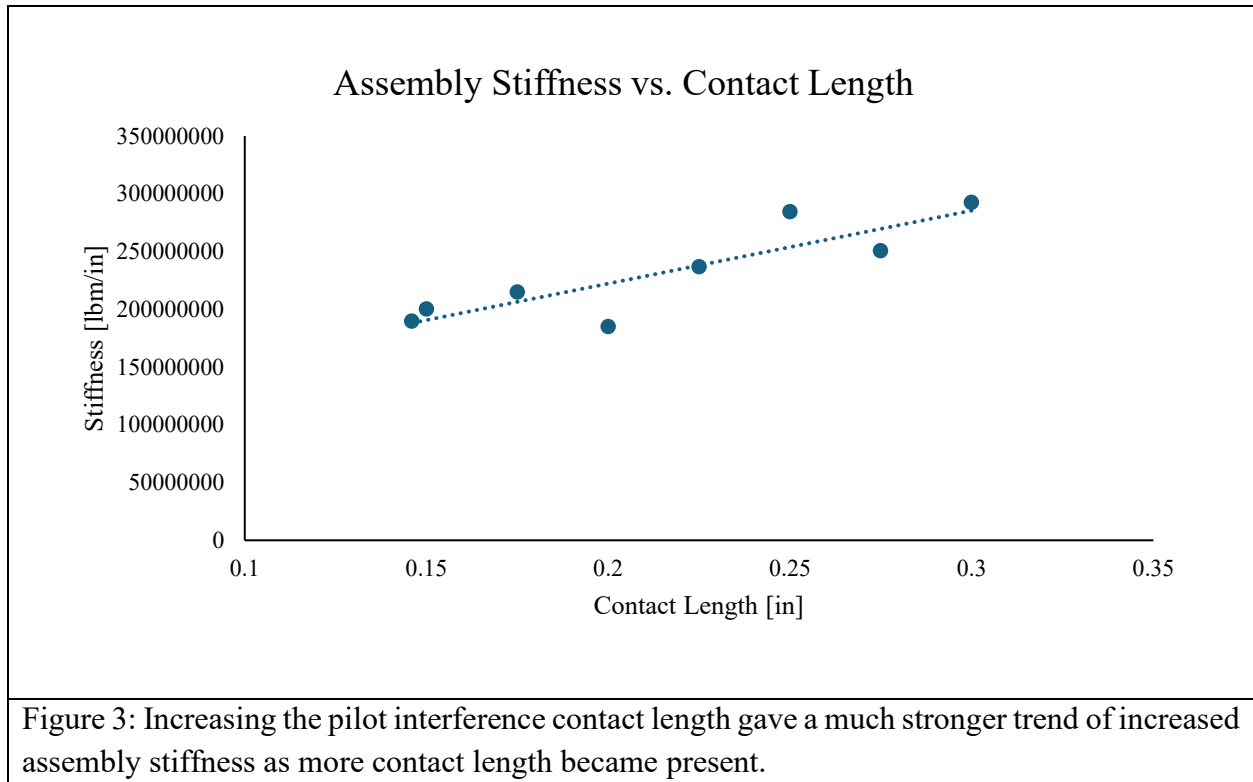
Results from the parametric study revealed that increasing dowel interference alone produced only minor increases in stiffness.



Increasing pilot diametral interference similarly resulted in modest stiffness gains.



In contrast, increasing pilot contact length produced a significant increase in natural frequency and therefore stiffness.



This result indicates that distributed contact area contributes more effectively to global stiffness than simply increasing interference magnitude. During the pilot contact length study, dowel and pilot interferences were maintained within light interference ranges to isolate the effect of contact length. The stiffness versus interference plots clearly demonstrated that contact length is the dominant parameter influencing assembly rigidity within the ranges tested.

Although finite element analysis provides strong predictive capability, experimental verification is required to validate results. Two complementary testing methods will be used. The first method involves modal impact hammer testing. The assembly will be mounted to a rigid base consistent with the boundary conditions applied in the finite element model. Accelerometers will be attached at strategic locations on the plates. The assembly will be struck using a calibrated modal impact hammer that records input force. The resulting acceleration response will be processed using a spectral analyzer to extract frequency response functions. Peaks in the frequency spectrum correspond to natural frequencies. Measured frequencies will be compared directly with finite element predictions, and relative accelerometer motion will be compared with predicted mode shapes.

The second method involves shake table testing. The assembly will be mounted to the shake table and subjected to a frequency sweep across a specified range. Accelerometers will measure the dynamic response of the system as excitation frequency increases. Resonance peaks will be identified from the response data. Extracted natural frequencies will again be used to

calculate stiffness by multiplying the squared frequency by assembly mass. Agreement between experimental and analytical stiffness values will validate the finite element model and confirm that interference induced preload is being accurately represented.

Two configurations were selected for physical testing to maximize observable stiffness differences. The first configuration represents a lower preload condition with relatively small diametral interferences and shorter pilot contact length. The second configuration represents a higher preload condition with increased dowel interference, increased pilot interference, and extended pilot contact length. Selecting both low and high preload states ensures that measurable differences in stiffness should be detectable during experimentation. These configurations correspond directly to specific design points evaluated in the finite element study, allowing direct analytical and experimental comparison.

In summary, the final design integrates vibration theory, parameterized finite element modeling, standards-based interference selection, and laboratory validation to quantify stiffness variation in rotor plate assemblies. The scaled assembly preserves critical contact mechanics while satisfying laboratory constraints. Preloaded modal analysis provides stiffness predictions grounded in fundamental physics. Experimental testing using both modal impact and shake table methods ensures verification of analytical results. This comprehensive approach provides Solar Turbines with validated insight into how interference fits and pilot geometry influence rotor plate stiffness and supports informed design decisions for future turbine assemblies.

Design Justification

Given the high tolerance nature of producing turbine engine rotor blades, we plan on minimizing the risk of tolerance stack up and propagation by milling our parts on a CNC mill. The methods for which we will test whether our physical model is within our decided tolerance is by using a CMM machine that can measure dimensions with a tolerance of ± 0.0001 . We will ensure that our physical model matches our CAD models using key dimensions. To do this, we will verify that the material properties of each model are as similar as possible by sourcing our stock from the same supplier and batch. We will also ensure proper machining by using the same machine, tools, and procedure. Utilizing these tools and processes, we are confident that we will be able to produce identical prototype models machined to our desired dimensions. We will then test each model on the shake table and verify that the mode shapes and natural frequencies align with anticipated results.

Table 1. Justification that our design will meet each of our specifications.

#	Spec	Specification Description	Justification
1	Rotor CAD model delivered	Deliver complete CAD for female and male plates that characterize the mating surface between compressor rotor plates and aft weldment. Fully constrained dimensions	Establishes the controlled geometry baseline for analysis, fabrication and testing.
2	Material Property definition	Provide accurate material properties used for analysis with datasheet and verification	Guarantees analysis inputs are traceable and repeatable.
3	Compressor Assembly CAD Package	Deliver full compressor assembly CAD that accurately characterizes the assembly. Supports bill of materials.	Allows for interface checks. Ensures proper part coupling and part count.
4	Ansys FEA Model	Build an Ansys model that can mesh and solve our chosen geometry based on assigned loads and boundary conditions	Provides an analytical tool to predict performance and guide design configurations and alteration
5	Prototype fabrication	Manufacture a prototype to drawing specifications with inspection results	Provides a physical article for dimensional testing and validating manufacturing plan
6	Modal Simulation	Create a model that can accurately calculate the natural frequency of a choice part	Predict dynamic behavior and supports design decisions before testing
7	Modal impact test	Conduct modal impact testing and report natural frequency and mode shapes.	Validates real world dynamics and suppliers' data to our experimental calculations. Also tests and verifies our testing procedure.
8	First Natural Frequency Requirement	Demonstrate first natural frequency meets a minimum of 186 Hz.	Prevents resonance issues and confirms stiffness meets system requirements
9	Test to analysis correlation	Quantify correlation between physical model and theoretical model values. Ensure values are within predefined tolerances	Confirms the model is predictive and supports confident design iterations.
10	Manufacturing process definition	Provide a tested manufacturing plan for each part and assembly	Reduces manufacturing error, streamlines production and reduces cost.

Manufacturing Plan

Table 2 shows a summary of the manufacturing plan for our three assemblies. A full detailed build plan for the assemblies can be found in Appendix E. The two chosen configurations for the interference fit assemblies will be manufactured using the same process. The female and male plate will both be machined via a 3-axis CNC mill in the Cal Poly Aero Hangar. The first operation will require four heavy duty vise jaw grips to hold the workpiece; the second operation will require soft jaws to be machined to hold the workpiece. One to two pairs of soft jaws will be machined to be reused for assembly. Steel tabs for mounting to a shake table will be cut using the waterjet in Mustang 60. Due to the water jet's poor tolerance, the holes on the tabs will be undersized and drilled on the drill press in the Aero Hangar. Four tabs will be waterjet and welded to the female plate as it is the bottom plate that will be directly on top of a shake table. A welding jig will be 3D printed to ensure that the tabs are welded onto the female plate in the correct positions to match the hole pattern on the shake table.

Before assembling, each male and female plate will be inspected on the CMM in the IME department to assess the critical interference dimensions. For the assembly of the interference fit configurations, a temperature gradient is necessary to assemble interference fits. We plan on utilizing an oven in the MATE department with the help of Professor Beaton. First, the male plate will be placed in the oven so that the dowel pin holes will expand. Meanwhile, the dowel pins will be placed in liquid nitrogen for shrinkage. The dowel pins will then be pressed into the hot male plate. Then, the female plate will be placed in the oven to expand. The male plate will then be pressed into the female plate using an arbor press.

The welded assembly has a similar but shorter manufacturing process. Both plates will be machined on the 3-axis CNC mill in the Aero Hangar. But the process excludes machining the dowel pin holes and pilot ledges. The plates will be MIG welded together.

Table 2. Manufacturing summary.

Subsystem	Component (Part Number)	Purchase (P) Modify (M) Build (B)	Raw Materials Needed	Where/ how procured?	Equipment and Operations
2 Interference Fit Configurations	Male plate (1100, 2100)	B	1018 alloy steel round stock	Sponsor to purchase from McMaster-Carr	CNC mill - 2 operations, soft jaws needed for 2nd op
	Female plate (1200, 2200)	B	1018 alloy steel round stock	Sponsor to purchase from McMaster-Carr	CNC mill - 2 operations, soft jaws needed for 2nd op
	Dowel pins (1300, 2300)	P	3/16" OD alloy steel dowel pin	Sponsor to purchase from McMaster-Carr	n/a
	Tabs (1110, 2110)	B	0.25" thick 1026 steel	Sponsor to purchase from McMaster-Carr	waterjet to cut profile, drill press to create hole, weld to attach to plate
	Shake table mounting bolts (1111, 2111)	P	3/8" OD, 1" length Grade 8 steel bolt	Sponsor to purchase from McMaster-Carr	n/a
1 Welded Configuration	Male plate (3100)	B	1018 alloy steel round stock	Sponsor to purchase from McMaster-Carr	CNC mill - 2 operations, soft jaws needed for 2nd op
	Female plate (3200)	B	1018 alloy steel round stock	Sponsor to purchase from McMaster-Carr	CNC mill - 2 operations, soft jaws needed for 2nd op
	Tabs (3110)	B	0.25" thick 1026 steel	Sponsor to purchase from McMaster-Carr	waterjet to cut profile, drill press to create hole, weld to attach to plate
	Shake table mounting bolts (3111)	P	3/8" OD, 1" length Grade 8 steel bolt	Sponsor to purchase from McMaster-Carr	n/a
Manufacturing tools	Soft jaws	B	6061-Al machinable 8"x2.5"x1" billet	Sponsor to purchase from McMaster-Carr	CNC mill – mill the profile of the machined plate
	Welding jig	B	PLA filament	Makerspace	Bambu printer

Design Verification

The design verification, or physical assemblies, of this project will be used to verify the Ansys model created to study the effects radial interference fits from the dowel pins and pilot fits have on overall system stiffness. As the Ansys model will be made to study many different conditions, three physical test assemblies will be manufactured to ensure the Ansys model is providing accurate results. Many specific checks will be run on the manufactured parts to ensure each part was made to the correct and necessary specifications for testing. Table 3 below includes these specification tests. Before the physical modal analysis is completed on the assemblies, multiple dimensional analyses are performed to import dimensions into the Ansys model that correctly match physical build. These assemblies will be used to verify the validity of the Ansys model; therefore, multiple tests of product sizing will be conducted before any other testing.

Table 3. Design Verification Plan

Project:		6RL-B	Sponsor:		Manny Aldana		Date:	2/13/26	
TEST PLAN									
Test #	Spec #	Test Description	Measurements	Acceptance Criteria	Facilities/ Equipment	Parts Needed	Member	TIMING	
								Start date	
1	1	Open, rebuild, and validate male and female plate models. Confirm mating surfaces and constraint status	Use rebuild/error log to check current status to ensure models are fully defined with key interface dimensions	Zero rebuild errors, all sketches are fully constrained.	SolidWorks	Digital CAD files	TBD	TBD	
2	2	Confirm properties match datasheet and are properly imported into Ansys	Elastic modulus, Poisson’s ratio, density, yield strength	Required properties are provided with units and source	Materials datasheet	none	TBD	TBD	
3	3	Open assembly, check mates and interference values. Compare assembly BOM with project BOM	Use rebuild/error log, check DOF, and interference values	Opens with zero errors, interference values match, BOM match	SolidWorks	Digital CAD files	TBD	TBD	

4	4	Import geometry and ensure proper meshing, correct loading values, and correct load locations are generated. Run a simulation properly	Mesh count, mesh type, solver completion status, run logs, boundary conditions, documented loads	Model meshes successfully, model completes successful simulation, values generated are within reason	Ansys	Digital CAD files	TBD	TBD
5	5	Manufacture prototype per drawing dimensions and manufacturing procedure. Inspect critical dimensions and record material certifications	Dimensional inspection using calipers and CMM. Check critical dimensions	All critical dimensions within drawing tolerance. Material properties match specifications. No major damage	CMM machine, Calipers, data sheets	Machined parts, drawing, fasteners,	TBD	TBD
6	6	Run modal analysis on selected part using defined boundary conditions and calculate mode shapes and verify they are within reason	Mode shapes, solvers settings, Boundary conditions definition, Mass and stiffness values	Modal analysis runs completely. Output mode shapes and natural frequencies.	Ansys model	Material data sheet	TBD	TBD
7	7	Perform impact hammer test, calculate/measure frequency response.	Utilize frequency response to determine measurement alignment.	Test produced data for coherence. Ensure data has/reduces noise	Shake table, systems laboratory, modal impact hammer	Prototype assembly, impact hammer, accelerometers	TBD	TBD
8	8	Confirm first natural frequency meets requirements under defined configurations.	experimental first natural frequency match Configuration and boundary conditions	Pass/fail: first natural frequency must be greater than 186 Hz	Shake table, systems laboratory, modal impact hammer	Prototype assembly, impact hammer, accelerometers	TBD	TBD

9	9	Compare model test results with FEA results for agreement. Quantity error	Percent difference per model between test and theoretical values.	All defined values are within +/- 15 percent. and are recorded	Ansys	FEA results, test results	TBD	TBD
10	10	Test the processing and manufacturing plan.	Verify tooling list, operations order, and operation time	At least one prototype build is executed using manufacturing and processing plans.	CNC mill, aero hanger	Drawings, Prototype manufacturing plans, inspection plan	TBD	TBD

Budget

The budget spending for this project is driven solely by the testing and verification assembly that will be used to verify the Ansys model. This test and verification assembly, also called the verification prototype, is made up of three sub-assemblies that represent three different configurations to be tested. Table 4 shows a list of materials that will be purchased to manufacture these three sub-assemblies. Most of the funds will be used for raw steel stock, with the remaining funds being used for tooling and purchased mechanical parts. This table does not include the exhaustive list of tooling needed to manufacture the assemblies. Other tooling and parts that were listed in the manufacturing plan will be provided by Cal Poly's machine shops. Although the project is under budget so far, it is predicted that there will be a need to order more stock or tooling as the process progresses through the remainder of winter quarter.

Table 4. Budget summary.

Item Description	Vendor	Vendor's Part #	iBOM #	Total Cost	Purchase Status
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	1100	\$74.87	On the way
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	1200	\$74.87	On the way
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	2100	\$74.87	On the way
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	2200	\$74.87	On the way
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	3100	\$74.87	Standby
Stock Steel 1018 Plates (7''D, 1.5''L)	McMaster-Carr	7786T267	3200	\$74.87	Standby
Dowel Pins (3/16''D)	McMaster-Carr	98372A117	1300	\$31.26	On the way
Carbide Drill Bit (11/64)	McMaster-Carr	27515A42	-	\$23.03	On the way
Reamer (0.1865)	McMaster-Carr	3087A52	-	\$25.95	On the way
Reamer (0.1855)	McMaster-Carr	2995A212	-	\$30.46	On the way
Soft Jaws	McMaster-Carr	1005N35	-	\$64.27	On the way
Plate Tab Stock (12''x12''x1/4'')	McMaster-Carr	6544K25	-	\$84.38	On the way
Bolts (3/8''-16, 1''L)	McMaster-Carr	92620A624	-	\$13.68	On the way
Total Budget:	\$4000				
Amount Spent:	\$618.31				
Planned Spending:	\$1000				
Remainder:	\$3000				

Appendix A: Design Analysis

Many calculations were performed to analyze the feasibility of the physical test models. With the given goals of the project, the team calculated the predicted hoop stress of the radial and axial interference fits, the resistive force capabilities of the radial fits combating the axial interference fits, the required bolt size to maintain the axial interference requirements, and more. An example of this can be seen in Figures A-1 and A-2. Overall, the decision to remove the study of axial interference on the plates was made after said calculations predicted the torque specs to be beyond the manufacturing capabilities of the team given the tools available. This can be seen in Figure A-3 where we calculated the tensile force the bolt would need to exert to create the required axial interference and its subsequent preload and torque specification. Further analysis of this design was completed in Ansys and is explained in further detail in the Ansys portions of this report.

Config. 1		
Input	Value	Units
A	3.5	in
B	2.345	in
C	2.342	in
D	1	in
E	1	in
F	0.83	in
G	0.833	in
H	0.1855	in
I	0.1875	in
J	0.5	in
E _o	2.97E+07	psi
E _i	2.97E+07	psi
v _o	0.29	-
v _i	0.29	-
μ	0.74	-

of Dowel Pins: 5
k 0.50 scaled version
Length of shear 0.837

Config 2.		
Input	Value	Units
A	3.5	in
B	3.345	in
C	3.342	in
D	1	in
E	1	in
F	0.83	in
G	0.831	in
H	0.1855	in
I	0.1875	in
J	0.5	in
E _o	2.97E+07	psi
E _i	2.97E+07	psi
v _o	0.29	-
v _i	0.29	-
μ	0.74	-

of Dowel Pins: 5
k 0.38 scaled version
Length of shear 0.837

Manny recommended Interferences		
Interf.	Value	Units
δ _{PR}	0.003	in
δ _{PA}	-0.001	in
δ _{DR}	0.002	in

solars tolerance
radial

Manny recommended Interferences		
Interf.	Value	Units
δ _{PR}	0.003	in
δ _{PA}	-0.001	in
δ _{DR}	0.003	in

solars tolerance
radial

Resulting Surface Areas		
SA	Value	Units
PR	2.457441738	in^2
PA	21.20881346	in^2
DR	0.294524311	in^2
PA_2	17.23152261	in^2
PA_3	6.572211831	in^2

outer area
face area
face cut area

Resulting Surface Areas		
SA	Value	Units
PR	3.548730495	in^2
PA	3.333151266	in^2
DR	0.294524311	in^2
PA_2	35.08833525	in^2
PA_3	7.432615519	in^2

outer area
face area
face cut area

	SciNot	Normal	Units
Pilot Radial Pressure:	1.05E+04	10504.97	psi
Dowel Radial Pressure:	3.14E+05	314015.63	psi

	SciNot	Normal	Units
Pilot Radial Pressure:	1.18E+03	1176.37	psi
Dowel Radial Pressure:	4.71E+05	471023.44	psi

net Forces		
Force	Value	Units
Pilot Radial	25815.4	lbs
Dowel Radial	462426.2	lbs

net Forces		
Force	Value	Units
Pilot Radial	4174.6	lbs
Dowel Radial	693639.3	lbs

Resulting Frictional Force: in axial direction		
Radial Fits:	361298.7334	lbs

Frictional Force: in axial direction		
Radial Fits:	516382.2868	lbs

axial interference		
pressure	Value	Units
Pilot axial	-3.57E+04	psi
Dowel Radial	0.0	lbs

axial force from interference		
Force	value	units
pilot axial with PA_2	-6.144E+05	lbf
pilot axial with step/PA_3	-2.343E+05	lbf

compare froces	
face	dowel
-2.34E+05	3.61E+05

good

bolt resultant

1.27E+05 lbf
126971.37

axial interference		
pressure	Value	Units
Pilot axial	-3.57E+04	psi
Dowel Radial	0.0	lbs

axial force from interference		
Force	value	units
pilot axial with PA_2	-1.254E+06	lbf
pilot axial with step/PA_3	-2.656E+05	lbf

compare froces	
face	dowel
-2.66E+05	5.16E+05

good

bolt resultant

2.51E+05 lbf
250740.07

Figure A-1: Example of axial vs radial interference fit forces comparison. Both configurations determine that the frictional force of the dowel pin from radial interference is enough to combat the axial interference of the pilot fit based on the dimensions given.

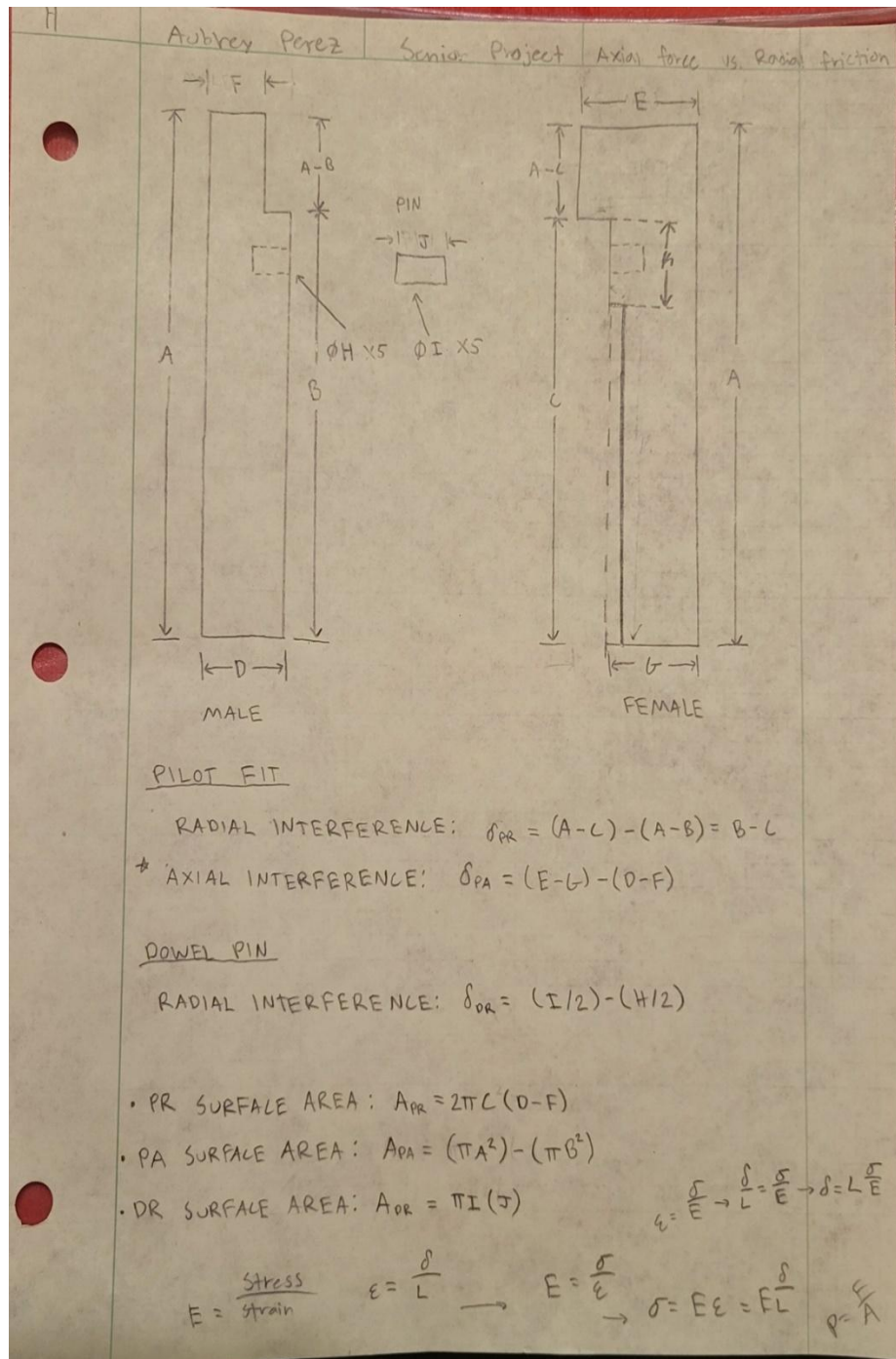


Figure A-2: Example dimensioning set up used for interference calculations, such as the ones shown in Figure A-1.

clamp member stress calculator
4 plates 2 washers and holes

Key

inputs calculated solutions

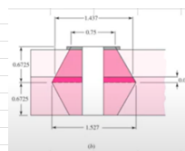
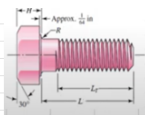
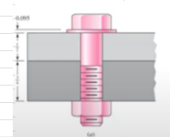


Table B-17

Grade or Class	Size Range	Endurance Strength
SAE 5	$\frac{1}{4}$ to $\frac{1}{2}$ in	18.0 kpsi
SAE 5	$\frac{3}{4}$ to 1 in	16.5 kpsi
SAE 7	$\frac{1}{4}$ to $\frac{1}{2}$ in	20.0 kpsi
SAE 7	$\frac{3}{4}$ to 1 in	18.5 kpsi
SAE 8	$\frac{1}{4}$ to $\frac{1}{2}$ in	21.5 kpsi
SAE 8	$\frac{3}{4}$ to 1 in	20.0 kpsi
SAE 9	$\frac{1}{4}$ to $\frac{1}{2}$ in	23.0 kpsi
SAE 9	$\frac{3}{4}$ to 1 in	21.5 kpsi
SAE 10	$\frac{1}{4}$ to $\frac{1}{2}$ in	24.5 kpsi
SAE 10	$\frac{3}{4}$ to 1 in	23.0 kpsi



hardware:		
Bolt	1.25, unc = 7, unf = SAE grade 8	
Max tensile force =	1.45E+05	145350
nut height	1.16 inches	
thread	7 threads per inch	
washer diameter theory	1.875 inches	
chosen washer diameter	2.5 inches	
bolt length calc	6.168571429 inches	
force trying to keep	9.15E+04 lbf	91455.3

Material props	E steel	3.00E+07 psi
	E cast iron,imu,	1.45E+07 psi
	E alumin	1.03E+07 psi

units	inches	
-------	--------	--

member	thickness	elastic modulus
top washer	0.165 in	3.00E+07
top cup	0.75 in	3.00E+07
top void	0.25 in	0.00E+00
male plate	1 in	3.00E+07
void 2	0.25 in	0.00E+00
female plate	1 in	3.00E+07
void3	0.25 in	0.00E+00
bottom cup	0.75 in	3.00E+07
bottom washer	0.165 in	3.00E+07

FATIGUE 4 pressure vessel

Max pressure	5.50E+05 psi
Min pressure	0 psi
pressure dia	0.8 in^2
#BOLTS	1
S_e	1.86E+04 psi

need to update value of D for metric problems, or undo

For solid metal, with hole

segment	thickness	Diameter	E	K	1/k
washer	1	0.165	1.875	3.00E+07	3.287E+08
t cup	2	0.75	2.065526	3.00E+07	7.22771E-09
void	3	0.25	2.931026	0.00E+00	1.32308E-07
t male	4	1	3.219526	3.00E+07	2.937E+08
void	5	0.125	4.373526	0.00E+00	2.99792E-07
void	6	0.125	4.517776	0	0
void	7	1	3.219526	3.00E+07	2.937E+08
female	8	0.25	2.931026	0	1.32308E-07
void	9	0.75	2.065526	3.00E+07	1.384E+08
cup2	10	0.165	1.875	3.00E+07	3.287E+08
washer	thickness chcl	4.58	member stiffness		1.690E+06

t tw = thickness top washer

pressure area	0.502654825 in^2
Net max load calc	2.76E+05 lbf
net max load used	9.15E+04
net min load	0.00E+00
max force per bolt plate assembly calc	9.15E+04 newtons
max force per bolt plate assembly chosen	9.15E+04
min force per bolt(P)	0.00E+00

With $\alpha = 30^\circ$, this becomes

$$k = \frac{0.5774\pi E d}{\ln \left(\frac{(1.155t + D - d)(D + d)}{(1.155t + D + d)(D - d)} \right)}$$

bolt properties		
bolt elastic modulus	3.00E+07 psi	
bolt length chosen	4.75 in	
bolt diameter	1.25 in	P_max
Strength_Ultimate	1.50E+05 psi	P_min
S_y (yield)	1.30E+05 psi	
s_p (Proof) calculated	1.11E+05 psi	
s_p (Proof) stated	1.10E+05 psi	
friction factor	0.2	

Dimensions	in	in
threaded length (8.13)	2.75	in
threaded length (8.14)	8.5	mm
unthreaded length	2	in
threaded length in grip	2.580	in
nominal area	1.22718463	in^2
threaded area	0.969	in^2

$$k_b = \frac{A_d A_t E}{A_d l_t + A_t l_d}$$

$$L_T = \begin{cases} 2d + \frac{1}{4} \text{ in} & L \leq 6 \text{ in} \\ 2d + \frac{1}{4} \text{ in} & L > 6 \text{ in} \end{cases} \quad (8-13)$$

$$L_T = \begin{cases} 2d + 6 & L \leq 125 \\ 2d + 12 & 125 < L \leq 200 \\ 2d + 25 & L > 200 \end{cases} \quad (8-14)$$

We need $F_i > 60\% F_p$
to prevent buckling out

bolt stiffness	6.989E+06 lbf/in	
C (joint compliance)	0.805	
preload F_i	5.330E+04	53295.000 lbf
F_b (F bolt exp)	1.269E+05	126943.470
torque	1.332E+04	13323.75 in-lbf
eta_p = proof bolt FOS	8.397E-01	
eta_not (opening FOS)	2.993E+00	
eta_L (proof load)	7.236E-01	
what bolt can take	1.45E+05	145350

Did a sanity check, looks good to me

copy and paste equation not cells		
spare(if accidently/check		
threaded length (8.13) EQ	2.75	
threaded length (8.14)	8.5	

Figure A-3: Bolt calculations for the case of needing a center bolt through the assembly to maintain the axial interference of the pilot fit.

Appendix B: Gantt Chart

The Gantt chart followed for this Critical Design Review can be seen below in Figure B-1. The majority of the documentation and milestones leading up to this design review can be found in this report. Since the Preliminary Design Review, a final Ansys model and testing design was created to achieve the end project goal. Through the months of January and February, the team created a manufacturing plan, testing and verification plan, budget and spending table, and detailed CAD to explain the final decisions made by the team for this project. The next phase in the project after this phase is Safety and Verification planning, which includes studying the safety of the planned testing methods and making alterations if required.

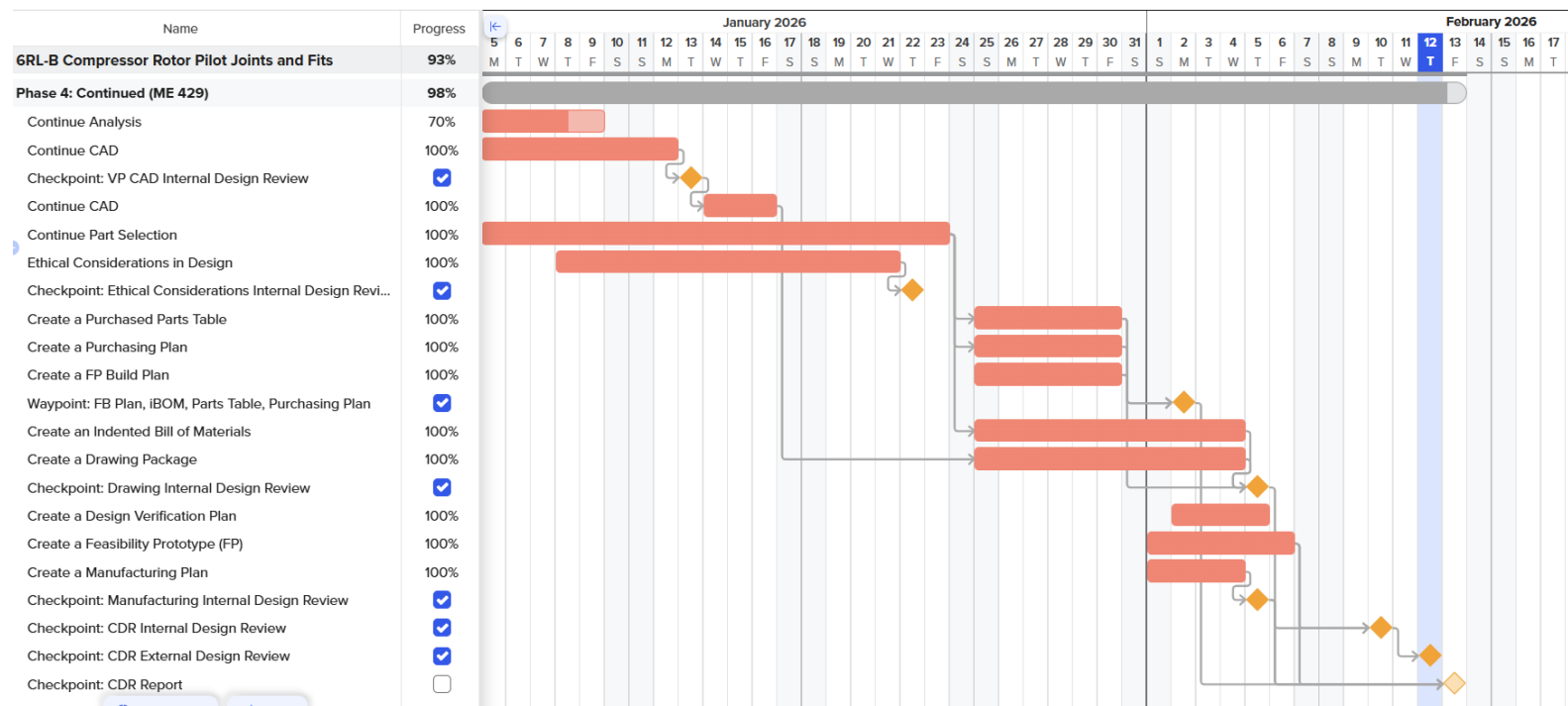


Figure B-1: Phase 4 Gantt chart leading up to Critical Design Review.

Appendix C: FMEA

Along with designing the Ansys model and physical testing assemblies, a failure mode and effects analysis (FMEA) was conducted to study ways the physical assemblies could fail. Most of the failure modes identified are preventable with careful manufacturing and frequent measurement analysis to make sure no complications arise. Nonetheless, the main points of potential failure are always a concern. These failure modes are listed below in Table C-1 along with actions to prevent such complications.

Table C-1: FMEA table of foreseeable failure modes during manufacturing of the testing assemblies. Due to having no high-RPN failure modes, no additional actions than these were written.

System / Function	Potential Failure Mode	Potential Effects of the Failure Mode	Severity	Potential Causes of the Failure Mode	Current Preventative Activities	Occurrence	Current Detection Activities	Detection	RPN
securely mount to shake table	tabs deform/detach	a) invalid data due to incorrect boundary conditions b) unsafe testing	6	1) poor weld penetration 2) waterjet tolerance error for holes	1) utilizing a welding jig 2) weld inspection 3) accurate drilling	3	visual weld inspection	2	36
maintain pilot radial interference	interference fit too large	a) cracking during press b) yielding c) residual stress deformations	8	1) machine tolerance stackup 2) holes didn't expand fully in the oven	1) CMM inspection 2) thermal expansion calculation 3) tolerance analysis	2	dimensional inspection, monitoring press fit	2	32
gather accurate modal data	incorrect accelerometer data	a) incorrect natural frequencies b) seemingly invalid ANSYS model c) incorrect conclusion of stiffness correlation	8	1) poor accelerometer mounting 2) cable interference 3) calibration error	1) proper mounting with threaded holes or glue 2) purchase very light accelerometer 3) lab procedure review	3	repeatability check, logic check	3	72
maintain concentric alignment	misalignment between male/female plates	1) eccentricity introduced in system 2) false modal response 3) out-of-plane modes	8	1) uneven heating in oven 2) dowel pin tilt 3) arbor press misalignment	1) controlled heating time 2) alignment fixture/method 3) concentricity inspection on CMM	3	CMM inspection post assembly, dial indicator runout check	3	72

Appendix D: Design Hazard Checklist

Along with the FMEA completed to find concerning failure points, a list of generic mechanical hazards was studied. From that list, the team analyzed if any of these hazards were applicable to this project. The list of hazards and applicability to this project can be seen below in Table D-1.

Table D-1: Table of generic mechanical hazards. The leftmost column explains whether the specific hazard is applicable to this project, and therefore something the team will need to take greater caution with.

Y	N	Category
		1. Will any part of the design create hazardous revolving, reciprocating, running, shearing, punching, pressing, squeezing, drawing, cutting, rolling, mixing or similar action, including pinch points and sheer points?
		2. Can any part of the design undergo high accelerations/decelerations?
		3. Will the system have any large moving masses or large forces?
		4. Will the system produce a projectile?
		5. Would it be possible for the system to fall under gravity, creating injury?
		6. Will a user be exposed to overhanging weights as part of the design?
		7. Will the system have any sharp edges?
		8. Will any part of the electrical systems not be grounded?
		9. Will there be any large batteries or electrical voltage in the system above 40 V?
		10. Will there be any stored energy in the system such as batteries, flywheels, hanging weights or pressurized fluids?
		11. Will there be any explosive or flammable liquids, gases, or dust fuel as part of the system?
		12. Will the user of the design be required to exert any abnormal effort or physical posture during the use of the design?
		13. Will there be any materials known to be hazardous to humans involved in either the design or the manufacturing of the design?
		14. Can the system generate high levels of noise?
		15. Will the device/system be exposed to extreme environmental conditions such as fog, humidity, cold, high temperatures, etc?
		16. Is it possible for the system to be used in an unsafe manner?
		17. Will there be any other potential hazards not listed above? If yes, please explain the reverse.

Description of Hazard	Planned Corrective Action	Planned Date	Actual Date
During the shake table tests, the assembly can be exposed to high lateral accelerations. Under strong lateral forces, there is a potential danger of the momentum of the assembly breaking the mounting tabs and becoming unconstrained.	To prevent this from happening, it will be ensured that the person planning to weld the mounting tabs to the plate will practice MIG welding to ensure proper weld penetration. A visual weld inspection test will be performed before each shake table test. The mounting tab holes will also be inspected to confirm they are not too large where the mounting holes would slide around during the shake table test.	Weld practice starting: 2/23/26 Weld and tab inspections during assembly practice	
Though not a strong projectile, there is a possibility that the interference fit pops the plates apart, which is a safety concern.	To prevent this, it is planned to alter the design to include a couple of bolts and nuts to loosely constrain the plates together in the case that they pop apart. The bolts will not tightly constrain the plates together due to that affecting the testing resulting.	2/20/26	
The assembly is considerably heavy and would need to be transported to and from the vibrations lab. Dropping the assembly could cause injuries.	To prevent this, a transportation method involving crates/boxes that at least two people carry will be used. These boxes/crates will be purchased via order form.	3/6/26	
If not properly placed on the shake table, the assembly can be incorrectly constrained and be a dangerous moving object during testing.	To prevent this, a testing procedure/manual will be created to ensure that the testing is performed safely.	2/20/26	

Appendix E: Manufacturing Plan

This appendix shows a detailed plan for the manufacturing and assembly process needed for each assembly configuration. Following the first prototype that is planned to be manufactured at the end of the winter quarter, this plan will be updated as needed as better techniques and methods will be found.

Male/Female Plate

CNC Mini Mill

1. Soft Jaws
 - a. Using an end mill, pocket out the profile of a 7" outer diameter circle with a depth of 0.25".
2. 1st Operation
 - a. Hold the round stock in the vise using four heavy duty vise jaw grips; the top half of the stock will be machined.
 - b. Using a face mill, face off the top.
 - c. Using an end mill, decrease the outer diameter of the stock to 7 inches.
 - d. Using a spot drill, a reamer, and a drill, drill a through clearance hole for a 1.5" bolt.
3. 2nd Operation
 - a. Flip the part over and put the machined side of the stock in the soft jaws.
 - b. Using a face mill, remove material from the top of the stock to produce a thickness of 1.25" inch.
 - c. Using an end mill, decrease the outer diameter of the stock to 7 inches.
 - d. Using an end mill, remove material to create the pilot interference ledge (male/female ledge). Using an end mill, remove material to create the second ledge.
 - e. Using a spot drill, a reamer, and a drill, drill the dowel pin holes with a diameter of 3/16" and a depth of 0.4".

Water Jet

4. Water jet the mounting tabs out of 0.25" thick 1026 steel. Undersize the water jet holes.

Drill Press

5. Drill 3/8" holes into the waterjet tabs.

3D Printer

6. Print a welding jig that will position the tabs onto the female plate and ensure the correct hole pattern.

Welding

7. Scotch Brite the tabs to weld-prep them.
8. Using the welding jig, weld the tabs onto the female plate.

CMM

Critical Dimensions to Check:

- Dowel pin hole diameter
- Dowel pin depth
- Pilot fit ledge depths and lengths
- Diameters of pilot ledges

Welded Plate

CNC Mini Mill

1. Soft Jaws
 - a. Using an end mill, pocket out the profile of a 7" outer diameter circle with a depth of 0.25".
2. 1st Operation
 - b. Hold the round stock in the vise using four heavy duty vise jaw grips; the top half of the stock will be machined.
 - c. Using a face mill, face off the top.
 - d. Using an end mill, decrease the outer diameter of the stock to 7 inches.
 - e. Using a spot drill, a reamer, and a drill, drill a through clearance hole for a 1.5" bolt.
3. 2nd Operation
 - f. Flip the part over and put the machined side of the stock in the soft jaws.
 - g. Using a face mill, remove material from the top of the stock to produce a thickness of one inch.
 - h. Using an end mill, pocket out the center.

Welding

4. Tack the two plates together.
5. Place on rotary table and rotate slowly as team member welds stationary with MIG torch.

Assembly – Interference Fits

1. Place the male plate into the MATE department's oven to heat up to 350 degrees Celsius for ~45 minutes.
2. Place the dowel pins in liquid nitrogen for ~10 minutes.
3. Using an arbor press, press the dowel pins into the hot male plate.
9. Place the female plate into the MATE department's oven to heat up to 350 degrees Celsius degrees for ~45 minutes.
10. Using an arbor press, press the male plate with the dowel pins into the hot female plate.

Appendix F: Drawing and Specifications Package

The drawing and specifications package is a group of drawings that show the planned-out manufacturing and assembly plan for the verification prototype. First, there is a copy of the iBOM for the three assemblies. The total cost for the three assemblies comes to around \$400. About \$200 more is estimated to be used for tooling. Around \$3000 remains in the total project budget, leaving room for extra purchases if needed. Next is the Purchased Parts table that includes the list of unmodified parts used in the three assemblies. Finally, the figures are drawings of the three assemblies, including the individual parts of the assemblies, as well as the assembled views and exploded views of each assembly. Figures E-1 through E-5 are the drawings for the first two assemblies, where the plates will be connected with dowel pins and a pilot fit, while Figures E-6 through E-8 are the drawings for the third assembly, where the plates are welded together.

Cost

Compressor Rotor Press Fit Validation Assembly Indented Bill of Material (iBOM)

Assy Level	Part Number	Descriptive Part Name				Qty	Part Cost	Total Cost	Source	URL	More Info
		Lvl0	Lvl1	Lvl2	Lvl3	Lvl4					
0		Final Assy									-
1	1000	Assembly 1									-
2	1100			Male Plate			1	\$ 55.80	\$ 55.80	Custom	-
3	1110				Mounting Tabs		4	\$ 5.00	\$ 20.00	Custom	-
4	1111					3/8"-16 Bolt	4	\$ 0.55	\$ 2.20	McMaster	https://wv
2	1200			Female Plate			1	\$ 55.80	\$ 55.80	Custom	-
2	1300			3/16" Dowel Pins			5	\$ 0.21	\$ 1.05	McMaster	https://w
1	2000	Assembly 2									-
2	2100			Male Plate			1	\$ 55.80	\$ 55.80	Custom	-
3	2110				Mounting Tabs		4	\$ 5.00	\$ 20.00	Custom	-
4	2111					3/8"-16 Bolt	4	\$ 0.55	\$ 2.20	McMaster	https://wv
2	2200			Female Plate			1	\$ 55.80	\$ 55.80	Custom	-
2	2300			3/16" Dowel Pins			5	\$ 0.21	\$ 1.05	McMaster	www.mcm
1	3000	Assembly 3									-
2	3100			Male Plate			1	\$ 55.80	\$ 55.80	Custom	-
3	3110				Mounting Tabs		4	\$ 5.00	\$ 20.00	Custom	-
4	3111					3/8"-16 Bolt	4	\$ 0.55	\$ 2.20	McMaster	https://wv
2	3200			Female Plate			1	\$ 55.80	\$ 55.80	Custom	-
Total Parts							40	\$403.50			-

Purchased Parts Table

Part Number	Part Name	Vendor	Vendor Part Number	Price	Notes	Link
1300, 2300	3/16" Dowel Pin	McMaster	98381A505	10.37	Pack of 50 3/16" pre-cut dowel pins	https://wv
1111, 2111, 3111	3/8"-16 Bolt	McMaster	92620A624	13.68	3/8"-16 course bolt	https://wv

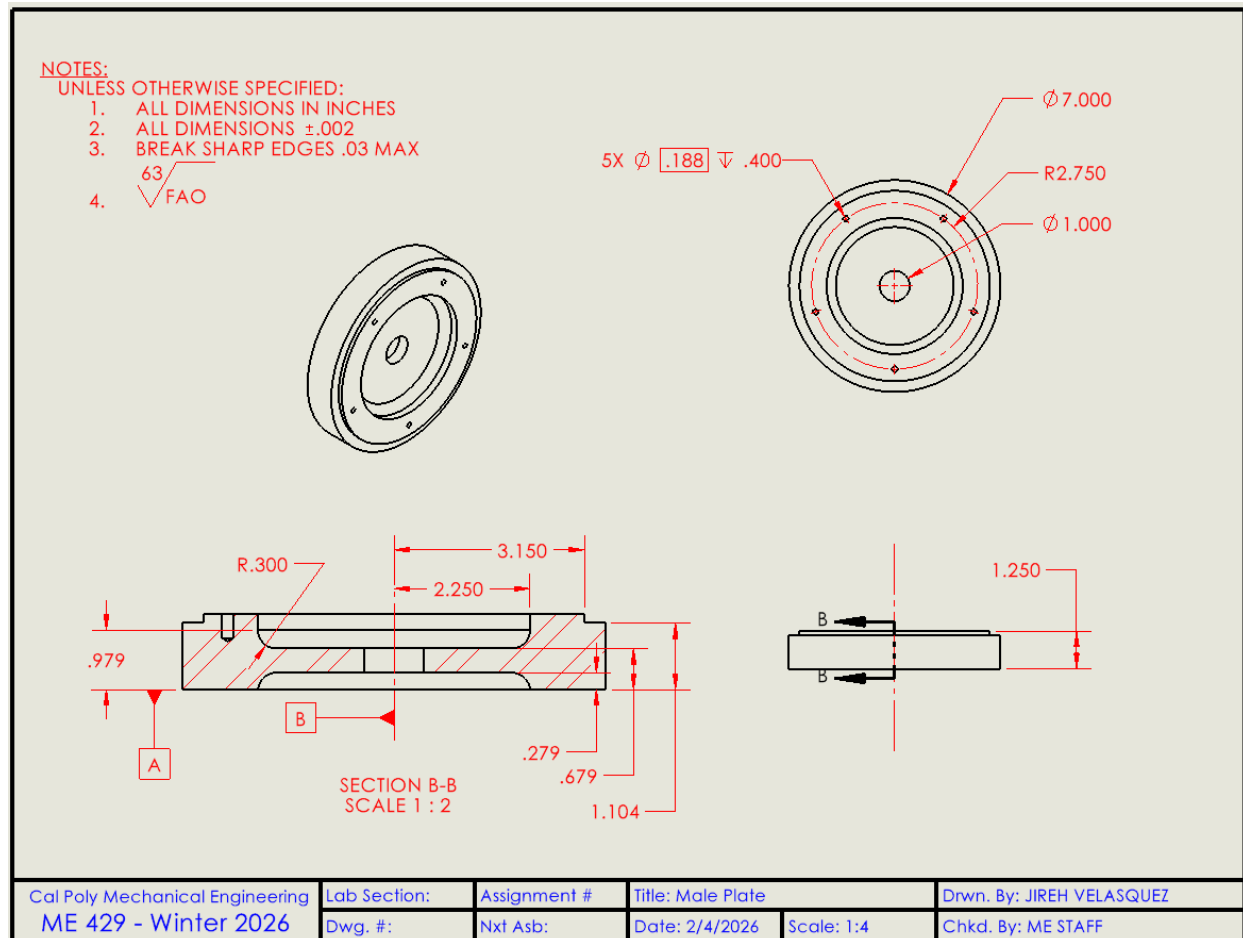


Figure E-1: Detailed drawing of the Male Plate for assemblies 1000 and 2000

Male Plate – Spec Sheet

Item	Specification
Title	Male Plate
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Jireh Velasquez
Checked by	Aubrey Perez
Date	2/4/2026
Scale	1:4 (sheet shows)
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Critical-to-fit (Outer lip / interface)

Item	Specification
Outer diameter (OD)	Ø7.000
Outer lip radius callout	R2.750
Step location (from centerline)	3.150 (to outer step), 2.250 (to inner step)
Lip/transition fillet	R0.300

Other key features

Item	Specification
Center bore/hole	Ø1.000
Hole pattern	5X Ø0.188, depth 0.400

Key heights (from section view)

Item	Specification
Section heights (as dimensioned)	0.279, 0.679, 0.979, 1.104

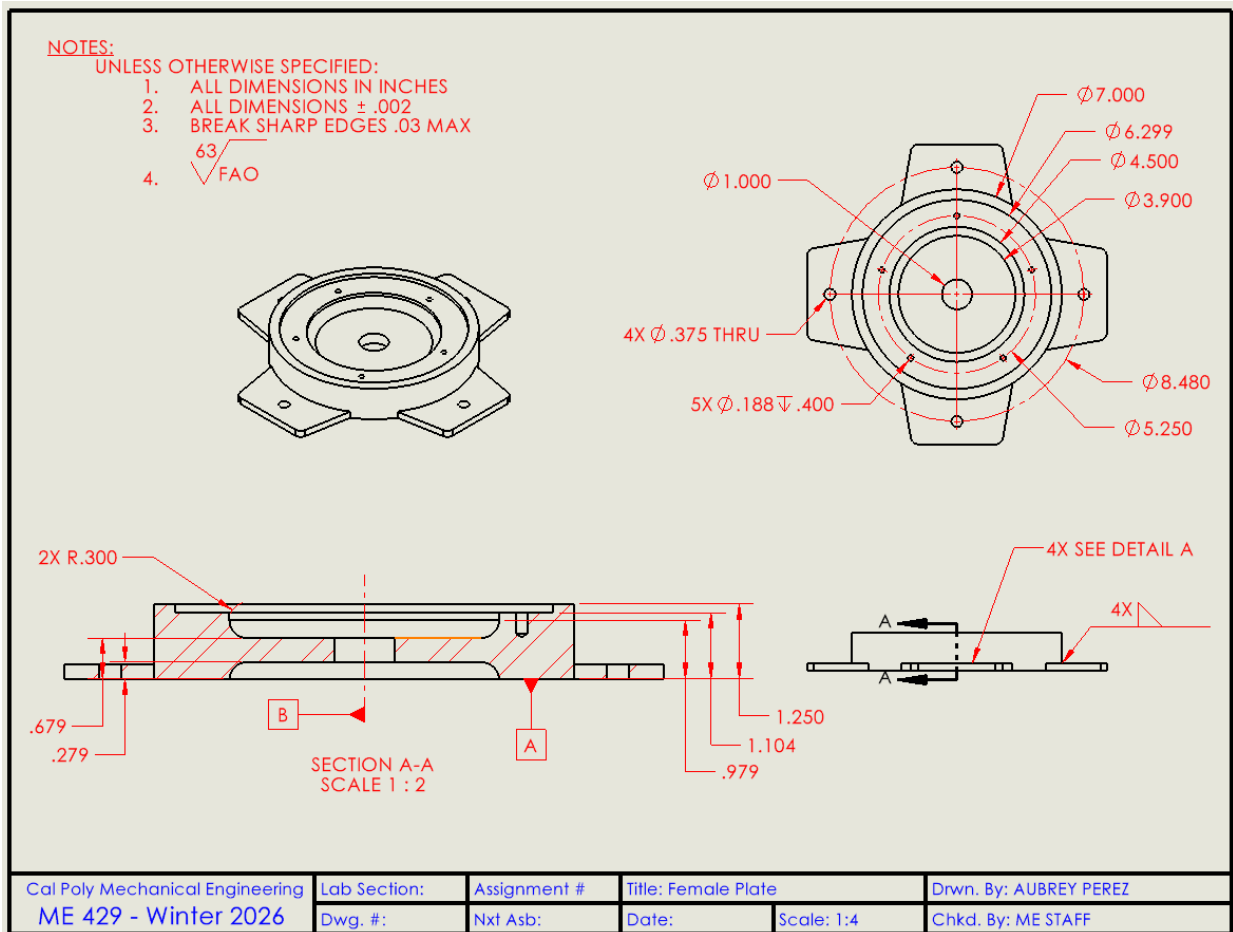


Figure E-2: Detailed drawing of Female Plate for assemblies 1000 and 2000

Female Plate – Spec Sheet

Item	Specification
Title	Female Plate
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Aubrey Perez
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:4
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	± 0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Critical-to-fit (Outer lip / interface)

Item	Specification
Outer diameter (OD)	$\varnothing 7.000$
Concentric lip/step diameters	$\varnothing 6.299$, $\varnothing 4.500$, $\varnothing 3.900$
Lip/transition fillets	2X R0.300

Other key features

Item	Specification
Center bore/hole	$\varnothing 1.000$
Thru holes	4X $\varnothing 0.375$ THRU
Hole pattern (blind)	5X $\varnothing 0.188$, depth 0.400
Tab outside diameter/extent	$\varnothing 8.480$
Bolt circle / pattern diameter	$\varnothing 5.250$

Key heights (from section view)

Item	Specification
Section heights (as dimensioned)	0.279, 0.679, 0.979, 1.104, 1.250

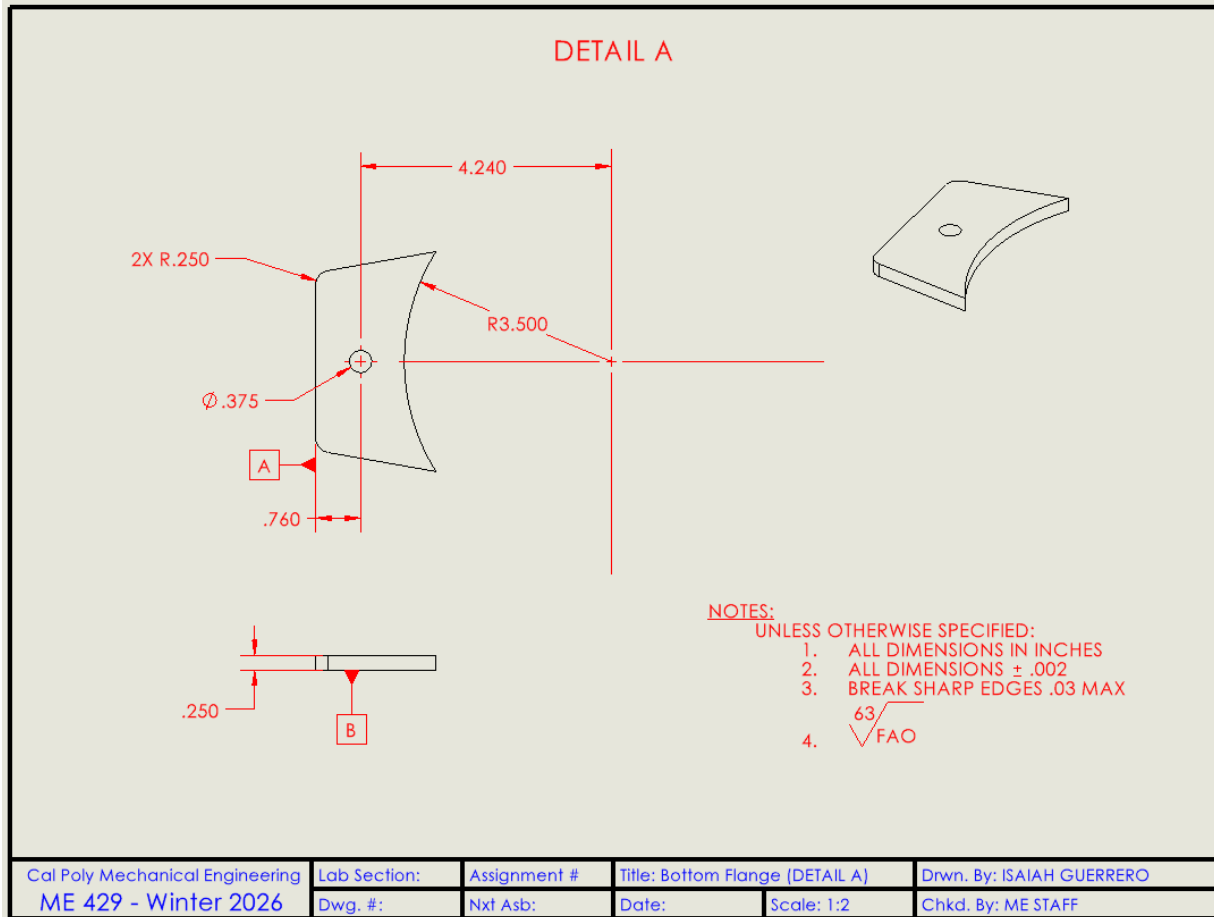


Figure E-3: Detailed drawing of the Bottom Flange attached to the Female Plate for assemblies 1000 and 2000

Bottom Flange (DETAIL A) – Spec Sheet

Item	Specification
Title	Bottom Flange (DETAIL A)
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Isaiah Guerrero
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:2
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	± 0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Critical-to-fit (Outer lip / interface)

Item	Specification
Interface note	Does not define cup lip, but affects assembled footprint and bolt-up

Other key features

Item	Specification
Overall length	4.240
Inner curve radius	R3.500
Corner/edge radii	2X R0.250
Hole	$\varnothing 0.375$
Hole location offset	0.760 (as shown)
Thickness	0.250

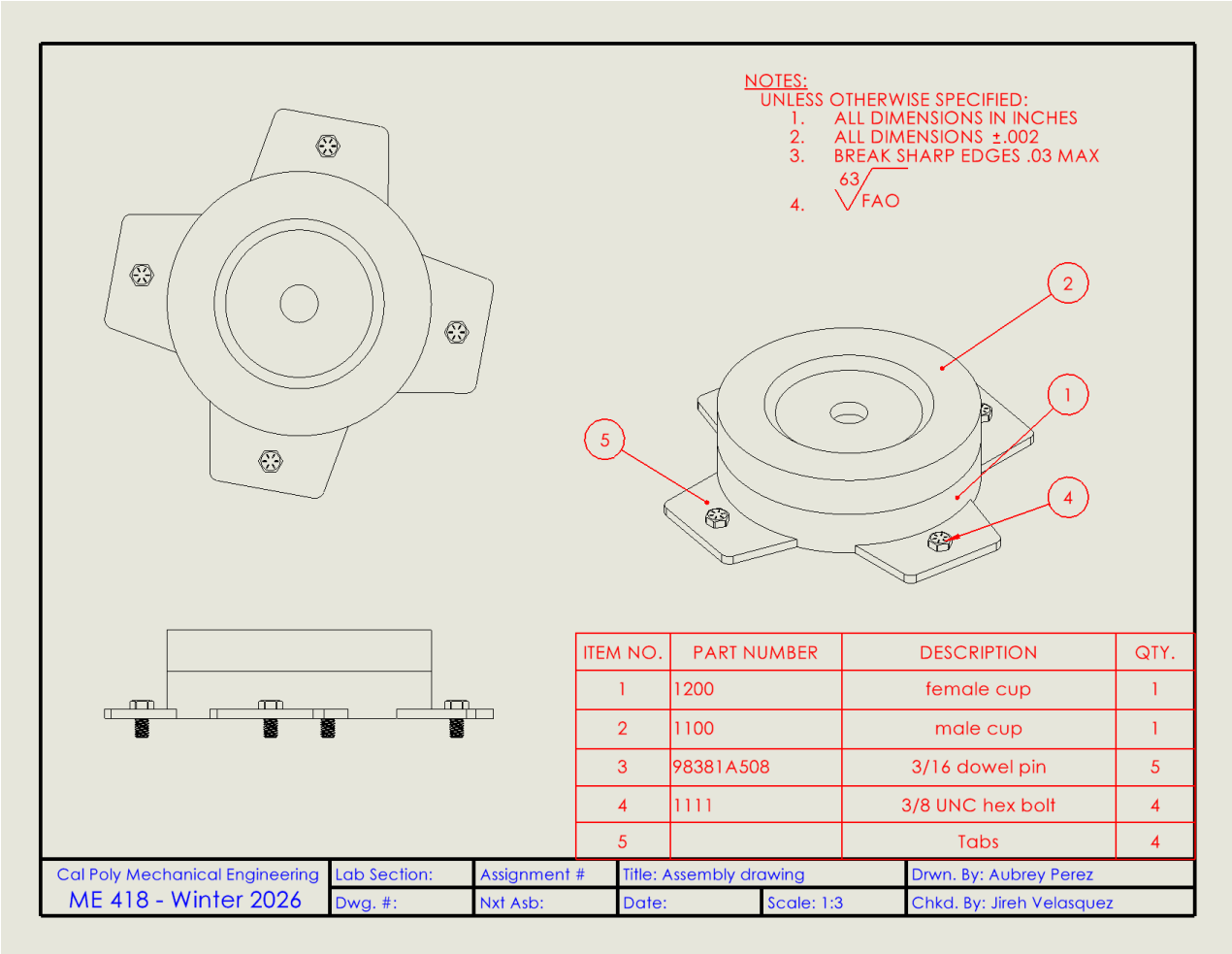


Figure E-4: Full assembly drawing for assemblies 1000 and 2000

Assembly Drawing – Spec Sheet

Item	Specification
Title	Assembly drawing
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Aubrey Perez
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:3
Course	ME 418 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	± 0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Bill of Materials (BOM)

Item No.	Part Number	Description	Qty
1	1200	female cup	1
2	1100	male cup	1
3	98381A508	3/16 dowel pin	5
4	1111	3/8 UNC hex bolt	4
5		Tabs	4

Critical-to-fit (Outer lip / interface)

Item	Specification
Critical-to-fit note	Verify cup outer lip OD/step diameters and step heights per part drawings before assembly

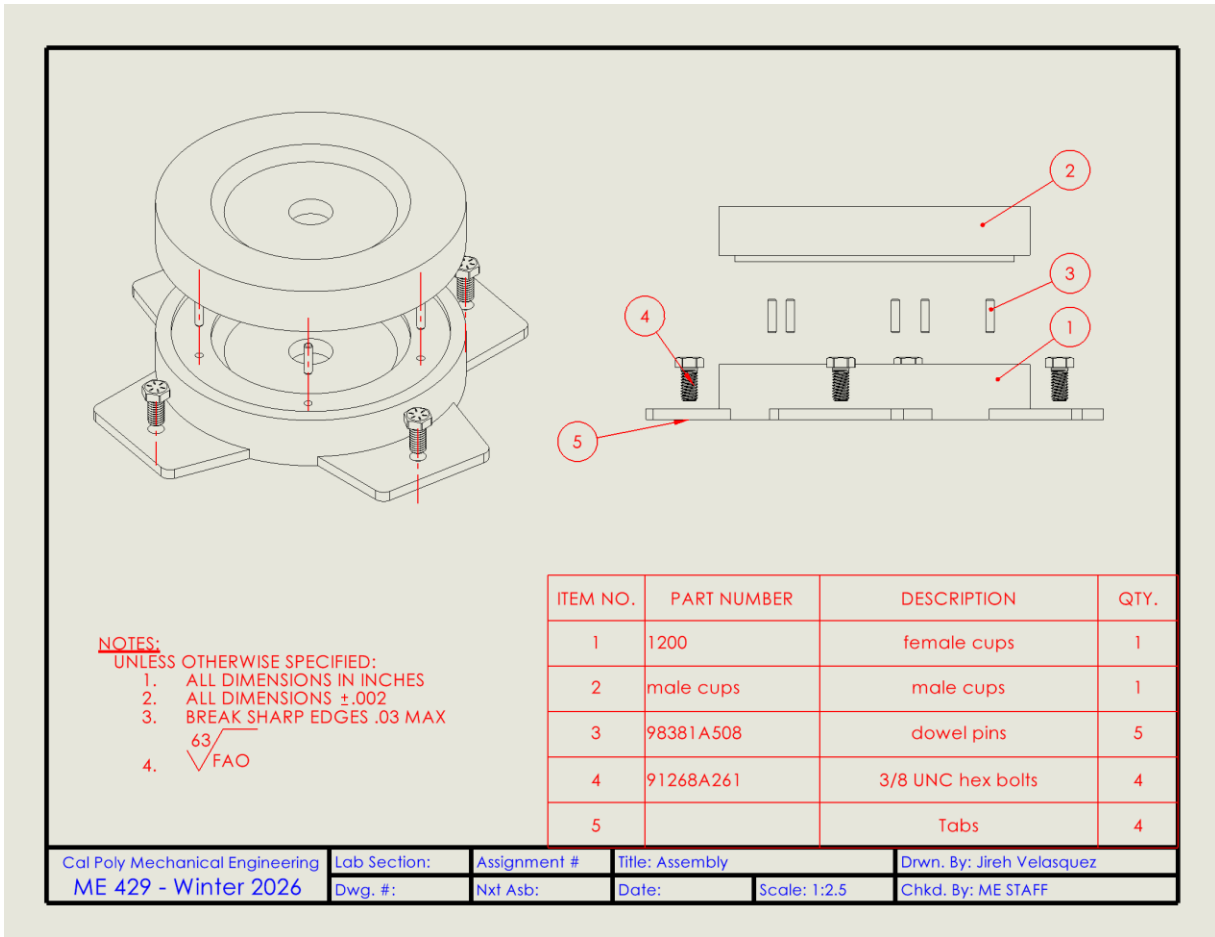


Figure E-5: Exploded assembly view for assemblies 1000 and 2000

Assembly (Exploded) – Spec Sheet

Item	Specification
Title	Assembly
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Jireh Velasquez (sheet shown)
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:2.5
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	± 0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Bill of Materials (BOM)

Item No.	Part Number	Description	Qty
1	1200	female cups	1
2		male cups	1
3	98381A508	dowel pins	5
4	91268A261	3/8 UNC hex bolts	4
5		Tabs	4

Critical-to-fit (Outer lip / interface)

Item	Specification
Critical-to-fit note	Same as assembly: fit depends on cup lip OD/step diameters and step heights from detail part prints

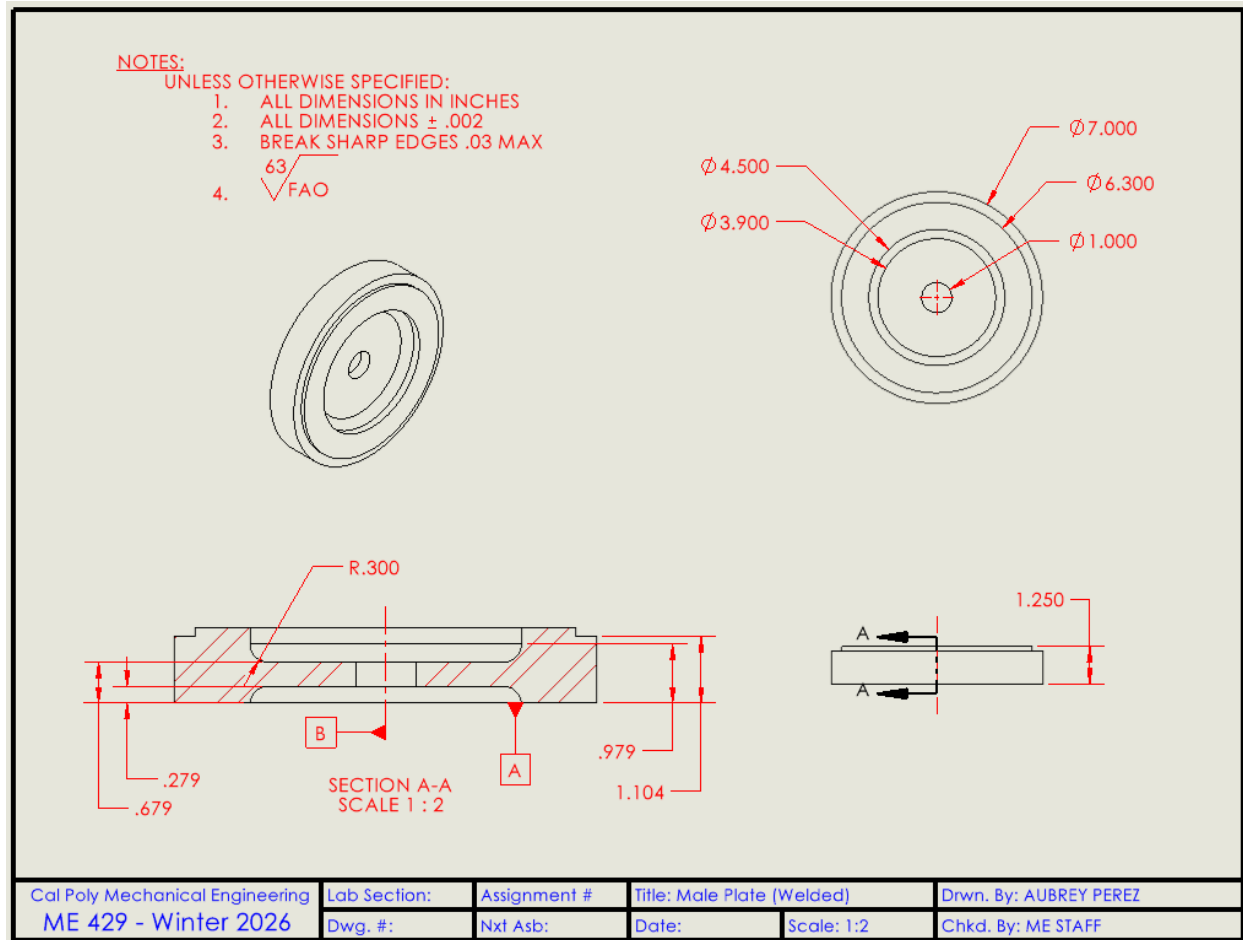


Figure E-6: Detailed drawing of Male Plate for assembly 3000

Male Plate (Welded) – Spec Sheet

Item	Specification
Title	Male Plate (Welded)
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Aubrey Perez
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:2
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Critical-to-fit (Outer lip / interface)

Item	Specification
Outer diameter (OD)	Ø7.000
Concentric lip/step diameters	Ø6.300, Ø4.500, Ø3.900
Lip fillet	R0.300

Other key features

Item	Specification
Center bore/hole	Ø1.000

Key heights (from section view)

Item	Specification
Section heights (as dimensioned)	0.279, 0.679, 0.979, 1.104

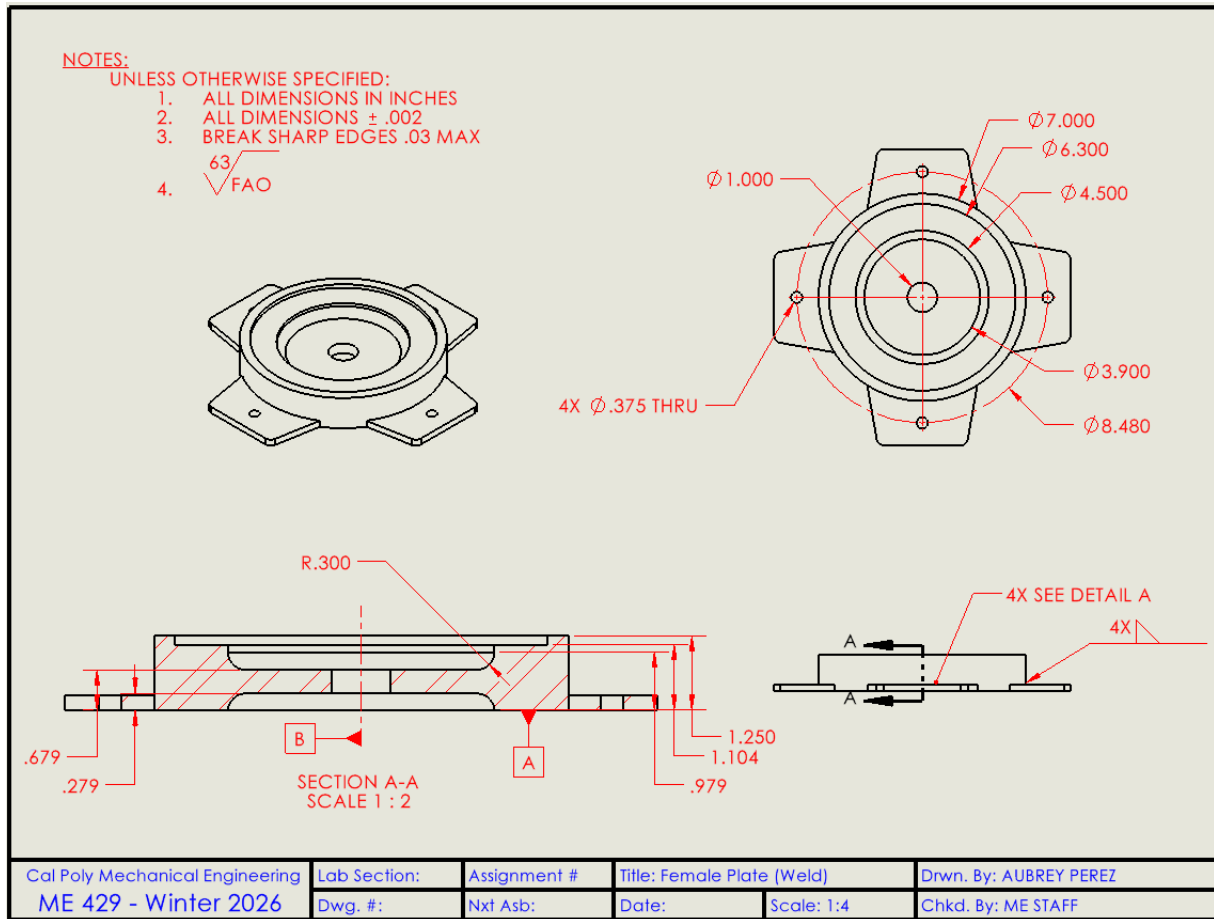


Figure E-7: Detailed drawing of Female Plate for assembly 3000

Female Plate (Weld) – Spec Sheet

Item	Specification
Title	Female Plate (Weld)
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Aubrey Perez
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:4
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	±0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Critical-to-fit (Outer lip / interface)

Item	Specification
Outer diameter (OD)	Ø7.000
Concentric lip/step diameters	Ø6.300, Ø4.500, Ø3.900
Tab outside diameter/extent	Ø8.480
Lip fillet	R0.300

Other key features

Item	Specification
Center bore/hole	Ø1.000
Thru holes	4X Ø0.375 THRU

Key heights (from section view)

Item	Specification
Section heights (as dimensioned)	0.279, 0.679, 0.979, 1.104, 1.250

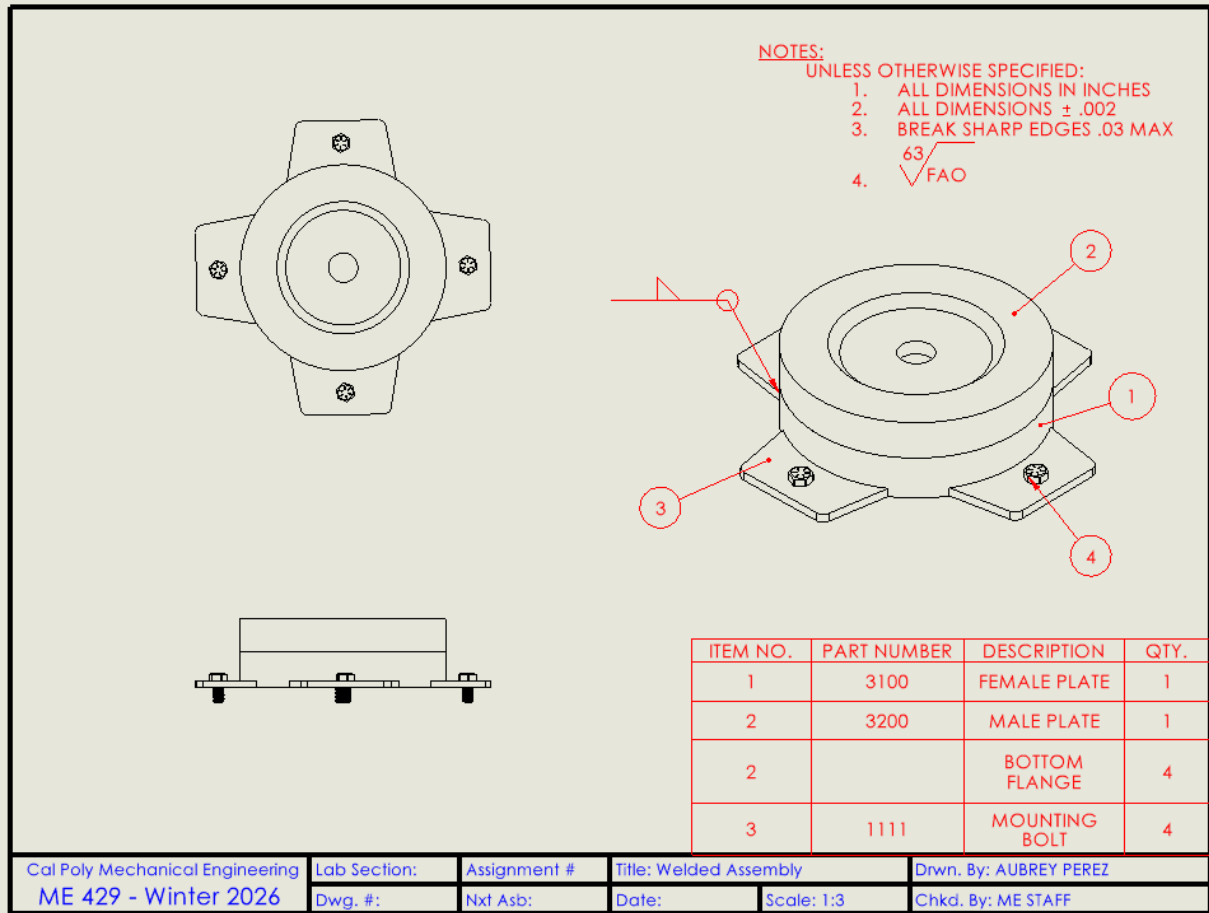


Figure E-8: Detailed welded assembly drawing for assembly 3000

Welded Assembly – Spec Sheet

Item	Specification
Title	Welded Assembly
Part number	See iBOM / BOM (if applicable)
Material	1018 Cold Rolled Steel
Drawn by	Aubrey Perez
Checked by	Jireh Velasquez
Date	2/12/2026
Scale	1:3
Course	ME 429 - Winter 2026

General notes

Item	Specification
Units	Inches
Default tolerance	± 0.002 (unless otherwise specified)
Edge break	Break sharp edges 0.03 max
Surface finish	63 (FAO)

Bill of Materials (BOM)

Item No.	Part Number	Description	Qty
1	3100	FEMALE PLATE	1
2	3200	MALE PLATE	1
3		BOTTOM FLANGE	4
4	1111	MOUNTING BOLT	4

Critical-to-fit (Outer lip / interface)

Item	Specification
Critical-to-fit note	Maintain plate lip OD/step diameters and lip heights from welded part drawings before weld-up/assembly

Compressor Rotor Joints and Interference Fit Study

Final Design Review

By:

Aubrey Perez

Isabela Jimenez

Isaiah Guerrero

Jireh Velasquez

Senior Project

Sponsor: *Solar Turbines*

California Polytechnic

University, San Luis Obispo

College of Engineering



6/4/2026

Project Overview Solar Turbines needed a comprehensive study tool on the effects radial interference fits have on turbine compressor rotor stiffness. Some of the weldments of the compressor have press-fit dowel pins and pilot joints to connect one weldment section to another. These fits needed to be assessed through a finite element modal analysis as well as practically tested through modal impacts on manufactured prototypes to investigate how the fits affected the stiffness of the turbine’s connected weldments. The experimental data was used to validate the finite element analysis (FEA) model tool that was to be presented to Solar Turbines.	Design Description The specifications for this design include the accurate modeling of radial interference effects on rotor stiffness from the pilot fit and dowel pins, along with qualitative verification from testing. The modeling and testing of only radial interferences was decided between Solar Turbines and the team due to complications that would arise with manufacturing an axial interference in the physical testing model. Other simplifications made to the design for the models, such as dowel pin size and lack of contours, allowed for feasible manufacturing, while also meeting the requirements on requested radial interference qualitative data.																								
Concept Description  Compressor plate final assembly and test setup The final prototype was three configurations of two rotor plates joined together. Two configurations included mating via radial interference fits, and one assembly that had the plates welded together to form a fully rigid joint. Four tabs were welded onto each configuration to allow the assemblies to be bolted to a shake table for fixed-free testing. These tabs were later utilized for free-free testing.	Design Verification <table><tr><th>Second Natural Frequency Comparison</th><th><i>Config 1</i></th><th><i>Config 2</i></th><th><i>Welded Config</i></th></tr><tr><td>ANSYS</td><td>4.397 kHz</td><td>4.826 kHz</td><td>5.892 kHz</td></tr><tr><td>Experimental</td><td>4.384 kHz</td><td>4.608 kHz</td><td>4.88 kHz</td></tr></table> <table><tr><th>Third Natural Frequency Comparison</th><th><i>Config 1</i></th><th><i>Config 2</i></th><th><i>Welded Config</i></th></tr><tr><td>ANSYS</td><td>5.655 kHz</td><td>5.765 kHz</td><td>7.592 kHz</td></tr><tr><td>Experimental</td><td>5.392 kHz</td><td>5.935 kHz</td><td>8.033 kHz</td></tr></table> Free-free results comparing experimental testing and ANSYS model The project was successful in validating a scaled-down ANSYS model with experimental results. Various testing revealed that an ANSYS model with a weak spring and a free-free experimental testing setup yielded the most similar results. The project also confirmed that while increasing radial interference in the joint slightly increased joint stiffness, a longer pilot length had the greatest effect in increasing joint stiffness.	Second Natural Frequency Comparison	<i>Config 1</i>	<i>Config 2</i>	<i>Welded Config</i>	ANSYS	4.397 kHz	4.826 kHz	5.892 kHz	Experimental	4.384 kHz	4.608 kHz	4.88 kHz	Third Natural Frequency Comparison	<i>Config 1</i>	<i>Config 2</i>	<i>Welded Config</i>	ANSYS	5.655 kHz	5.765 kHz	7.592 kHz	Experimental	5.392 kHz	5.935 kHz	8.033 kHz
Second Natural Frequency Comparison	<i>Config 1</i>	<i>Config 2</i>	<i>Welded Config</i>																						
ANSYS	4.397 kHz	4.826 kHz	5.892 kHz																						
Experimental	4.384 kHz	4.608 kHz	4.88 kHz																						
Third Natural Frequency Comparison	<i>Config 1</i>	<i>Config 2</i>	<i>Welded Config</i>																						
ANSYS	5.655 kHz	5.765 kHz	7.592 kHz																						
Experimental	5.392 kHz	5.935 kHz	8.033 kHz																						

Overview

The scope of this project involved creating an analysis model to study and predict how interference fits in a rotor assembly affected the stiffness of Solar Turbines' turbine compressor. Half of the forward weldment, half of the aft weldment, and dowel pins were modeled in ANSYS along with the necessary connections between the components. A contact stress and modal analysis were performed on various assemblies iterating the pilot interference fit between the rotor plates, the interference fit between the rotor plates and the dowel pins, and the pilot ledge length to determine the correlation between those variables and the stiffness of the assembly. To further refine and validate this ANSYS model, three different assemblies were manufactured: a welded configuration, a low-interference configuration, and a high-interference configuration. To compare the stiffness of the physical assemblies to the ANSYS model, modal impact testing was performed on each configuration to obtain the natural frequencies and compare them to the mode shapes of the ANSYS model.

Concept Description and Justification

Our senior project is a study on the effects that the radial interference between the dowel pins and the rotor plates, the interference of the pilot ledge between the plates, and the pilot ledge length have on the overall stiffness of the Solar Turbines rotor plate assembly. We chose to isolate and test the effects of these critical dimensions because we initially believed they would have a large impact on stiffness and help limit the scope of our study due to limited time and resources. To better understand the complex dynamics of this system, we developed an ANSYS model to perform contact stress and modal analysis. Subsequently, we used the ANSYS model to iterate the chosen critical dimensions to find critical relationships between features and assembly natural frequency that would help us find the optimal path to increase the system's stiffness. Once a trend was identified, we then manufactured two prototype configurations of different interference values and pilot ledge lengths to test and validate the ANSYS model as well as the identified trend between the critical dimension and stiffness. In addition, we also manufactured a fully welded configuration to use as a baseline reference for which we could compare both interference configurations to the stiffest configuration possible. This also aided further proof of a valid ANSYS model that could be used to predict assembly stiffness. This information was then presented to Solar Turbines to help guide their efforts in future turbine redesigns to increase system stiffness.

Some design decisions that were made regarding the physical production models include scaling down the models, using 1018 steel for the rotor plates, and adding a one-inch center hole. We decided to scale down their model from an original 20 inches to 6.9 inches in diameter due to our restricted manufacturing capabilities and work-holding size limits. 1018 steel was chosen rather than aluminum due to the increased natural frequency range of aluminum in comparison to steel. Due to aluminum having a much lower density than steel, it had a much higher natural frequency, which was proven through a modal analysis in ANSYS. It would have dramatically affected our capability to test the assembly's natural frequencies by exceeding the frequency range of the accelerometer and spectrum analyzer. An added benefit of using 1018 steel is that its material properties are comparable to Inconel, which is the material Solar Turbines uses for their plates. Machining Inconel was not within the manufacturing capability of the machines we had access to. Lastly, we decided to model and manufacture our prototypes with a one-inch center hole to replicate the center hole that Solar Turbines has in their rotors for the main shaft of their turbine.

Another design decision that was made for this study was the removal of axial interference. There were many reasons for the removal of this aspect from our study, the driving factor being the high difficulty in manufacturing a scaled down model that would have required an extremely high torque value. This would have also required specialized tools that we did not have access to and were not appropriately equipped to operate. The implications of this design decision were further explored in the next steps section of this report.

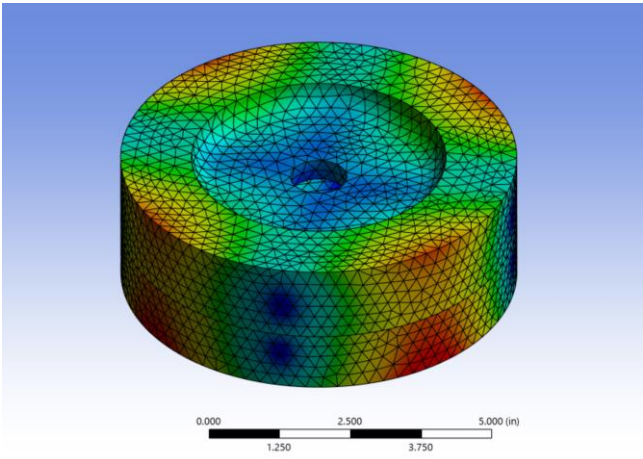


Figure 1. Final model

Outline of Schematic B8: Parameters				
	A	B	C	D
1	ID	Parameter Name	Value	Unit
2	Input Parameters			
3	Geometry (A1)			
4	P11	Dowel_radius	0.10938	in
5	P12	Hole_Radius	0.10908	in
6	P16	Male_Pilot_Length	0.151	in
7	P17	Female_Pilot_Length	0.151	in
8	P18	Female_Radial_Inteference	3.1538	in
9	P19	Male_Radial_Interference	3.1543	in
*	New input parameter	New name	New expression	
11	Output Parameters			
12	Static Structural (B1)			
13	P20	Mesh Nodes	1.1079E+05	
14	P21	Stress Probe Maximum Normal - Y Axis	927.98	psi
15	P22	Stress Probe 2 Maximum Normal - Y Axis	32852	psi
*	New output parameter		New expression	
17	Charts			
18	Parameter Parallel Chart 0			
19	Parameter Parallel Chart 1			
20	Parameter Parallel Chart 2			
21	Parameter Parallel Chart 3			
22	Parameter Parallel Chart 4			

Figure 2. Parameters of final model

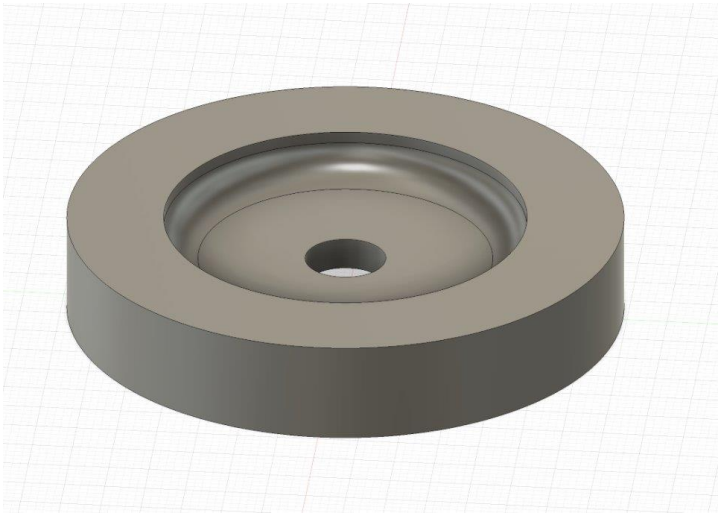


Figure 3. Verification Prototype

Implementation

Initial ANSYS Analysis Objectives

Solar Turbines tasked the team with studying the relationship between pilot fit interference, dowel pin interference, and compressor rotor assembly stiffness. The primary challenge became determining a method to confidently collect data and draw conclusions regarding how dimensional changes to the rotor joint affected overall assembly stiffness.

ANSYS was selected as the primary analysis tool used to guide design decisions throughout the project. The ultimate goal was to manufacture a configuration that was theoretically less stiff, manufacture a configuration that was theoretically more stiff, and manufacture a configuration that was theoretically the stiffest by making two to have interference fits and one to have welds, respectively. These were then used to experimentally validate whether changes in interference fits could measurably alter assembly stiffness. Blindly manufacturing multiple configurations without first developing a simulation model could have resulted in designs that were difficult to manufacture, assemble, or test, ultimately wasting both time and resources.

Additionally, a validated ANSYS model would provide long-term value to Solar Turbines. If simulation results could be shown to correlate with experimental testing, future design iterations could be evaluated through simulation prior to manufacturing. This would allow design decisions to be made more efficiently while reducing the number of physical prototypes required for validation.

The primary objective of the initial ANSYS model was to identify trends between pilot interference, dowel pin interference, pilot ledge contact length, and natural frequency. Natural frequency was selected as the primary metric because it is directly proportional to stiffness through the fundamental vibration relationship between stiffness, mass, and frequency. Since the mass of the assemblies remained relatively constant, increases in natural frequency were interpreted as increases in assembly stiffness. By identifying these trends, the team hoped to establish meaningful design directions that could later be validated through physical testing.

Initial Fixed-Free ANSYS Workflow

The ANSYS workflow consisted of creating the assembly geometry, performing a Static Structural analysis to determine contact stresses generated by the interference fits, and then transferring the resulting pre-stress information into a Modal analysis. This workflow was specifically designed to replicate the planned physical testing procedure as closely as possible. The physical assemblies would ultimately be manufactured, press fit together using pilot and dowel interferences, then be subjected to modal testing. The simulation process mirrored these same steps by first generating contact stresses and then evaluating the resulting natural frequencies.

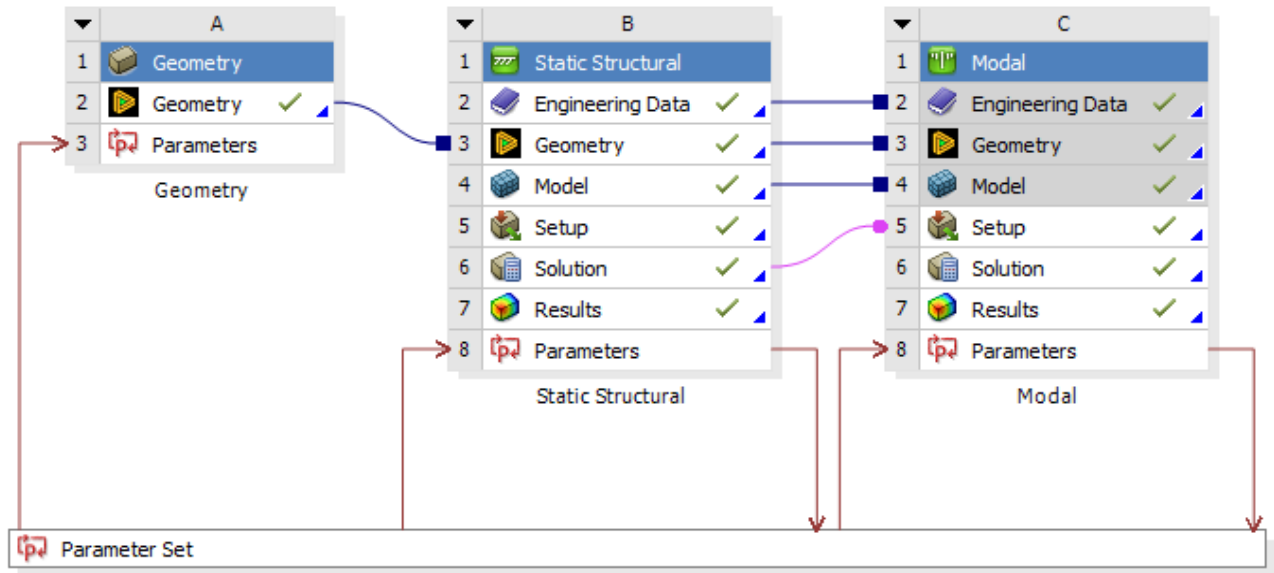


Figure 4. ANSYS workflow with parametric table setup.

A major advantage with using ANSYS was the parameterization capability available within Workbench. The parameter manager creates an Excel-style table where geometric dimensions can be varied while output parameters are automatically recorded. Because the project required a large number of design iterations, this feature significantly reduced analysis time. Instead of manually modifying geometry, rerunning analyses, and recording results for every configuration, the geometry and output frequencies could be parameterized and solved automatically. This allowed relationships between interference fit dimensions and natural frequency to be quickly visualized and evaluated.

Table 1. Iterated Dimensions

Iterations		
Pilot Length	Hole Radius	Female Radius
0.146	0.0937	3.1494
0.15	0.09365	3.1493
0.175	0.0936	3.1492
0.2	0.09355	3.1491
0.225	0.0935	3.149
0.25	0.09345	3.1489
0.275	0.0934	3.1488
0.3	0.09335	3.1487
0.325	0.0933	3.1486
0.35	0.09325	3.1485
0.375	[-]	3.1484
0.4	[-]	3.1483
[-]	[-]	3.1482
[-]	[-]	3.1481
[-]	[-]	3.148
[-]	[-]	3.1479

[-]	[-]	3.1478
[-]	[-]	3.14775

The Static Structural analysis was responsible for calculating the contact stresses generated by the interference fits. Contact and target surfaces were assigned to the pilot fit interfaces and dowel pin interfaces, allowing ANSYS to solve for the resulting stresses created by the interference conditions. Frictional contacts were applied between all mating steel surfaces using a coefficient of friction of 0.2, a commonly accepted value for steel-on-steel contact¹.

ANSYS Mechanical utilized the Augmented Lagrange contact formulation to solve all frictional contact interfaces within the model. The Augmented Lagrange method combines aspects of the penalty method and Lagrange multiplier methods to enforce contact constraints while maintaining robust numerical convergence². Contact forces are initially calculated using a penalty stiffness approach, and iterative corrections are then applied to minimize penetration between contacting surfaces. This contact formulation is the default method used by ANSYS for many structural contact problems and is well-suited for interference fit analyses because it accurately captures contact pressure distributions while maintaining reasonable solution times. The resulting contact pressures generated by the pilot and dowel interferences were subsequently transferred into the prestressed modal analysis to evaluate their effect on assembly stiffness and natural frequency.

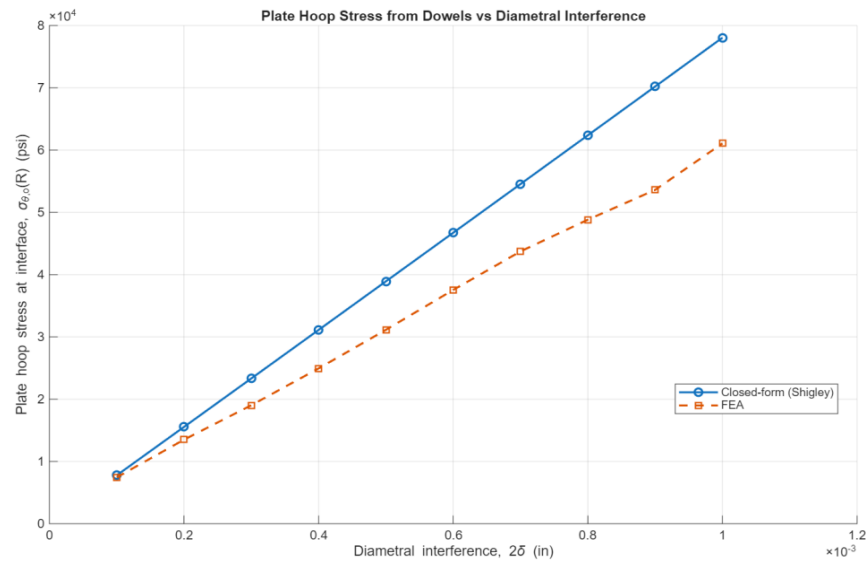


Figure 5. Hoop Stress Generated from Interference Fit Dowel Pins

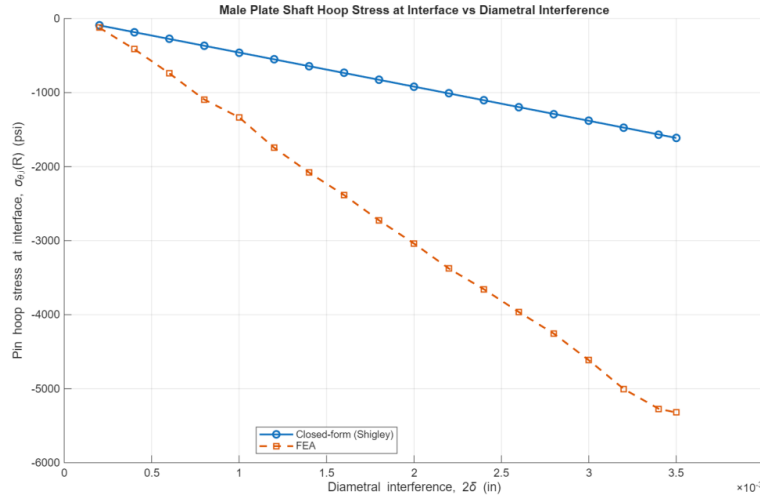


Figure 6. Hoop Stress Generated from Interference Fit Pilot Feature

To verify that the contact stresses produced by ANSYS were reasonable, the resulting hoop stresses were compared to analytical interference fit calculations obtained from Shigley's Mechanical Engineering Design³. Agreement between the analytical calculations and ANSYS results provided confidence that the interference fits were being modeled correctly. Equation 1 shows this estimation with σ being stress, p being interface pressure, d_o being a hub diameter, and d being a shaft diameter.

$$\sigma = p \frac{d_o^2 + d^2}{d_o^2 - d^2} \quad (1)$$

The resulting stress state was then transferred into a prestressed Modal analysis. Pre-stressing the model was important because the project hypothesis assumed that increasing interference fit contact pressures would increase assembly stiffness and therefore increase natural frequency. By incorporating the static stress state into the modal solution, the effects of interference fit stiffness could be captured within the natural frequency results.

The initial fixed-free modal analysis focused primarily on the first mode shape and its associated natural frequency. The objective was to determine whether changes in interference fit dimensions produced measurable changes in assembly stiffness. The resulting trends were then used to guide future design decisions and manufacturing efforts.

Fixed-Free Boundary Condition Development

The original testing plan involved utilizing a vibration shake table to perform modal testing on the rotor plate assemblies. To support this testing method, tabs were designed and welded onto the perimeter of one of the plates for each assembly. These tabs would allow the assembly to be mounted while leaving the majority of the structure unconstrained and accessible for testing.

Based on this planned testing methodology, a fixed-free boundary condition was selected for the initial ANSYS model. The underside of the mounting tabs was assigned a fixed support in ANSYS while the remainder of the assembly remained unconstrained. At the time, this approach was believed to

adequately represent the intended experimental setup.

As testing plans evolved, however, limitations associated with the shake table became apparent. The combined weight of the steel assemblies required the shake table to operate near its maximum load capacity, restricting the achievable excitation frequencies to approximately 3000 Hz. Because the anticipated natural frequencies of the assemblies exceeded this range, the shake table was ultimately abandoned in favor of modal impact testing using an instrumented hammer. Although the testing methodology changed, the initial fixed-free ANSYS model remained valuable for identifying design trends and establishing a baseline understanding of the assembly behavior.

Initial Model Geometry and Contact Definitions

The initial ANSYS model included the complete rotor plate assembly geometry, including both plates, dowel pins, and mounting tabs. No geometry was excluded from the analysis because the team wanted to represent all features that were present during physical testing. Including all components allowed the simulation to more closely represent the planned experimental configuration.

While later refinements would improve the overall accuracy of the simulation, the initial fixed-free model represented the team's first complete attempt at estimating stiffness trends associated with the selected design variables.

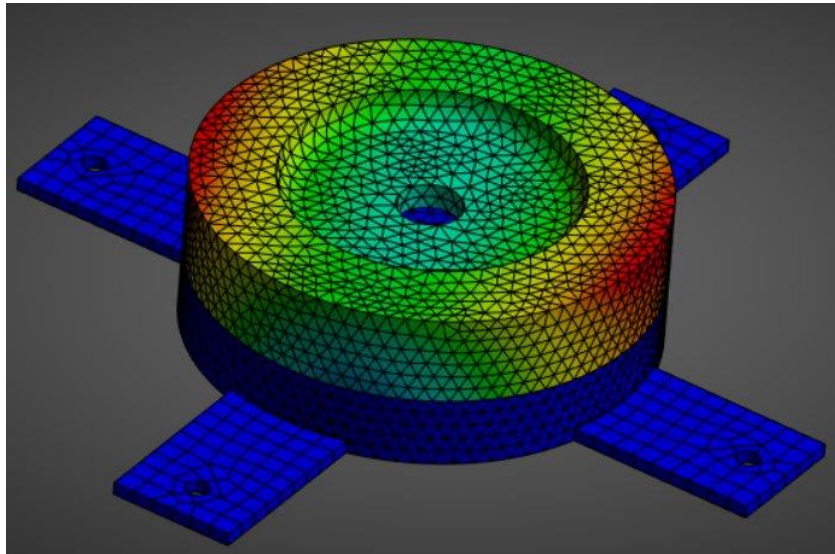


Figure 7. Initial ANSYS model and mesh of compressor plate assembly.

Parametric Study Setup

Three primary design variables were selected for investigation: dowel pin interference, pilot interference, and pilot ledge contact length. These variables were selected directly based on guidance from Solar Turbines and represented the primary dimensions of interest within the project scope.

The dowel pin interference study was performed by varying the hole diameters rather than the dowel pin diameters themselves. This approach provided greater flexibility during manufacturing

because hole dimensions could be more easily changed compared to the dowel pin sizes. Pilot interference was studied by varying the diameter of the female pilot feature. Pilot ledge contact length was included because increasing engagement length increases the contact area between mating components and was therefore expected to better resist relative motion.

Axial pilot interference was intentionally excluded from the study. Creating realistic axial preload conditions would have required specialized fasteners, hydraulic tooling, and preload verification methods that were beyond the available resources and tooling capabilities of the project.

Table of Design Points				
	A	B	C	D
1	Name ▾	P11 - Dowel_radius ▾	P12 - Hole_Radius ▾	P16 - Male_Pilot_Length ▾
2	Units	in ▾	in ▾	in ▾
3	DP 0 (Current)	0.09375	0.0937	0.146
4	DP 1	0.09375	0.09365	0.146
5	DP 2	0.09375	0.0936	0.146
6	DP 3	0.09375	0.09355	0.146
7	DP 4	0.09375	0.0935	0.146
8	DP 5	0.09375	0.09345	0.146
9	DP 6	0.09375	0.0934	0.146
10	DP 7	0.09375	0.09335	0.146
11	DP 8	0.09375	0.0933	0.146
12	DP 9	0.09375	0.09325	0.146

Figure 8. Image of ANSYS Parametric Feature that Allows Quick Iterations

Initial Fixed-Free Results

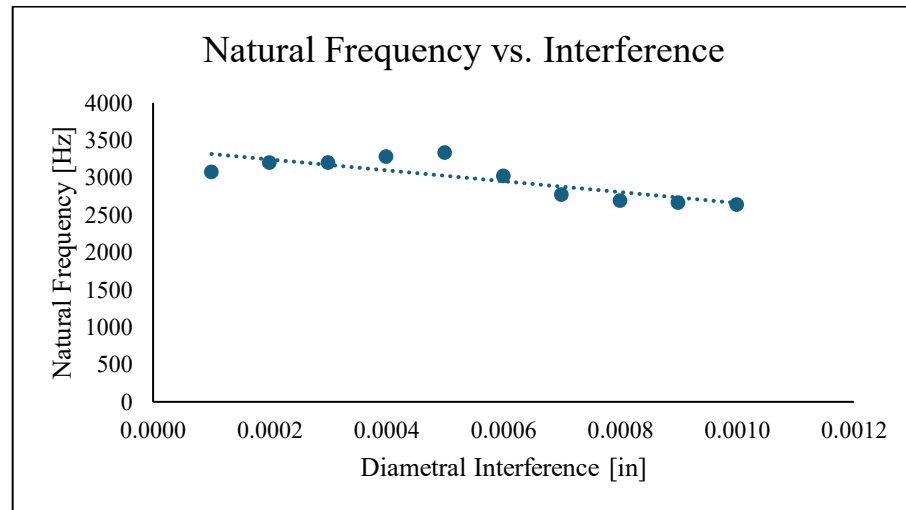


Figure 9: Dowel Pin Interference Results

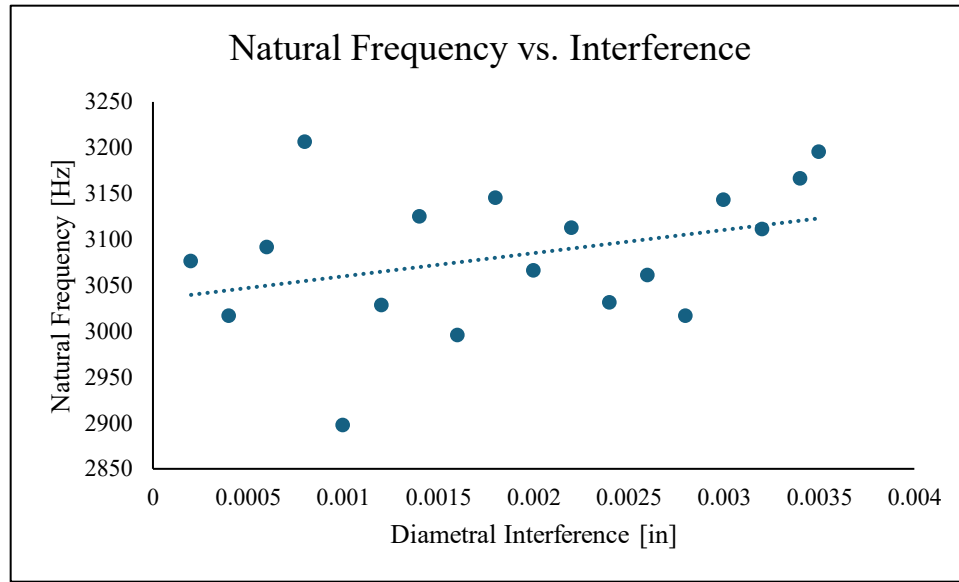


Figure 10: Pilot Interference Results

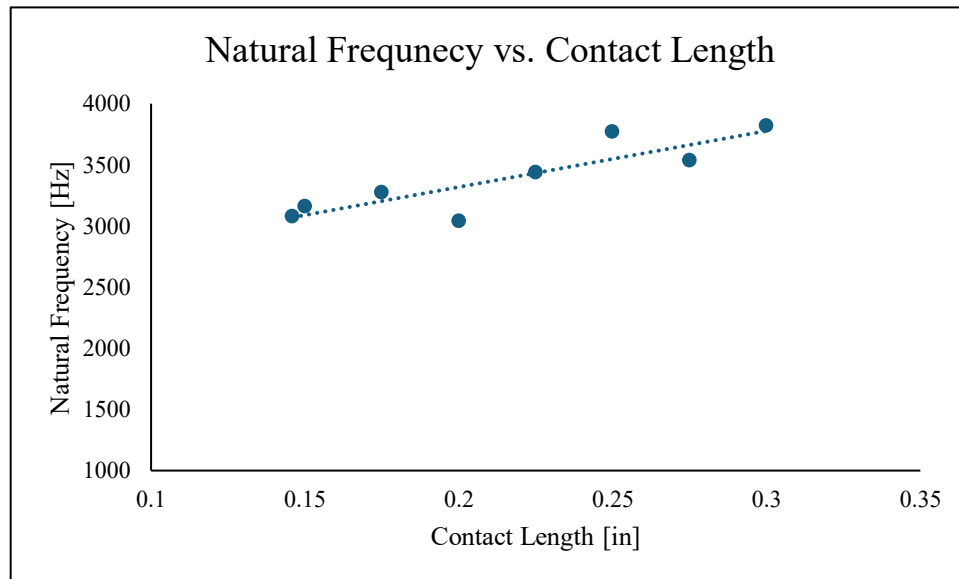


Figure 11: Pilot Ledge Length Results

The initial fixed-free analysis revealed varying levels of influence among the three design variables. Increasing dowel pin interference produced only a weak increase in natural frequency. This result was initially surprising because increased interference was expected to increase assembly stiffness. One possible explanation is that the dowel pins represented a relatively small portion of the overall assembly, causing the resulting stresses to remain localized near the pin locations without significantly affecting global assembly stiffness.

Increasing pilot interference produced a slightly stronger trend than the dowel pin study, although the relationship remained relatively weak. One possible explanation is that the pilot interference values remained within a standard locational interference fit range and may not have generated sufficient contact pressure to significantly alter assembly stiffness. Additional investigation

using larger interference values may provide further insight into this relationship.

Pilot ledge contact length demonstrated the strongest relationship with natural frequency. As the engagement length increased, the natural frequency consistently increased as well. This result supported the original hypothesis that increasing the available contact area between mating components would more effectively restrict relative motion and increase assembly stiffness.

Initial Design Conditions

The results obtained from the fixed-free ANSYS model were used to guide the selection of the physical test configurations. Configuration 1 was intentionally designed with lower interference values, while Configuration 2 incorporated larger pilot and dowel interferences in an effort to increase assembly stiffness. Pilot ledge length was treated as a more experimental variable and was increased based on the trends observed during the parametric study.

Several practical manufacturing constraints influenced the final design selections. Excessive interference values raised concerns regarding assembly feasibility, available tooling capacity, and the positional tolerances achievable during machining. To minimize these risks, interference values were generally maintained within LN1-LN2 style interference fit ranges while still providing measurable differences between configurations.

Although the fixed-free model proved valuable for identifying design trends and guiding prototype selection, it relied on several assumptions that would later be revisited following experimental testing. Nevertheless, the initial analysis successfully provided the information necessary to proceed into manufacturing and establish the first round of physical validation testing.

Manufacturing

Each rotor plate in each configuration was machined out of 7", 1018 round steel stock on a TM-1 CNC mill. Custom aluminum soft jaws were first manufactured to be able to securely grip the stock. Fusion 360 CAM software was utilized to create the toolpaths for each operation on the plates. Due to the complexity of the project and precision needed, we decided to first manufacture a prototype male plate before continuing to manufacture the final prototype models. This allowed us to determine the tolerance capability of the CNC mills, prove the CAM process, and reduce the manufacturing time of one plate from seven hours to four hours.

Two CNC operations were needed to manufacture each plate, and both utilized the soft jaws. The first operation faced off the top of the stock, reduced the diameter to 6.9 inches, drilled out the center hole, and machined a center pocket. The second operation required the stock to be flipped over and recentered. It then reduced the plate thickness to 1.25 inches, drilled out the dowel pins holes, and machined the pilot ledge and pockets. The plates in the welded configuration did not have the second operation performed on the stock. Due to the welded configuration not requiring dowel pin holes or a pilot ledge, making the top and bottom of the plates identical.

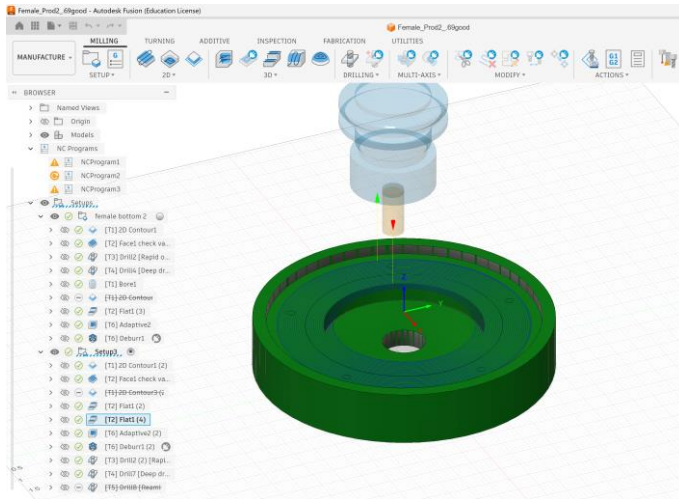


Figure 12. CAM program for rotor plates



Figure 13. CNC milling rotor plates

After each plate was machined, a CMM was used to confirm that the critical dimensions were within tolerance; these critical dimensions included the dowel pin hole diameter and depth, the pilot ledge length, and the pilot diameter. After inspecting one of the plates, we found that the CNC mill oversized the dowel pin holes, resulting in clearance rather than interference. We decided to purchase larger dowel pins rather than re-machine the plate and to update the CAM to undersize the holes. The dowel pin holes were enlarged to the correct diameter manually using a drill press and reamers and repeatedly checking the diameter.

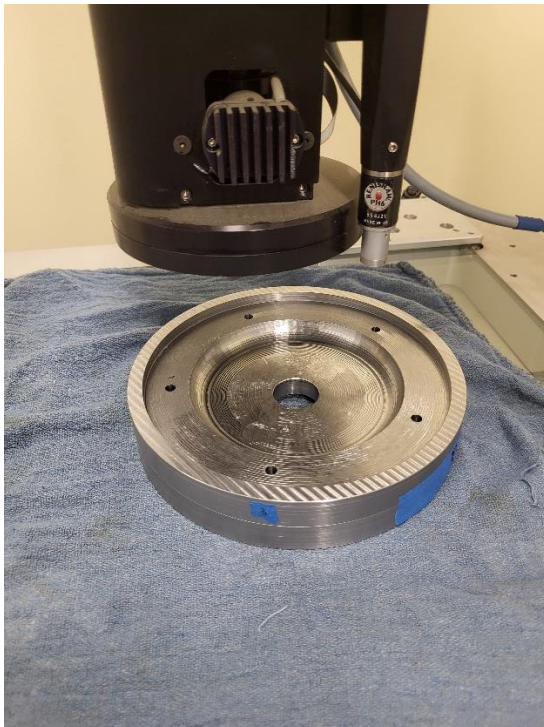
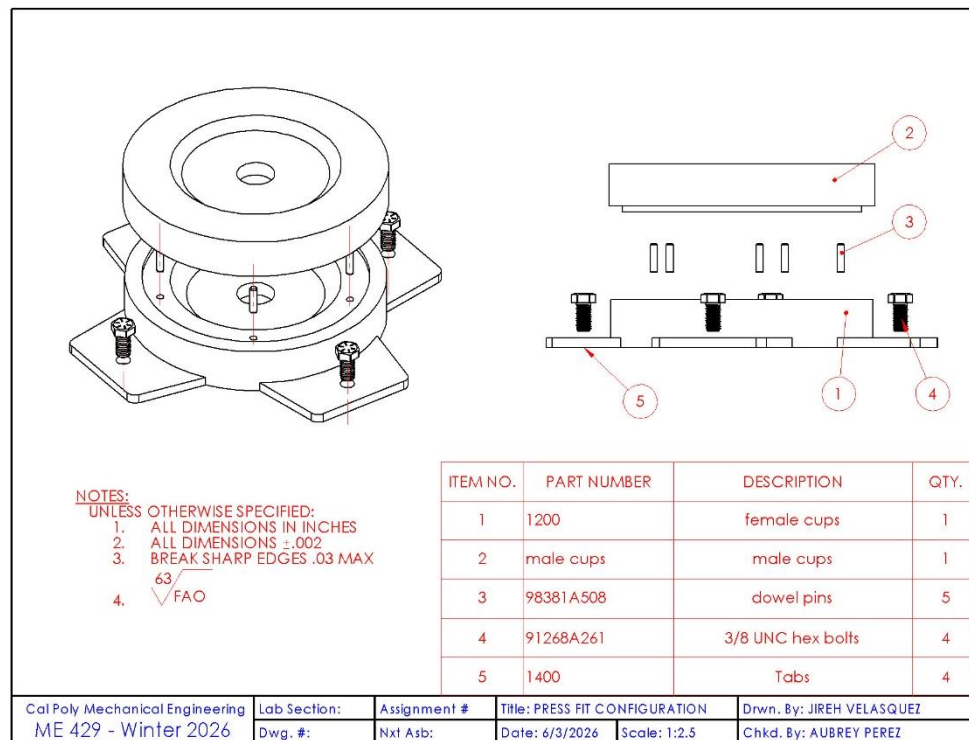


Figure 14. CMM of female plate**Figure 15.** Reaming and upsizing holes

To assemble the welded configuration, the two identical plates were welded together. The assembly process for the interference configurations was more involved. To be able to press the male and female plates of each assembly together, a temperature gradient was created. First, the male plate was placed in an oven provided by the school's materials engineering department for 75 minutes at 200°C to thermally expand the dowel pin holes while the dowel pins were placed in a freezer. Then an arbor press was used to press the dowel pins halfway into the male plate holes. The female plate was then placed in the oven for the same temperature and duration while the male plate with the pressed dowel pins was placed in the freezer. The plates were then pressed together with a hydraulic press, using a scrap aluminum plate to distribute the load more evenly. A drawing of this assembly can be seen below in Figure 16.

**Figure 16.** Configuration 1 and 2 assembly drawing.

Tabs were used to bolt the assemblies to the shake table. These tabs were water jet from 0.25" alloy steel sheets. Each assembly had four tabs. To ensure that the tabs were positioned correctly to match the shake table's hole pattern, a welding jig was 3D-printed and was intended to be used to weld the tabs to the female plate. The tabs were unintentionally welded to the male plate instead, but ANSYS was adjusted to accommodate this change. After each weld, the tabs were visually inspected to ensure that they were flat at the bottom of the plate and that minimal warping occurred. More detailed images of the manufacturing process can be found in the Appendix A figures.



Figure 17. Welding tabs onto the plate assemblies.

Design Verification: Prototype Testing and ANSYS Model Refinement

The initial testing of the plate assemblies began by setting up the testing procedure as seen in Figure 18 below. This procedure consisted of bolting each assembly to the shake table and recording modal frequencies with a uniaxial accelerometer placed with wax in the axial direction. As mentioned before, initially the shake table was going to be used to excite the assemblies, however it was discovered later in the ANSYS analysis that the first natural frequencies of the assemblies were too high for the shake table to excite. Therefore, the testing procedure was limited to using only a modal impact hammer. The data from the accelerometer was processed with a spectrum analyzer using 3-point averages from the modal impact hammer. In this testing phase, only roving hammer tests were conducted.

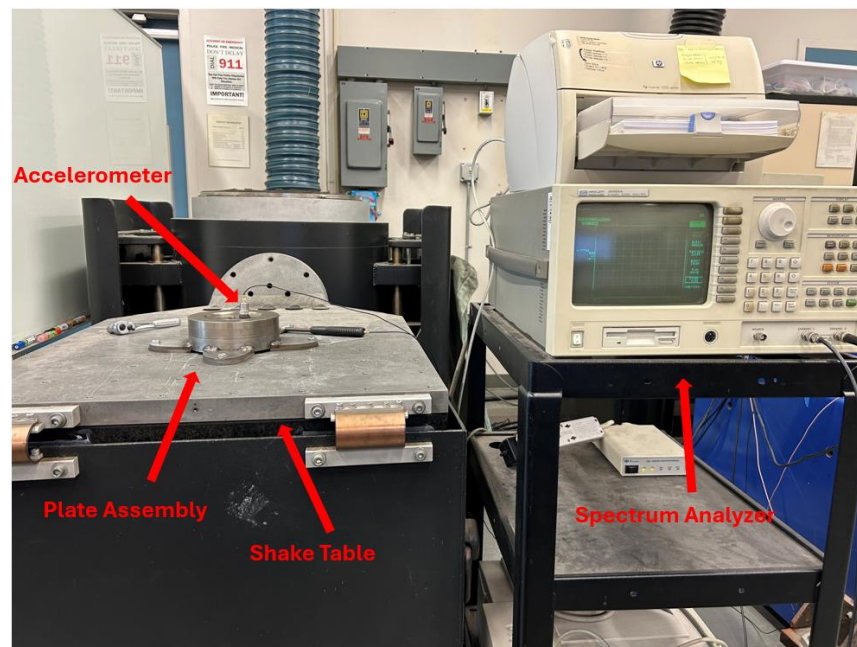


Figure 18. Testing setup for fixed-free testing.

The results of the fixed-free testing are shown below in Table 2. What was found from these results was that the ANSYS model appeared to be underestimating the natural frequency of the interference fit plate assemblies. Furthermore, there were other, much lower, natural frequency peaks being picked up from the spectrum analyzer, which was theorized to be from random noise or the tabs of the plate assembly.

Table 2. Testing results of the fixed-free condition.

<u>Natural Frequency Comparison</u>	Configuration 1	Configuration 2	Welded Configuration
ANSYS	3.198 kHz	3.135 kHz	7.52 kHz
Experimental	4.72 kHz	4.912 kHz	7.06 kHz

In addition to the poor relationship between the ANSYS model and experimental tests, the directionality of the fixed boundary conditions in the ANSYS model were producing inconsistent values. As mentioned earlier, the mounting tabs were welded onto the male plate rather than the female plate, making the assemblies not match the ANSYS model. Despite the ANSYS model being changed to reflect this manufacturing error, the comparisons still did not match the physical testing results and had entirely different results than the previous ANSYS model. Due to these inconsistencies and unexpected trends, the team reconsidered the testing procedure and decided to proceed with a free-free examination after consulting Solar Turbines.

To refine the ANSYS model to better represent the behavior of the physical configurations, it was updated to perform a free-free modal analysis. The fixed support on the assembly was removed and instead, weak springs were enabled for the model in order to converge to a solution. Frictional contacts were still placed between the faces of the plates in the interface with a coefficient of 0.2, while the solver still used the Augmented Lagrange method. With the tabs included in the assembly, the free-free modal analysis showed natural frequencies that were much lower than what the fixed-free modal impact testing showed. We concluded that the ANSYS model was finding the natural frequencies of the tabs and not the plate assembly, so the tabs were removed. Table 3 shows the resulting frequency values for four modes for each configuration.

Table 3. First three natural frequencies from ANSYS analysis.

Configuration	Mode 6 (hz)	Mode 7 (hz)	Mode 8 (hz)	Mode 9 (hz)
1	0.196267677	4275.014956	4397.17501	5655.72032
2	0.236836986	4268.660265	4826.45801	5765.51508
Welded	7.55E-04	5891.8	7395.6	7592

For a free-free modal analysis, the first six modes are zero, so we recorded the values for Mode 6 to ensure that they were near zero. We considered Modes 7-9 to be the first three natural frequencies for each assembly. For Mode 7, the first natural frequency, the values between Configuration 1 and Configuration 2 were almost the same. When looking at Mode 7 shape, the plates appeared to move in-phase with each other without separation. We concluded that in this mode shape, increasing the interference and the pilot ledge length would not have any effect on restricting the assembly movement, which explained why the natural frequency values were very close. For Modes 8-9, Configuration 2 had higher frequency values than Configuration 1. When looking at the two mode shapes, the plates appeared to be separating from each other; changing the interferences and the pilot ledge contact length would make a difference in restricting the plate movement in these two mode shapes, which would explain why Configuration 2 had higher frequency values. The ANSYS animation of these deflections can be visualized below in Figure 19. The welding configurations had much higher natural frequency values

than the interference configurations, meaning welding the plates together made the assembly much stiffer.

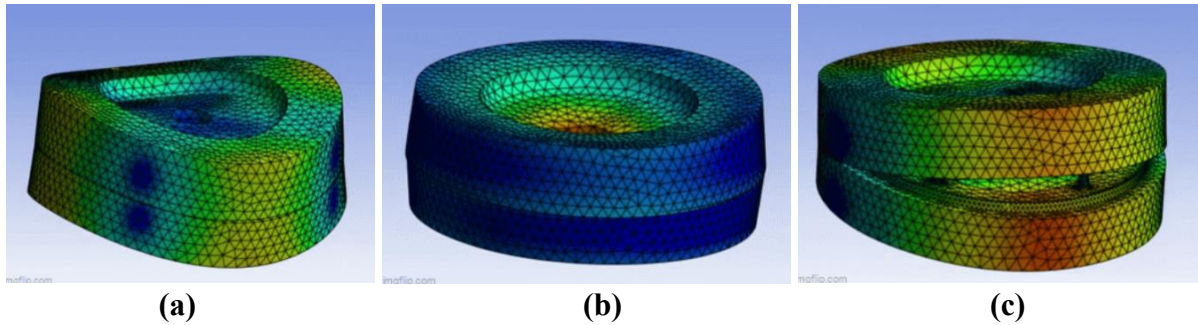


Figure 19. First mode shape (a), second mode shape (b), and third mode shape (c) suggested by refined ANSYS model.

The same iterations for the dowel pin interference, pilot interference, and pilot ledge length were used on the free-free modal analysis. The resulting trends were similar to the ones found in the fixed-free modal analysis. Appendix B shows the graphs for Modes 7-9 of how the natural frequency changes when increasing those dimensions. Increasing the dowel pin interference or the pilot interference did not have a large effect on increasing the stiffness of the assembly. Increasing the length of the pilot ledge did significantly increase the assembly stiffness.

After revising the ANSYS model into a free-free configuration, the team was able to hang the plate assemblies in the air to represent the weak spring boundary condition added to the model. For this, the team used a bungee cord and suspended the plate assemblies from the ceiling. Similar to the fixed-free testing, the modal frequencies of the plate assemblies were measured by an accelerometer stuck onto the plates with wax and fed into the spectrum analyzer. The collected data was still processed by 3-point averages from the modal impact hammer. Unlike the initial fixed-free testing, the team conducted both roving hammer and roving accelerometer tests for this setup. The testing setup can be seen below in Figure 20.

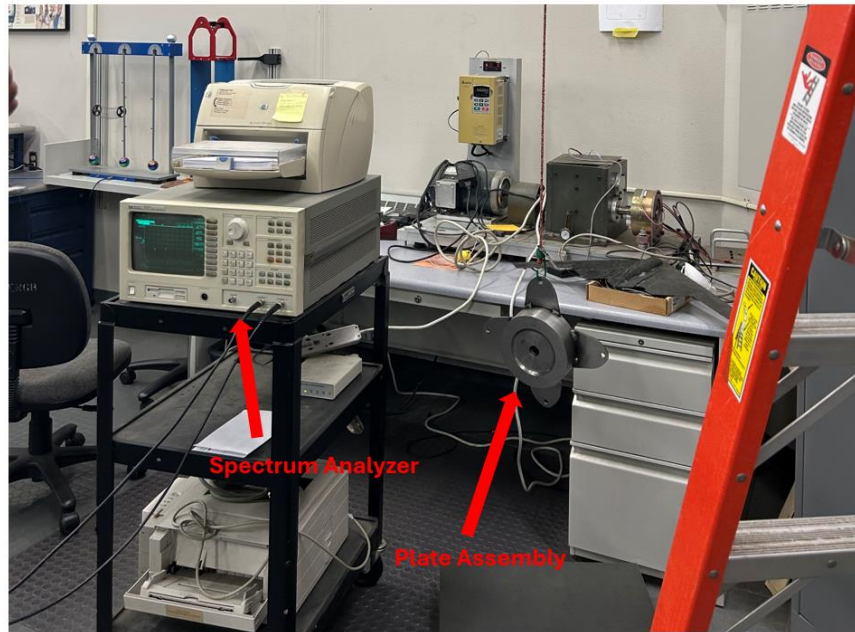


Figure 20. Testing setup for free-free testing.

When looking at the new results from the free-free testing, we found multiple improvements in the data. First, the data collected by the spectrum analyzer became significantly more similar to the ANSYS results, which can be seen below in Tables 4 and 5. Second, more mode shapes were observed compared to the fixed-free testing, which furthermore validated what was expected based on the ANSYS model. Finally, the data appeared cleaner, helping the team see these results.

Table 4. Testing results for the second natural frequency of the free-free condition.

<u>Second Natural Frequency Comparison</u>	Configuration 1	Configuration 2	Welded Configuration
ANSYS	4.397 kHz	4.826 kHz	5.892 kHz
Experimental	4.384 kHz	4.608 kHz	4.88 kHz

Table 5. Testing results for the second natural frequency of the free-free condition.

<u>Third Natural Frequency Comparison</u>	Configuration 1	Configuration 2	Welded Configuration
ANSYS	5.655 kHz	5.765 kHz	7.592 kHz
Experimental	5.392 kHz	5.935 kHz	8.033 kHz

These results proved to be a significant improvement from the initial fixed-free testing, and so was considered the final results used for the conclusion and discussion of this project.

Conclusion and Next Steps

A parametric ANSYS model was successfully developed along with three different rotor plate assemblies used for testing and validation. Our ANSYS model results closely matched the experimental trends from the modal impact testing, validating our simulation approach. Differences between the simulation and testing results were likely caused by imprecise manufacturing tolerances due to lack of precise tooling and testing inconsistencies. From our ANSYS model and testing results, we concluded that increasing the pilot ledge length was the biggest contributor to an increase in natural frequency and subsequently an increase in overall assembly stiffness. Whereas increases in dowel pin and pilot fit interference values provide minimal gains in stiffness.

After completing this study and identifying key correlations between lip length and stiffness, future work should include a more in-depth ANSYS analysis with sensitivity tests for each configuration and variable evaluated. This would help increase the credibility of the study while also identifying any unexpected results that may have gone unnoticed in the initial analysis. Additional future steps include developing a more accurate model with fewer design simplifications. This would involve accounting for axial interference, bolt preload, and multiple configurations assembled in series to investigate how larger assemblies interact. Finally, thermal loads should also be incorporated into both the ANSYS model and experimental testing to more closely replicate the actual operating conditions of Solar Turbines' compressor plates.

Appendix A: Manufacturing Pictures

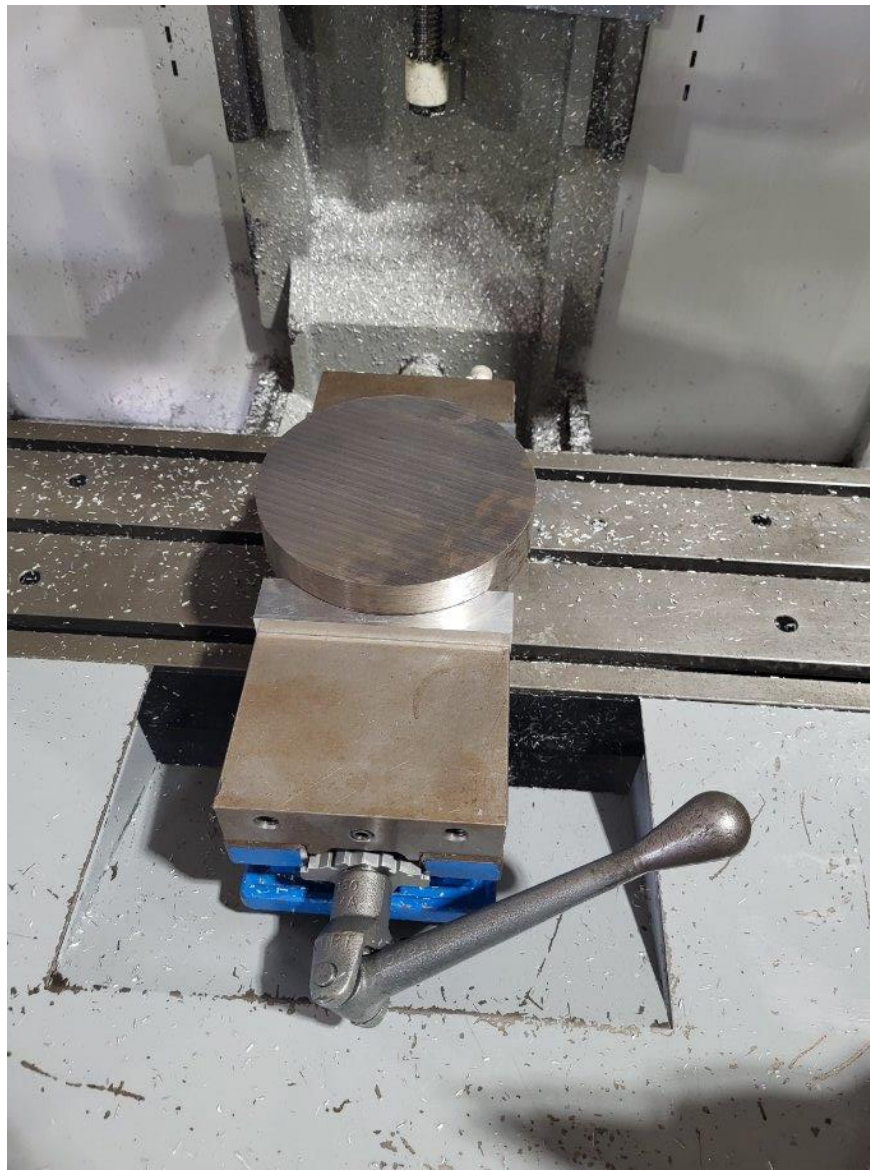
This appendix illustrates more in-depth pictures taken during the manufacturing process of the project. These pictures illustrate all the processes we underwent to create the production models which include CNC, drilling, and assembly.



Milling vise jaws for the two plate operations.



Using Haas CNC mill.



Raw stock plate being held with custom vise jaws.



Raw stock plate undergoing first operation.



Raw stock plate undergoing first operation.



Finished plate in CNC mill.



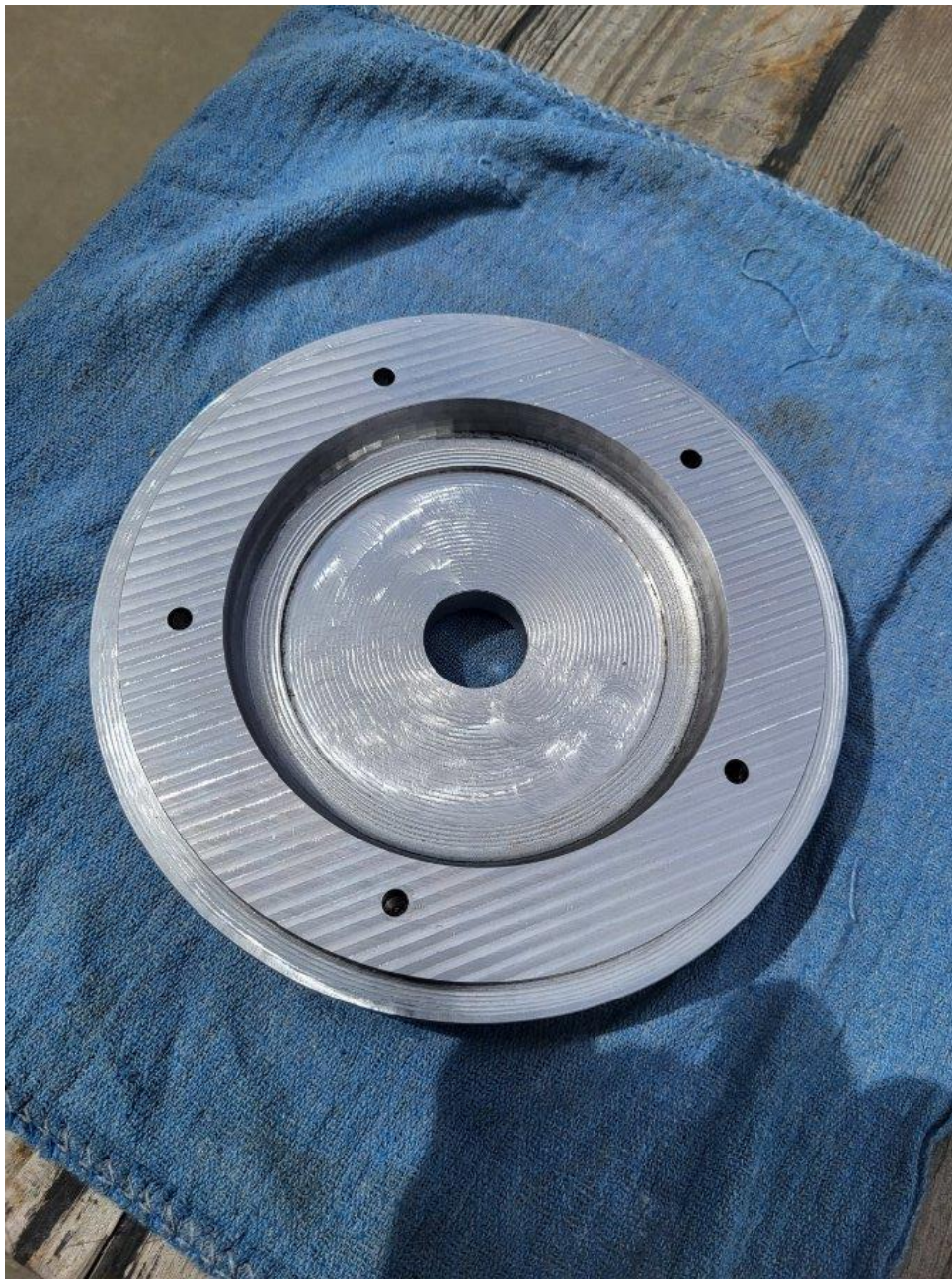
Close-up of machined rotor plate.



Drilling holes for dowel pins.



Close-up view of welded configuration plate.

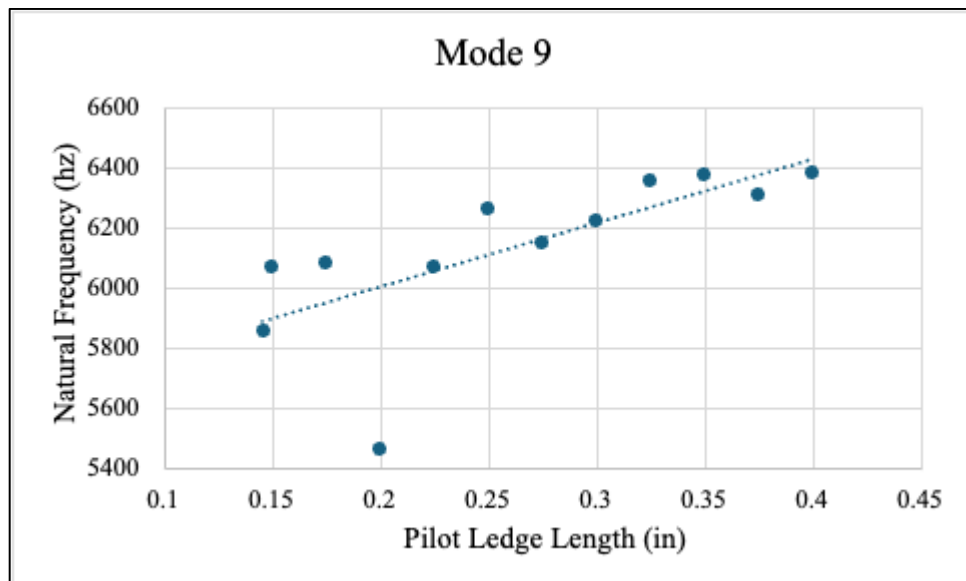
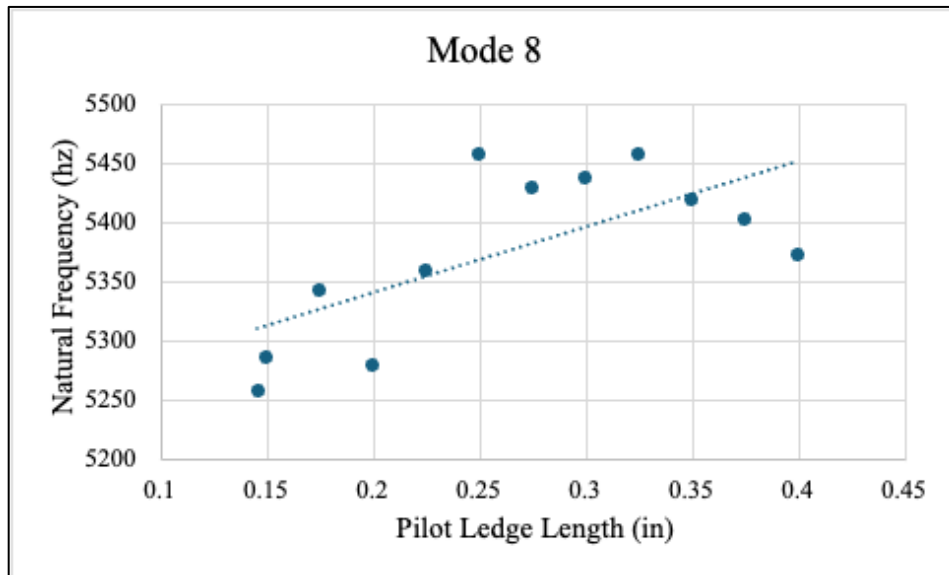
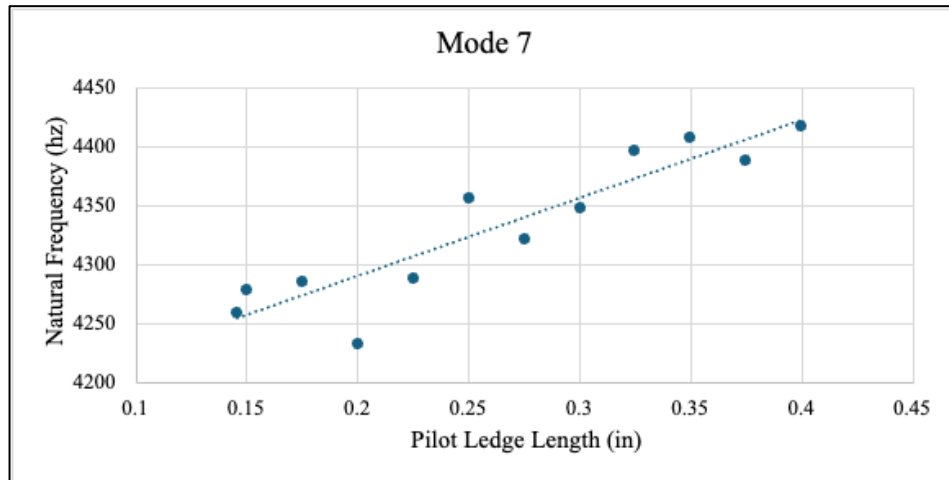


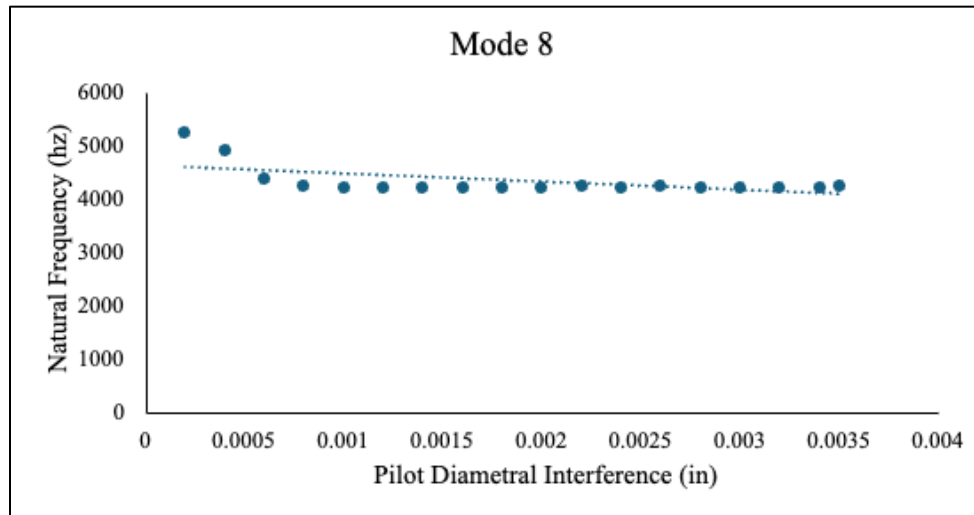
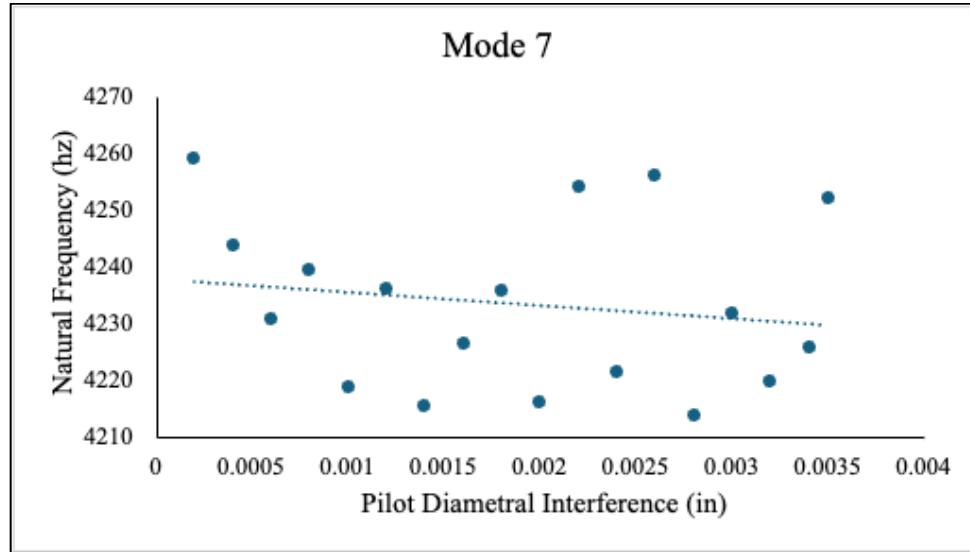
Close-up view of male plate with holes drilled out.

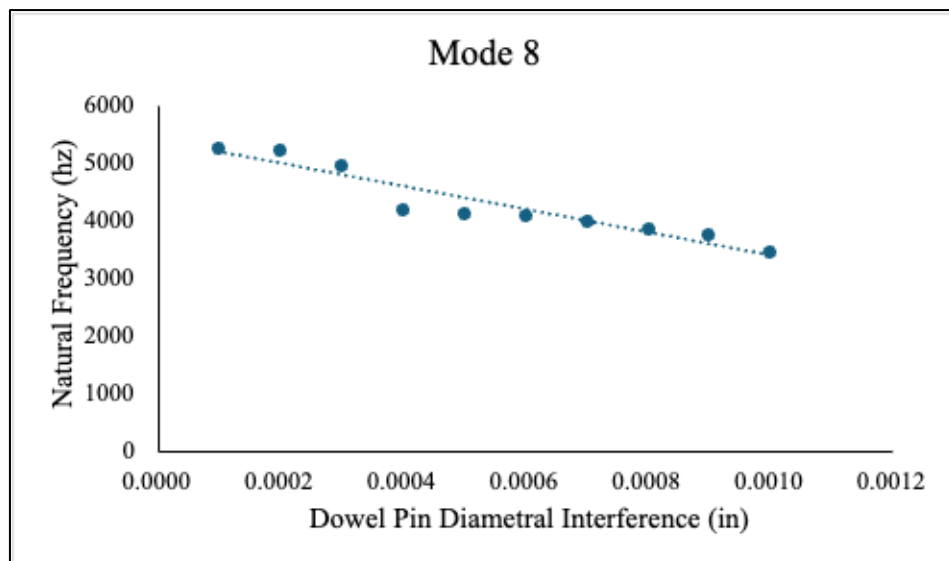
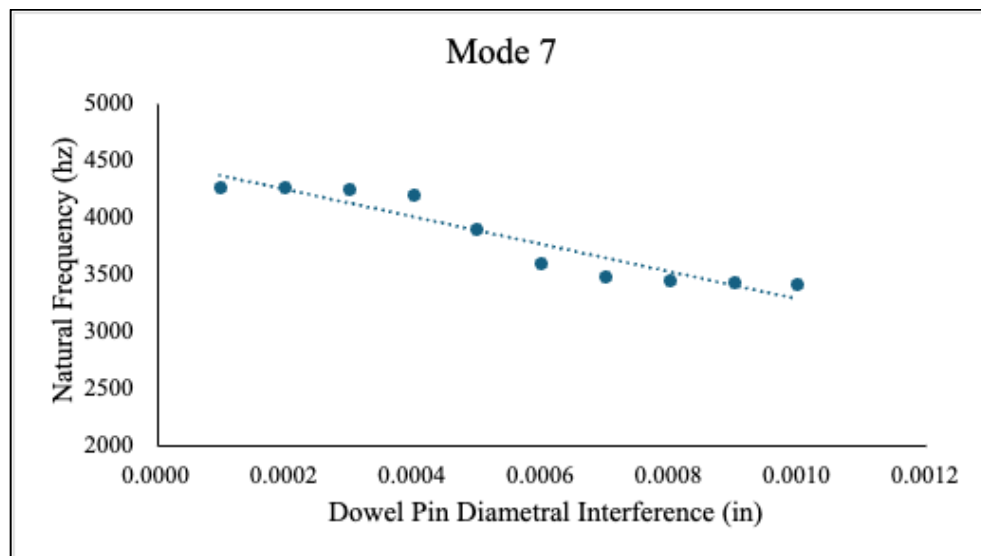
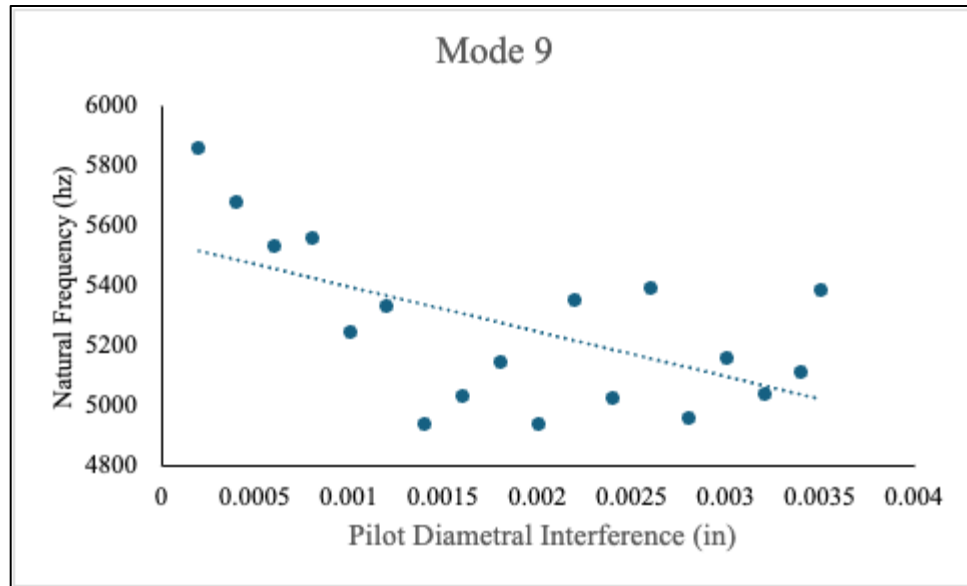


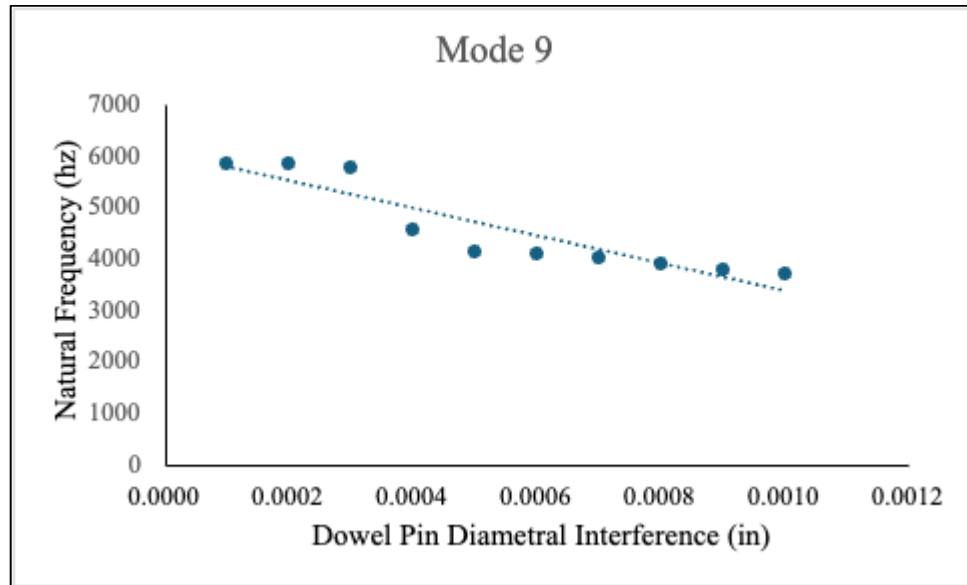
Close-up view of female plate with holes drilled out.

Appendix B: Free-Free ANSYS Iteration Results









Appendix C: Final Budget

The final budget spreadsheet for all tooling, material, and equipment bought to complete this project. Most of the expense came from the tooling required to machine the verification prototype.

Item No.	Description	Part Number	Date	Supplier	Unit Cost	Quantity	Total	Funding Source
1	Low-Carbon Steel Disc, 7"D, 1.5"L	7786T267	2/10/2026	McMaster-Carr	\$ 74.87	4	\$ 299.48	Solar Turbines
2	Carbon Steel Dowel Pins, 3/16"D	98372A117	2/10/2026	McMaster-Carr	\$ 31.26	1	\$ 31.26	Solar Turbines
3	Carbide Drill Bit, 11/64"D	27515A42	2/10/2026	McMaster-Carr	\$ 23.03	1	\$ 23.03	Solar Turbines
4	High-Speed Steel Reamer, 0.1865"D	3087A52	2/10/2026	McMaster-Carr	\$ 25.95	1	\$ 25.95	Solar Turbines
5	High-Speed Steel Reamer, 0.1855"D	2995A212	2/10/2026	McMaster-Carr	\$ 30.46	1	\$ 30.46	Solar Turbines
6	6061 Aluminum Vise Jaws, 8"W, 1.5"L	1005N35	2/10/2026	McMaster-Carr	\$ 64.27	1	\$ 64.27	Solar Turbines
7	Low-Carbon Steel Sheet, 12x12", 0.25"W	6544K25	2/10/2026	McMaster-Carr	\$ 84.38	1	\$ 84.38	Solar Turbines
8	Torx Head Screws, 3/8"-16, 1"L	92610A624	2/10/2026	McMaster-Carr	\$ 13.68	1	\$ 13.68	Solar Turbines
9	Low-Carbon Steel Disc, 7"D, 1.5"L	7786T267	2/22/2026	McMaster-Carr	\$ 74.87	4	\$ 299.48	Solar Turbines
10	High-Speed Steel Reamer, 0.1870"D	2995A213	2/22/2026	McMaster-Carr	\$ 33.49	1	\$ 33.49	Solar Turbines
11	High-Speed Steel Reamer, 0.1872"D	2777A231	2/22/2026	McMaster-Carr	\$ 32.10	1	\$ 32.10	Solar Turbines
12	Multipurpose Aluminum Rod, 0.5"D	8974K28	2/22/2026	McMaster-Carr	\$ 6.70	1	\$ 6.70	Solar Turbines
13	Carbide Square End Mill, 4 Flute, 0.5"D	8923A43	2/22/2026	McMaster-Carr	\$ 112.10	1	\$ 112.10	Solar Turbines
14	Countersink Drill-Point, HSS, 2"L	2915A13	2/22/2026	McMaster-Carr	\$ 7.91	1	\$ 7.91	Solar Turbines
15	HSS Shank Drill Bit	KFDRSD1	2/22/2026	DrillsandCutters	\$ 25.95	1	\$ 25.95	Solar Turbines
16	Work Gloves, M, L, XL	91165T16	2/22/2026	McMaster-Carr	\$ 20.34	3	\$ 61.02	Solar Turbines
17	HSS Tap, M3x0.5mm, 5/8" Thread L	8305A12	2/22/2026	McMaster-Carr	\$ 11.42	1	\$ 11.42	Solar Turbines
18	6061 Aluminum Vise Jaws, 6"W, 2"L	1005N24	2/24/2026	McMaster-Carr	\$ 58.58	2	\$ 117.16	Solar Turbines
19	Carbide Square End Mill, 4 Flute, 0.5"D	8923A43	3/3/2026	McMaster-Carr	\$ 112.10	1	\$ 112.10	Solar Turbines
20	Carbide Ball End Mill, 4 Flute, 3/8"D	8887A462	3/3/2026	McMaster-Carr	\$ 52.02	1	\$ 52.02	Solar Turbines
21	Carbide Square End Mill, 4 Flute, 0.5"D	8928A561	4/7/2026	McMaster-Carr	\$ 67.59	2	\$ 135.18	Solar Turbines
22	6061 Aluminum Vise Jaws, 6"W, 2"L	1005N24	4/7/2026	McMaster-Carr	\$ 58.58	1	\$ 58.58	Solar Turbines
23	HSS Reamer, 0.1865"D	3087A52	4/21/2026	McMaster-Carr	\$ 27.24	1	\$ 27.24	Solar Turbines
24	Carbide Ball End Mill, 4 Flute, 3/8"D	8887A462	4/21/2026	McMaster-Carr	\$ 52.05	1	\$ 52.05	Solar Turbines
25	Low-Carbon Steel Disc, 7"D, 1.5"L	7786T267	4/21/2026	McMaster-Carr	\$ 74.87	1	\$ 74.87	Solar Turbines
26	Low-Carbon Steel Sheet, 12x12", 0.25"W	6544K25	4/21/2026	McMaster-Carr	\$ 84.38	1	\$ 84.38	Solar Turbines
27	Welding Anti-Splatter Spray	KR106AE01	4/24/2026	Amazon	\$ 21.99	1	\$ 21.99	Solar Turbines
28	HSS Reamer, 0.2140"D	8803A411	5/5/2026	McMaster-Carr	\$ 26.72	1	\$ 26.72	Solar Turbines
29	4140 Alloy Steel Dowel Pin, 7/32"D, 5/8"L	98381A387	4/27/2026	McMaster-Carr	\$ 2.49	15	\$ 37.35	Solar Turbines
30	Cobalt Steel Reamer, 0.2180"D	8324A72	4/27/2026	McMaster-Carr	\$ 45.18	1	\$ 45.18	Solar Turbines
31	Carbide Steel Reamer, 0.2170"D	8307A405	4/27/2026	McMaster-Carr	\$ 61.00	1	\$ 61.00	Solar Turbines
32	0.25x0.25-0.5 Carbide 4Fl	93029221	3/10/2026	MSC	\$ 54.41	1	\$ 54.41	Solar Turbines
TOTAL							\$2,122.91	

Table of all purchases.

Appendix D: User Manual

The following appendix consists of a full user manual for the ANSYS model. This user manual will be helpful for understanding and making changes to the ANSYS model for the benefit of whomever wishes to use it.

1. Introduction

The Compressor Rotor Joints and Fits Study is a senior project completed by Cal Poly mechanical engineering students and overviewed by Solar Turbines. The project consists of creating a reliable Ansys model that simulates the modal frequencies of the compressor's rotor plates held together via pilot interference fits and dowel pins. The goal of creating an Ansys model was to be able to study the effect increased levels of interference had on the rotor plate's overall stiffness, aiming to increase plate stiffness overall.

The accuracy of the Ansys model has been validated with physical testing, of which any minor discrepancies between the physical model and Ansys model are explained below.

2. System Requirements

2.1 Hardware Requirements

The computational performance of ANSYS simulations is highly dependent on system hardware. Insufficient hardware may result in long solve times or solver failure.

Minimum requirements:

- **Memory (RAM):**
 - At least **16 GB RAM** is required to run the model reliably
- **Processor (CPU):**
 - Multi-core processors are recommended for improved solver performance
 - ANSYS recommends high clock speed CPUs for structural simulations
- **Graphics (GPU):**
 - A dedicated GPU is recommended for improved visualization and post-processing

These recommendations are consistent with ANSYS guidelines for medium-scale finite element models.

2.2 Software Requirements

The model was developed using a specific ANSYS environment, and compatibility must be maintained.

Required software:

- **ANSYS 2025**
- Required analysis systems:
 - Static Structural
 - Modal (Prestressed Modal Analysis)

Model capabilities and constraints:

- The system must support meshes of **at least 125,000 nodes**
- Lower mesh densities may result in inaccurate stress and frequency predictions

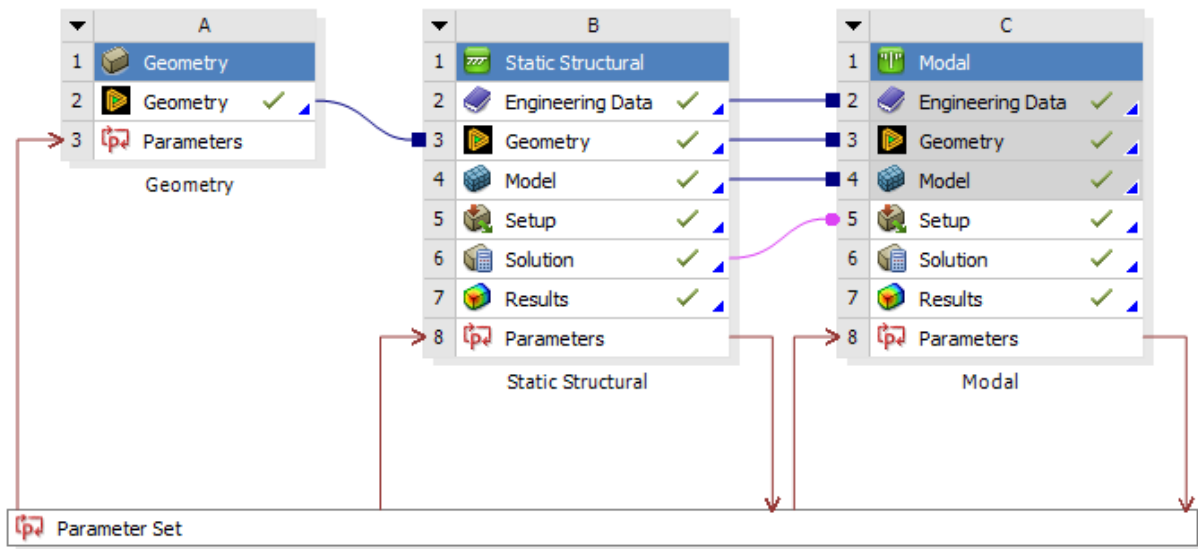


Figure 1. ANSYS Workbench Setup

3. Safety Information and Model Validation

Although this system is a simulation, improper use presents a significant engineering risk due to the potential for incorrect conclusions.

3.1 Risks Associated with Improper Use

Failure to follow the defined setup procedure may result in:

- Non-physical stress values
- Incorrect natural frequency predictions
- Misleading conclusions that could impact design decisions

3.2 Required Validation Procedure

Before relying on simulation results, the following validation steps must be performed:

- Conduct **closed-form theoretical calculations** to estimate:
 - Tangential stress around dowel pins and pilot fit
 - Expected stress magnitudes
- Compare simulation results to theoretical values:
 - Results should match within a reasonable order of magnitude

- Large discrepancies indicate setup errors and singularities
- Only proceed to modal analysis if:
 - Static structural results are physically reasonable
 - Contact stresses align with theoretical expectations

3.3 Proper Use of Results

- Simulation results should **not** be used without verification
- Engineering judgment must always accompany numerical results
- The model is intended as a **supplement to analysis**, not a replacement

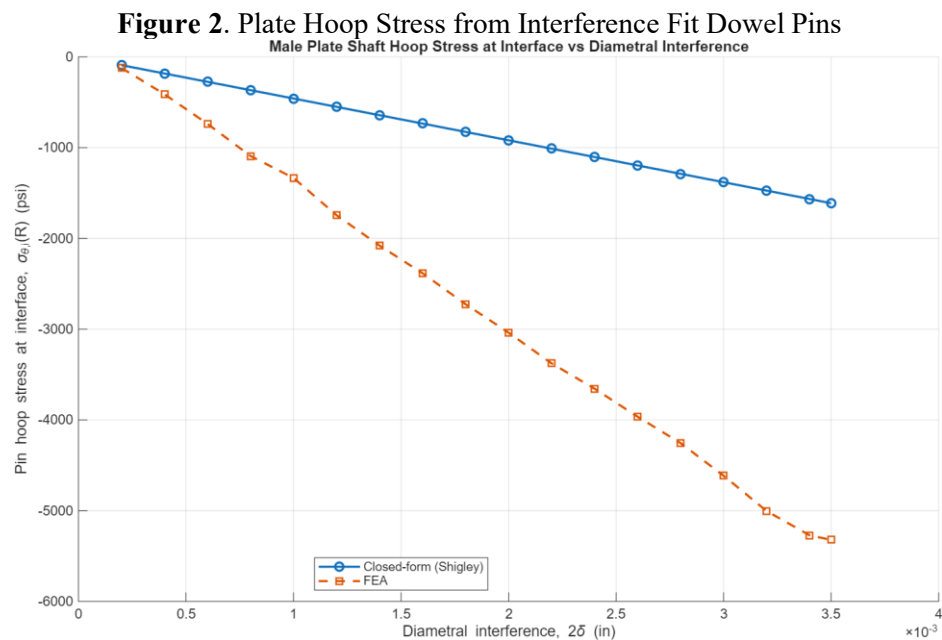
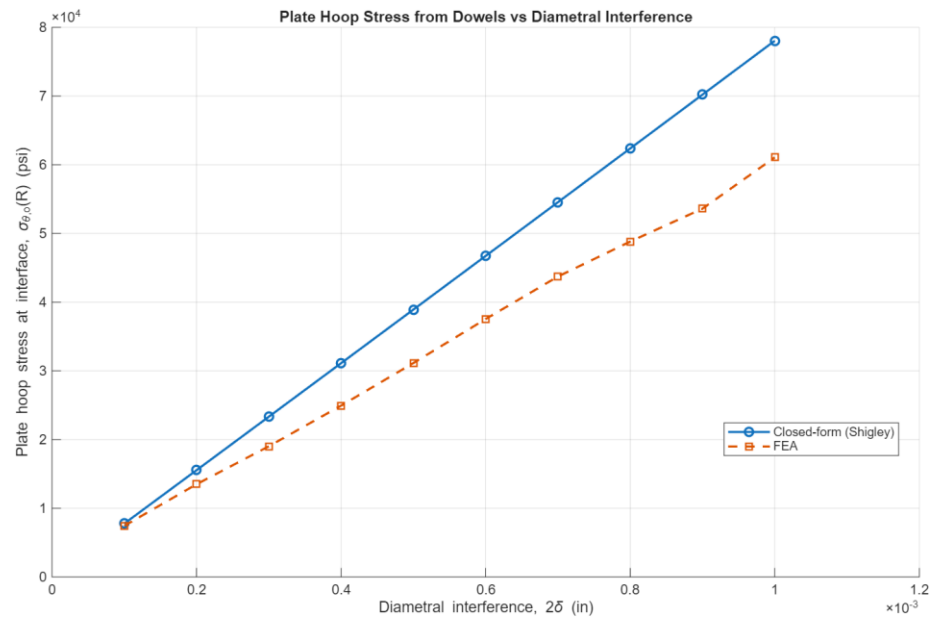


Figure 3. Male Pilot Feature Hoop Stress from Interference Fit

4. File Structure and Organization

Proper file handling is critical for successful model transfer and operation.

4.1 Recommended File Format

- **.wbpz (ANSYS Archive File):**
 - Contains all necessary project files
 - Preserves links between geometry, mesh, and solution data
 - Ensures compatibility across different systems

4.2 Limitations of .wbpj Files

- The **.wbpj** file references local file paths from the original system
- When transferred:
 - Links to geometry and data files may break
 - The model may fail to open or solve

For this reason:

- The project **must be shared as a .wbpz file**
- This ensures the model can be opened and executed without errors

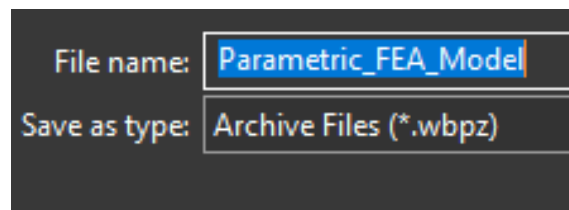


Figure 4. WBPZ File Type to Share ANSYS Files

5. Getting Started

5.1 Opening the Model in ANSYS Workbench

To begin using the model:

- Open **ANSYS Workbench**
- Load the provided **.wbpz** file

Upon successful loading, the following systems should be visible and connected:

- Geometry
- Static Structural
- Modal Analysis

Additionally:

- A **parameterization system** should be visible
- This indicates the model is properly configured for iterative studies

5.2 Geometry and Parameterization

The geometry is fully parameterized to allow efficient testing of different interference fits.

Steps to access and modify geometry:

- Double-click **Geometry** to open DesignModeler
- The plate assembly will be displayed

To modify parameters:

- Select **Parameters** from the toolbar
- A table will appear at the bottom of the screen
- Modify geometric dimensions as needed

Important notes:

- Parameterization allows rapid testing of design variations
- Each change must be updated by clicking **Generate**
- Failure to regenerate will result in outdated geometry

After completing changes:

- Save and exit DesignModeler
- Return to ANSYS Workbench

ID	Parameter Name	Value	Unit
1	Input Parameters		
2	Geometry (A1)		
3	P11	Dowel_radius	0.09375 in
4	P12	Hole_Radius	0.0937 in
5	P16	Male_Pilot_Length	0.146 in
6	P17	Female_Pilot_Length	0.146 in
7	P18	Female_Radial_Interference	3.1494 in
8	P19	Male_Radial_Interference	3.1495 in
9	New input parameter	New name	New expression
10	Output Parameters		
11	Static Structural (B1)		
12	P20	Mesh Nodes	1.164E+05
13	Model (C1)		
14	P7	Total Deformation Reported Frequency	3076.4 Hz
15	New output parameter	New name	New expression
16	Charts		
17	Nodes		
18	Plot Dowel Stress		
19	Natural Frequency		

Name	Units	P11 - Dowel_radius	P12 - Hole_Radius	P16 - Male_Pilot_Length	P17 - Female_Pilot_Length	P18 - Female_Radial_Interference
DP 1 (Current)	in	0.09375	0.0937	0.146	0.146	3.1494
DP 2	in	0.09375	0.0937	0.15	0.15	3.1494
DP 3	in	0.09375	0.0937	0.175	0.175	3.1494
DP 4	in	0.09375	0.0937	0.2	0.2	3.1494
DP 5	in	0.09375	0.0937	0.225	0.225	3.1494
DP 6	in	0.09375	0.0937	0.25	0.25	3.1494
DP 7	in	0.09375	0.0937	0.275	0.275	3.1494
DP 8	in	0.09375	0.0937	0.3	0.3	3.1494
DP 9	in	0.09375	0.0937	0.325	0.325	3.1494
DP 10	in	0.09375	0.0937	0.35	0.35	3.1494
DP 11	in	0.09375	0.0937	0.375	0.375	3.1494

Figure 5. Parameter Table Interface

6. Model Setup and Verification

Proper verification of the model setup is critical before running any simulation.

6.1 Contact Conditions

Contact definitions significantly affect both stress and modal results.

Verification steps:

- Expand the “**Contacts**” section in the Static Structural model
- Review each defined contact region

Correct setup should include:

- Contacts between:
 - Dowel pins and plates
 - Pilot interference regions

Contact properties:

- Type: **Frictional**
- Coefficient of friction: **0.2 (steel-on-steel)**

Warning signs of incorrect setup:

- Large flat faces being highlighted as contacts
- Missing contact regions

These errors will result in inaccurate natural frequency predictions.

6.2 Mesh Verification

The mesh must be inspected to ensure numerical accuracy.

Mesh characteristics:

- Type: **Tetrahedral Quadratic Elements**
 - Reduces stress singularities
 - Improves accuracy near contact regions

Inspection procedure:

- Hide components to isolate critical regions
- Examine mesh near dowel pin holes

A correct mesh should:

- Follow a **circular pattern** around holes
- Avoid distorted or inward-pointing elements

If mesh irregularities are present:

- Remeshing is required before proceeding

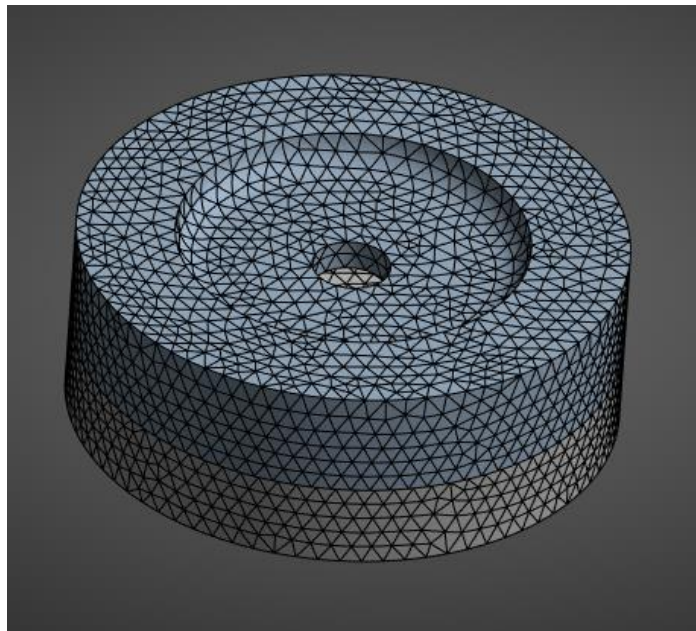


Figure 6. Tetrahedral Quadratic Mesh Appearance of Assembly

6.3 Boundary Conditions

Boundary conditions must replicate real-world testing conditions.

Required setup:

- Turn on **weak springs** and adjust stiffness as needed to aid convergence
- Range of 500 – 2000 lbf/in is recommended

This represents:

- A **Free – Free** set up as this is how the test should be physically tested

Incorrect boundary conditions will:

- Produce unrealistic deformation
- Lead to incorrect modal results

Solver Controls	
Solver Type	Program Controlled
Weak Springs	On
Spring Stiffness	Manual
Spring Stiffness Value	50. lbf/in
Solver Pivot Checking	Program Controlled
Large Deflection	On

Figure 7. Turn On Weak Springs in “Analysis Settings”

6.4 Material Properties

Material properties must be verified before running the analysis.

Key parameters:

- Elastic Modulus
- Poisson’s Ratio
- Density

These values:

- Directly affect both stress and modal results
- Must match the material being analyzed

7. Running the Simulation

7.1 Static Structural Analysis

The static structural analysis must be completed first.

Procedure:

- Solve the static structural model
- Review stress results

Validation criteria:

- Stress values should remain within realistic ranges
- Extremely high values (e.g., >100,000–1,000,000) indicate:
 - Mesh singularities
 - Improper contact definitions

If issues are detected:

- Refine mesh and re-run analysis

7.2 Modal Analysis Setup

The modal analysis must incorporate prestress effects.

Required configuration:

- Analysis type: **Prestressed Modal Analysis**
- Prestress source: **Static Structural**

This ensures:

- Contact stresses influence natural frequency results

7.3 Solving the Model

- Run the full model for a single configuration
- Verify results before performing parametric studies

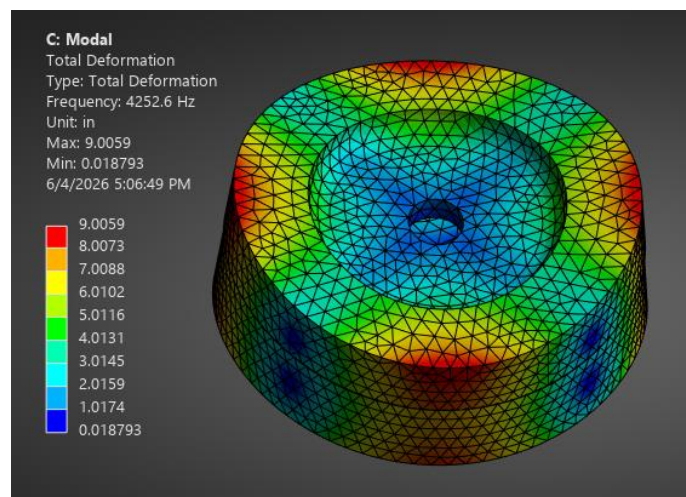


Figure 8. A Successful Run Will Yield the Mode Shape Asked For

8. Parameterization and Iterative Analysis

The parameterization tool enables efficient evaluation of multiple configurations.

Using the Parameter Table

Steps:

- Open **Parameters** in Workbench
- Modify:
 - Geometry dimensions
 - Parameterizing dimensions must be done in Geometry
 - Interference values
 - Output parameters (e.g., natural frequency)
 - Adding output parameters must be done in ANSYS Mechanical
- Add multiple design points

To execute:

- Click “**Update All Design Points**”

Benefits:

- Automates multiple simulations
- Significantly reduces analysis time compared to manual iteration

Table of Design Points						
	E	F	G	H	I	J
	P17 - Female_Pilot_Length	P18 - Female_Radial_Inteference	P19 - Male_Radial_Interference	P7 - Total Deformation Reported Frequency	P20 - Mesh Nodes	Retain
	in	in	in	Hz		
1	0.146	3.1494	3.1495	3076.4	1.1646E+05	☒
2	0.15	3.1494	3.1495	3158.9	1.1631E+05	☐
3	0.175	3.1494	3.1495	3273.3	1.1675E+05	☐
4	0.2	3.1494	3.1495	3036.7	1.1961E+05	☐
5	0.225	3.1494	3.1495	3436.4	1.2087E+05	☐
6	0.25	3.1494	3.1495	3765.7	1.2174E+05	☐
7	0.275	3.1494	3.1495	3534.5	1.2233E+05	☐
8	0.3	3.1494	3.1495	3819.1	1.258E+05	☐
9	0.325	3.1494	3.1495			☐
10	0.35	3.1494	3.1495			☐
11	0.375	3.1494	3.1495			☐
12	0.4	3.1494	3.1495			☐
13						☐
14						☐
*						☐

Figure 9: Design Point Table

9. Troubleshooting

Common issues and solutions include:

- **Unrealistic stress values**
 - Cause: Mesh singularities
 - Solution: Refine mesh
- **Incorrect modal results**
 - Cause: Improper contacts or boundary conditions
 - Solution: Recheck setup
- **File errors when opening**
 - Cause: Use of .wbpj file
 - Solution: Use .wbpz archive

10. Model Limitations

This model includes several inherent limitations:

- Results are sensitive to:
 - Contact definitions
 - Mesh quality
 - Boundary conditions
- Simplifications may not capture:
 - All real-world behaviors
 - Manufacturing tolerances

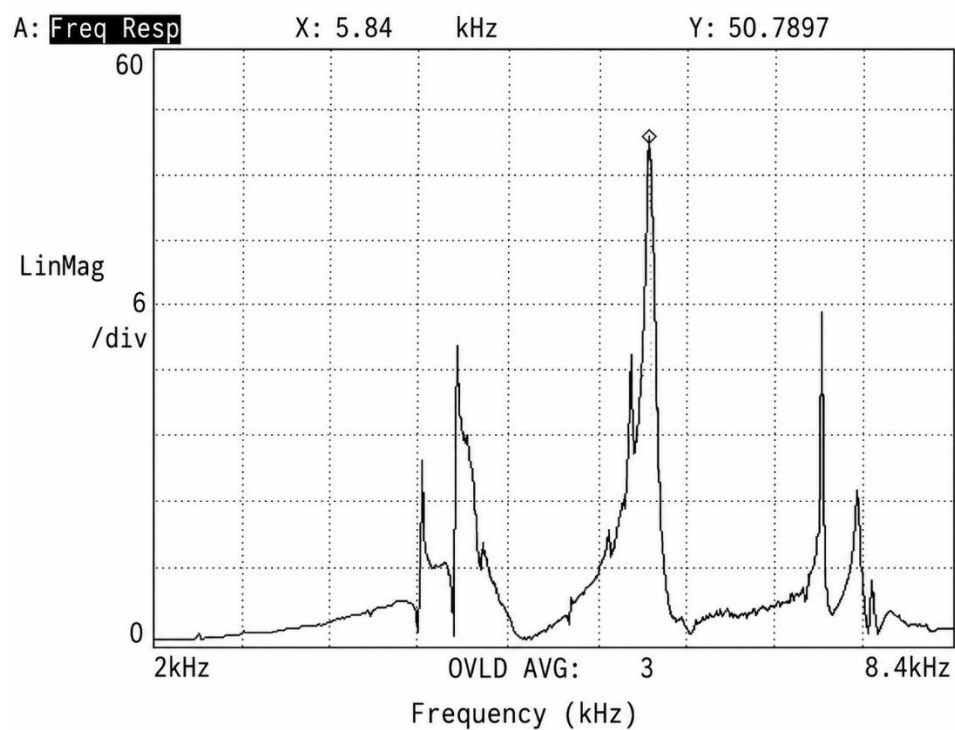
The model should be used as a **supporting analysis tool**, not a standalone decision-making method.

11. Conclusion

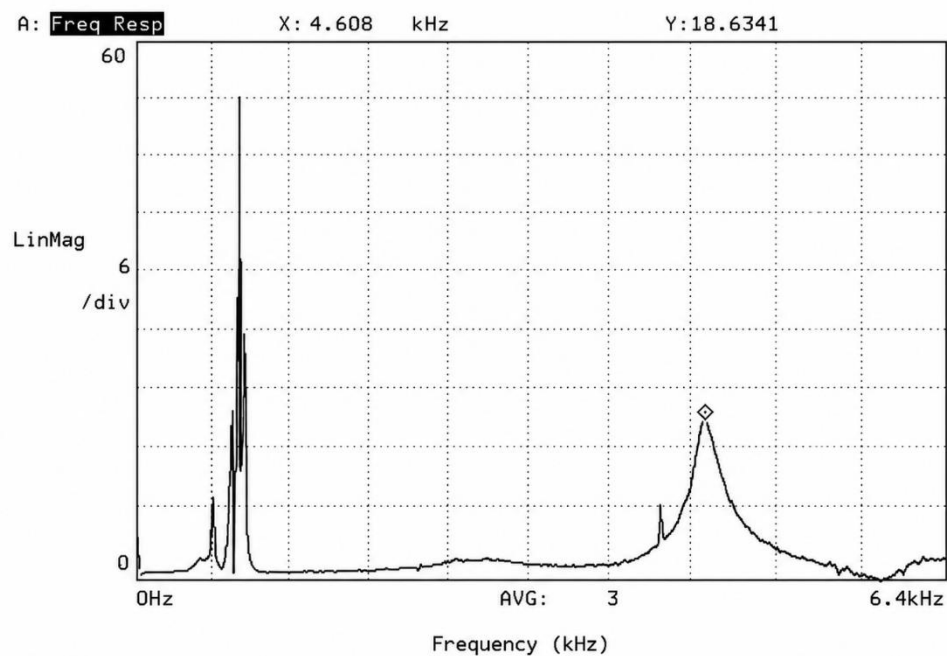
This ANSYS model provides a structured and efficient method for analyzing the compressor plate assembly when used correctly. Accurate results depend on careful setup, validation, and parameter control. Users must follow all procedures outlined in this manual to ensure reliable and meaningful outcomes.

Appendix E: Experimental Test Results

The test results shown in this appendix are from the free-free testing conducted after the fixed-free testing that was deemed invalid. These results include direct bode plots from the spectrum analyzer along with tables of numerical results.



Configuration 1 bode plot from free-free testing.



Configuration 2 bode plot from free-free testing.

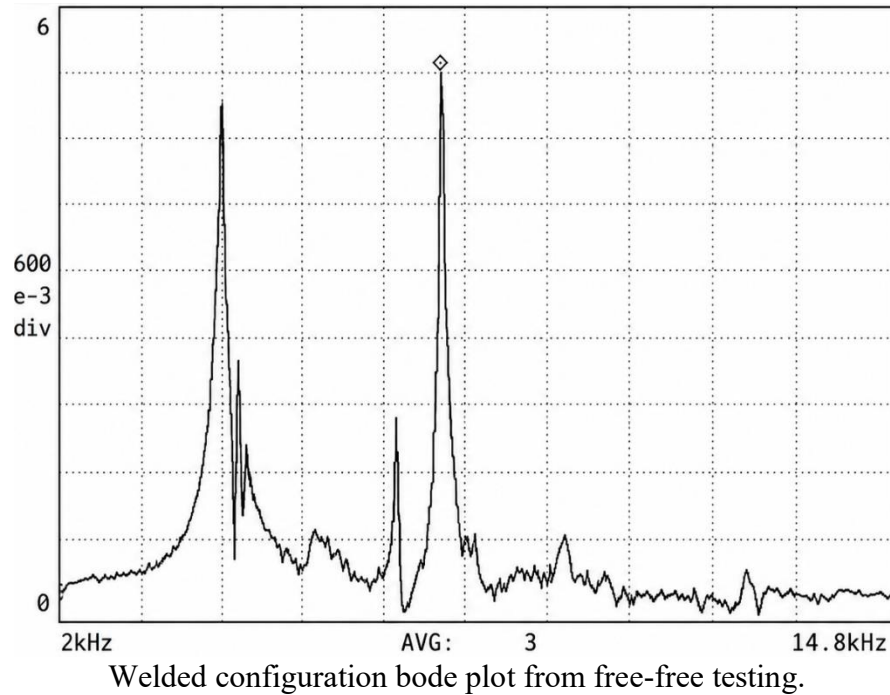


Table of ANSYS suggested values for first (mode 7), second (mode 8), and third (mode 9) frequencies.

Configuration	Mode 6 (hz)	Mode 7 (hz)	Mode 8 (hz)	Mode 9 (hz)
1	0.196267677	4275.014956	4397.17501	5655.720325
2	0.236836986	4268.660265	4826.45801	5765.515084
Welded	7.55E-04	5891.8	7395.6	7592

Raw testing results for second natural frequency.

Second Mode				
Natural Frequency [kHz]				
Test #	Config 1	Config 2	Welded	Measurement Type
1	4.384	4.608	4.880	Stationary
2	4.247	4.501	4.603	Roving Hammer
3	4.555	4.443	4.720	Roving Accel.

Raw testing results for third natural frequency.

Third Mode				
Natural Frequency [kHz]				
Test #	Config 1	Config 2	Welded	Measurement Type
1	5.888	-	8.016	Stationary
2	5.392	5.935	8.033	Roving Hammer
3	5.349	5.877	7.931	Roving Accel.

Second natural frequency comparison chart.

<u>Second Natural Frequency Comparison</u>	Configuration 1	Configuration 2	Welded Configuration
ANSYS	4.397 kHz	4.826 kHz	5.892 kHz
Experimental	4.384 kHz	4.608 kHz	4.88 kHz

Third natural frequency comparison chart.

<u>Third Natural Frequency Comparison</u>	Configuration 1	Configuration 2	Welded Configuration
ANSYS	5.655 kHz	5.765 kHz	7.592 kHz
Experimental	5.392 kHz	5.935 kHz	8.033 kHz

Appendix F: References

A list of all references used for this report in the order they are referenced in the paper.

- [1] Pijpers, R. J. M., and H. M. Slot. "Friction Coefficients for Steel to Steel Contact Surfaces in Air and Seawater." *Journal of Physics Conference Series*, vol. 1669, no. 1, Oct. 2020, p. 012002. <https://doi.org/10.1088/1742-6596/1669/1/012002>.
- [2] 9.6.2. *Contact Formulation Theory*.
ansyshelp.ansys.com/public/account/secured?returnurl=/Views/Secured/corp/v242/en/wb_sim/ds_contact_theory.html.
- [3] "Richard Budynas, Keith Nisbett _ Shigley's Mechanical Engineering Design _ McGraw Hill Education (2019) .Pdf." *Google Docs*, drive.google.com/file/d/1ftQZlCfV6aSIoenQ0z_zvwT4sByot1DV/view.

Compressor Rotor Joints and Fits Interference Study

Aubrey Perez – Bela Jimenez – Isaiah Guerrero – Jireh Velasquez

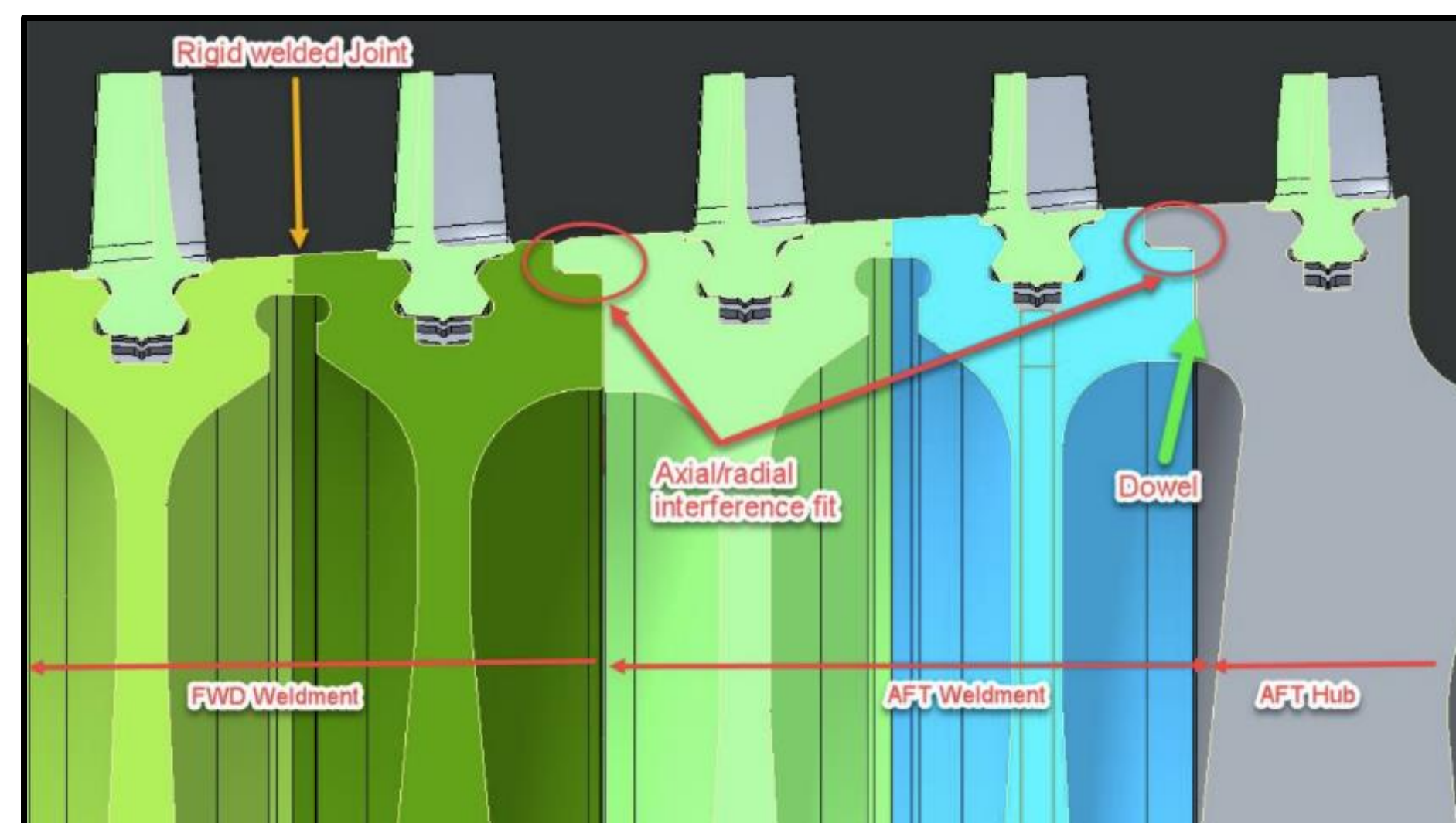
Background

Solar Turbines

- Solar Turbines uses stacked rotor plates in their compressors
- Plates are joined using interference pilot fits and dowel pins
- The plates have experienced radial shifting after compressor operation
- Stiffness changes are predicted to affect natural frequency response and radial shifting

The Problem

Rotor plates shifting radially under operating loads potentially affect compressor stability and vibration behavior.



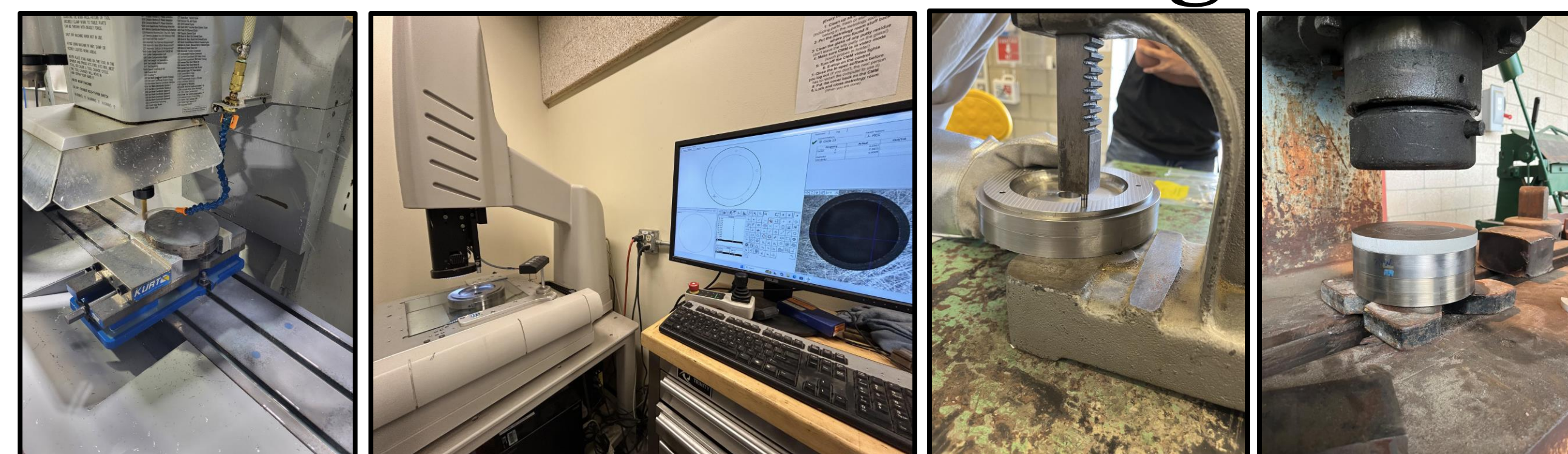
*Side schematic of rotor plates' connection with pilot fits and dowel pins

Objective

Investigate how interference-fit joints affect compressor assembly stiffness with three configurations:

- Fully welded configuration
- Low interference configuration (0.001in – 0.002in) with short pilot ledge overlap
- High interference configuration (0.002in – 0.004in) with long pilot ledge overlap
- Perform modal analysis in ANSYS to predict effects on stiffness from varying interference fits and pilot ledge lengths
- Manufacture and test a welded configuration and two interference configurations to validate ANSYS simulation

Manufacturing



Machining plates on CNC mill

CMM to measure critical dimensions

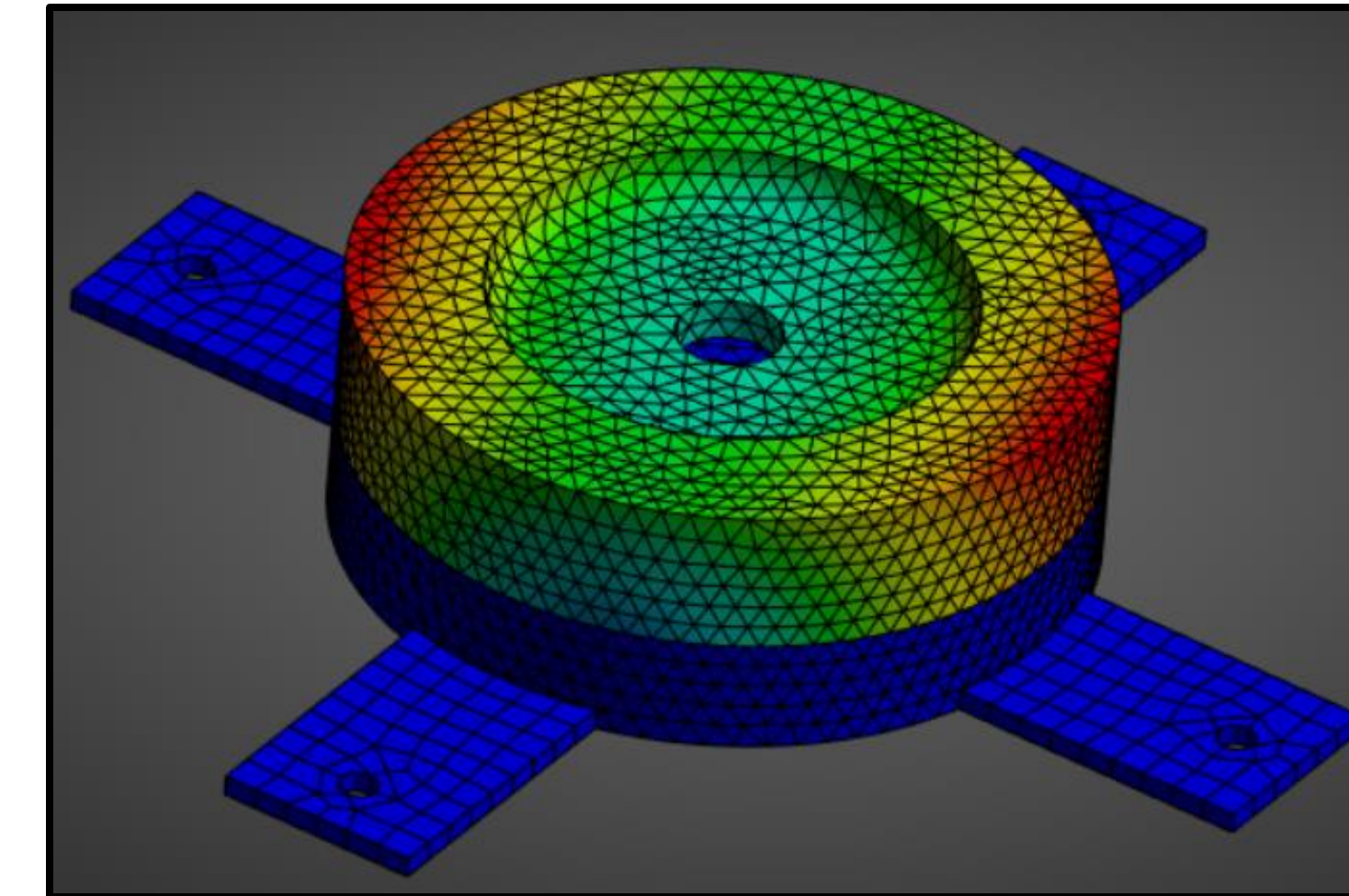
Pressing in dowel pins with arbor press

Pressing both plates with hydraulic press

Modeling & Testing

ANSYS Model

A model of two plates with all necessary connections was created in ANSYS for each configuration. Each model was then put through a modal analysis to theoretically determine the first 3 mode shapes. Having these 3 natural frequencies allowed us to draw conclusions on design decisions.

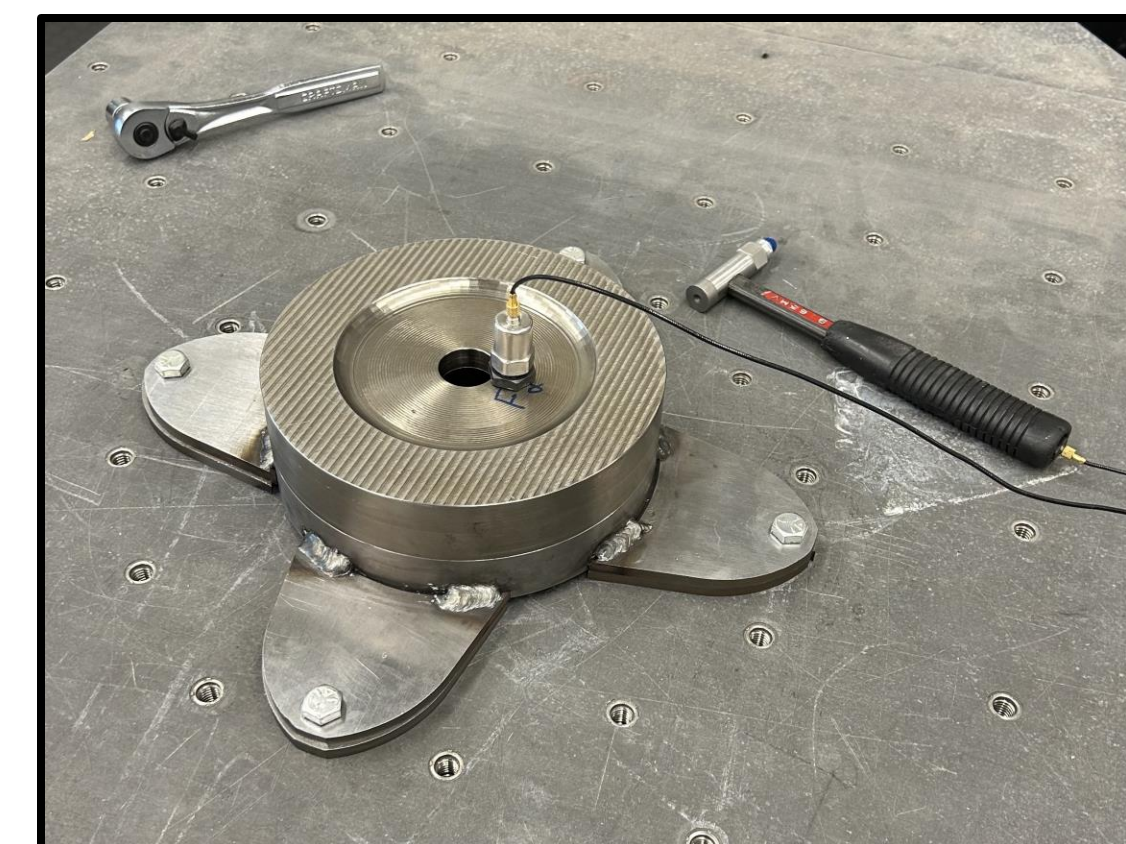


Ansys model of a plate configuration

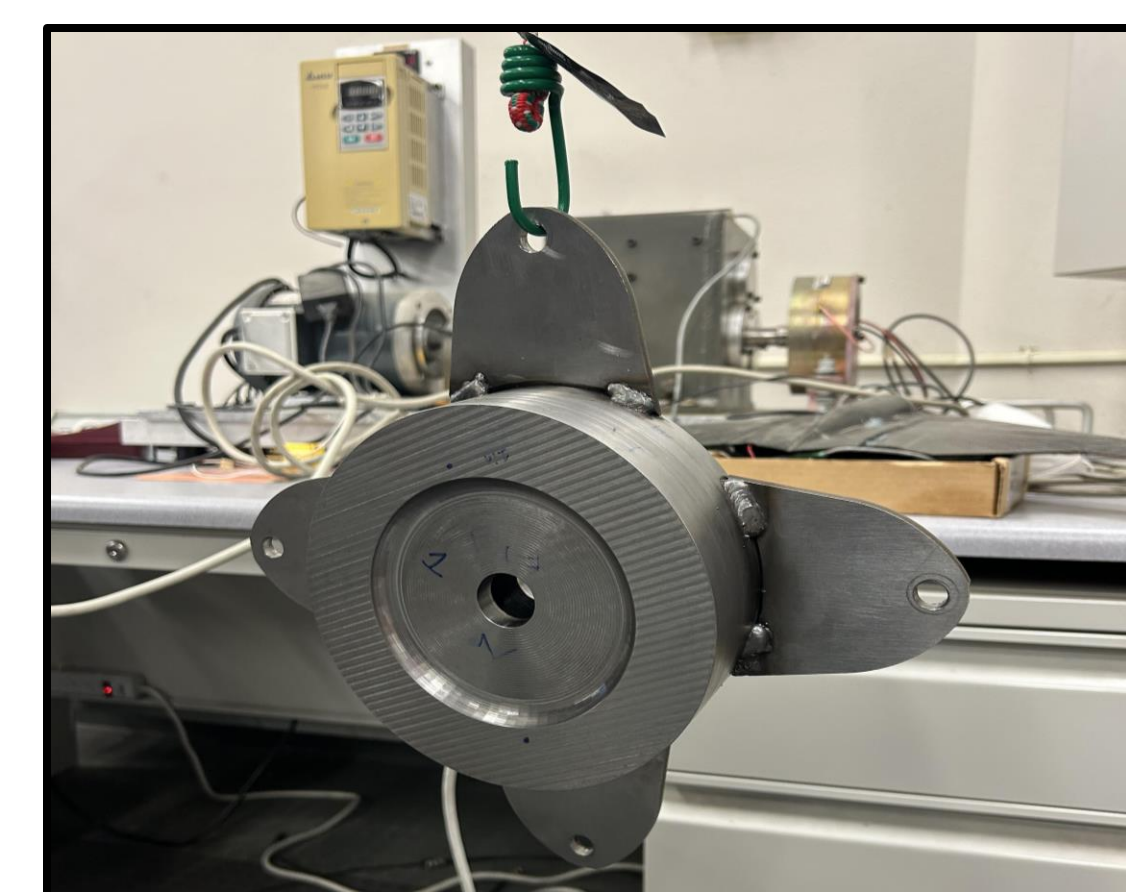
Physical Testing

The ANSYS simulation for mode shapes was validated and refined through modal impact hammer testing. The three plate configurations were put through two testing conditions:

- Free-Fixed Condition
- Free-Free Condition



Free-fixed testing condition

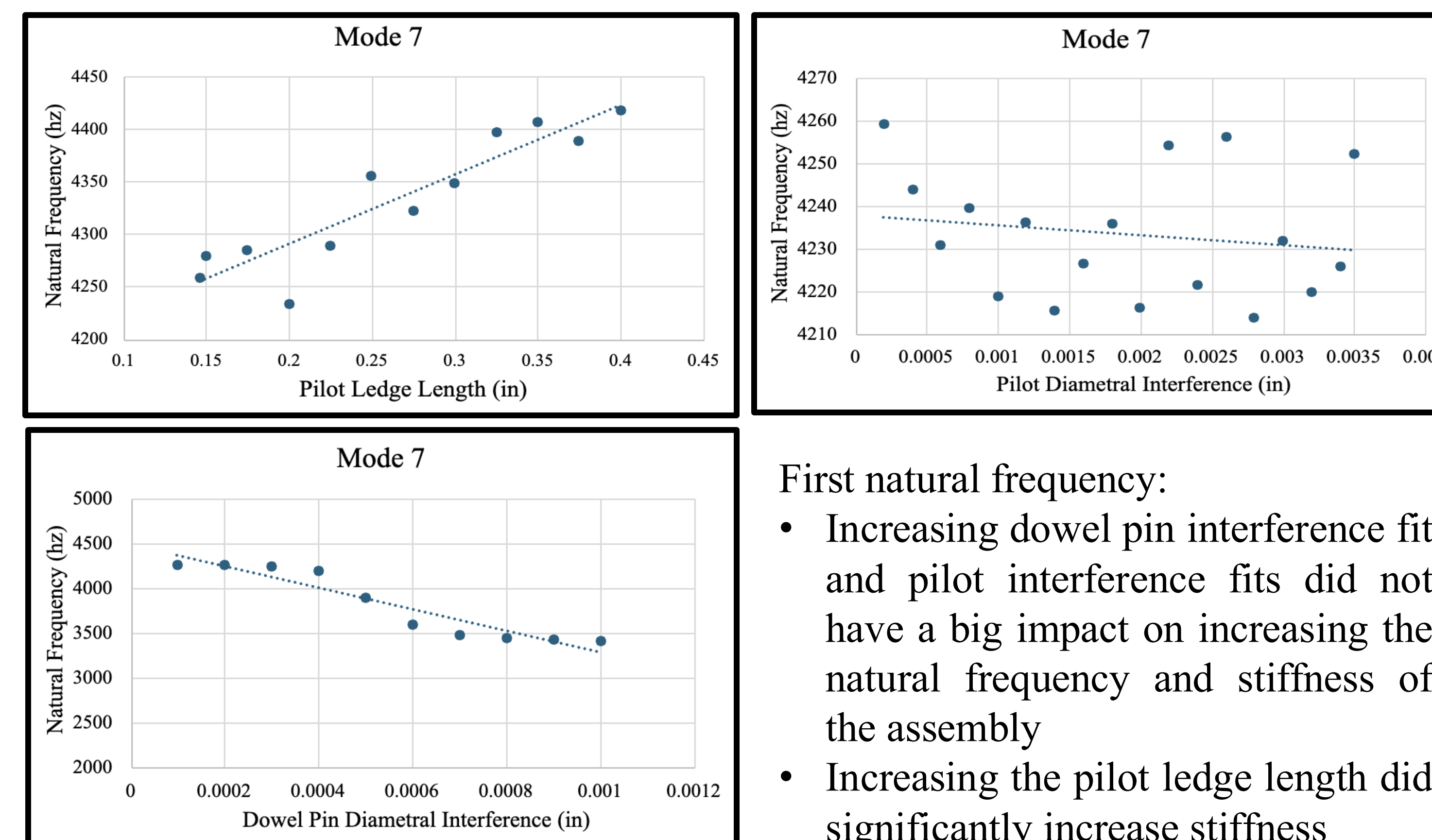


Free-free testing condition

A spectrum analyzer was used to read the natural frequencies from the impact hammer and accelerometer, allowing for a direct comparison with the ANSYS model. Three types of tests were conducted for each condition:

- Stationary Accelerometer/Hammer
- Roving Hammer/Fixed Accelerometer
- Roving Accelerometer/Fixed Hammer

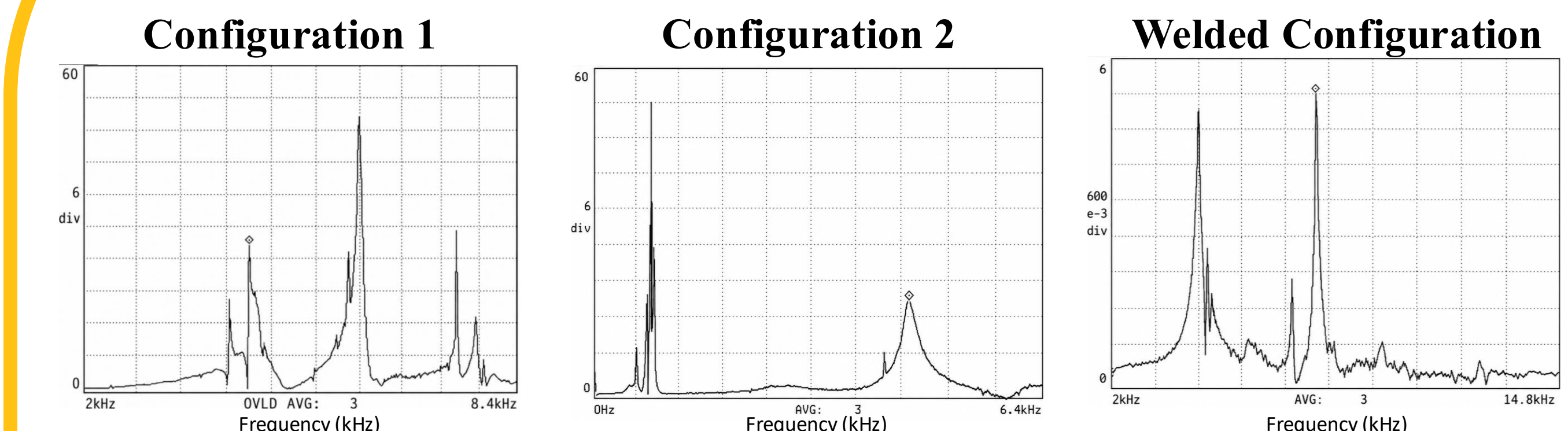
ANSYS Iterations



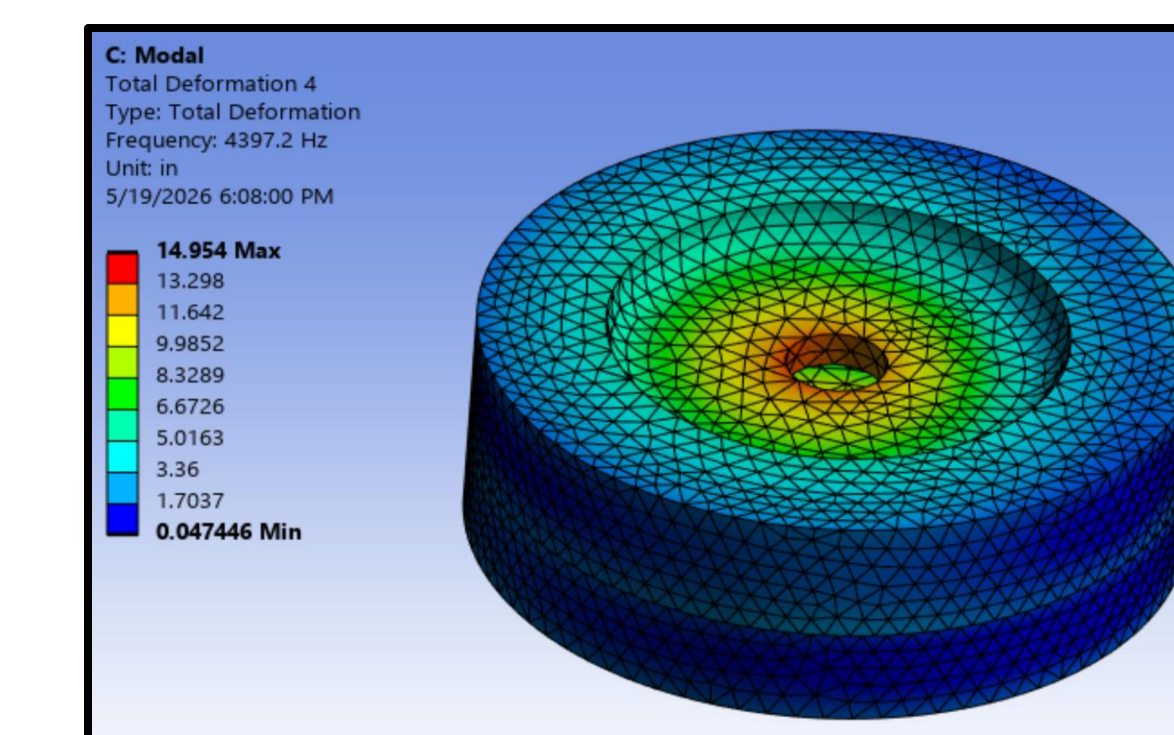
First natural frequency:

- Increasing dowel pin interference fit and pilot interference fits did not have a big impact on increasing the natural frequency and stiffness of the assembly
- Increasing the pilot ledge length did significantly increase stiffness

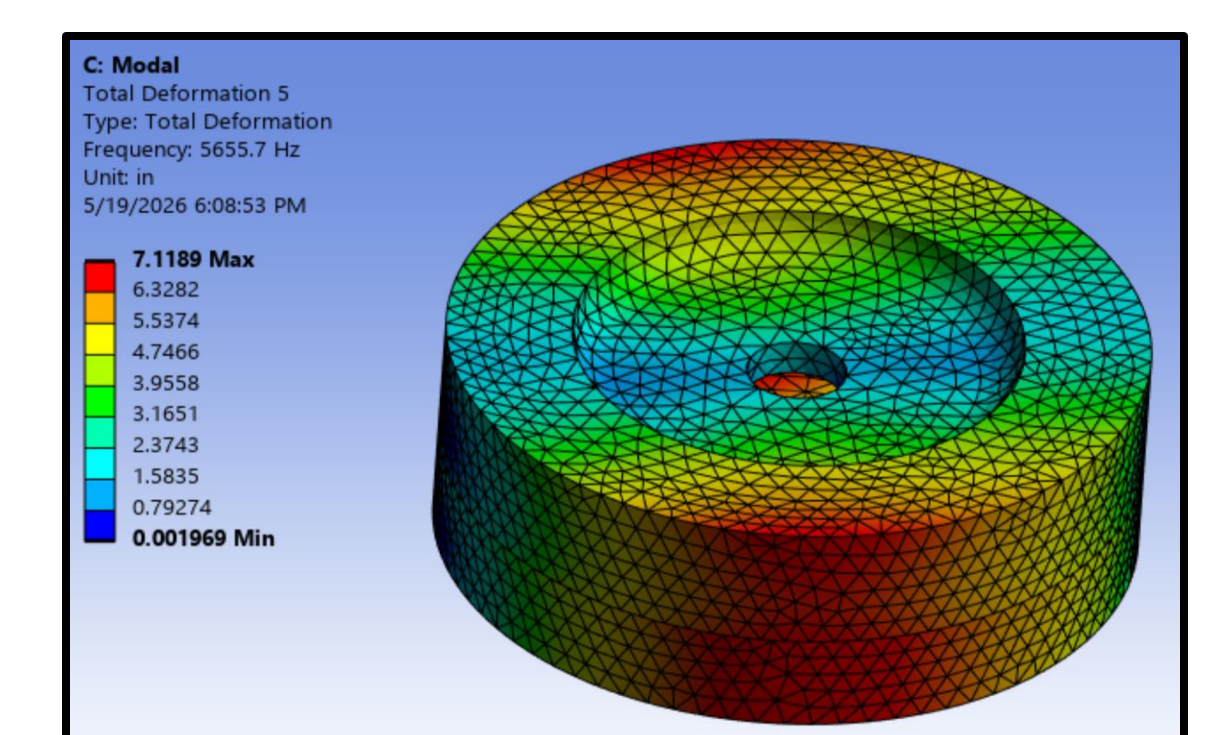
Results – Free-Free



Second Natural Frequency Comparison	Configuration 1	Configuration 2	Welded Configuration
ANSYS	4.397 kHz	4.826 kHz	5.892 kHz
Experimental	4.384 kHz	4.608 kHz	4.88 kHz
Percent Difference	0.29%	4.62%	18.78%
Third Natural Frequency Comparison	Configuration 1	Configuration 2	Welded Configuration
ANSYS	5.655 kHz	5.765 kHz	7.592 kHz
Experimental	5.392 kHz	5.935 kHz	8.033 kHz
Percent Difference	4.76%	3.06%	5.64%



Second Mode Shape



Third Mode Shape

Conclusion

- Increasing pilot ledge length was the main contributor to the increase in natural frequency and overall assembly stiffness
- Configuration 2 consistently produced higher frequencies than Configuration 1 in both ANSYS and experimental testing.
- The welded configuration produced the highest stiffness and natural frequencies overall.
- ANSYS predictions closely matched experimental trends, validating the simulation approach.
- Differences between simulation and testing were likely caused by manufacturing tolerances, material variation, and testing inconsistencies.